

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Electronics and Communication Engineering

CLASS NOTES

ANALOG ELECTRONICS

Course Code: EC031302 | L-T-P: 3-0-2 | Credits: 04

Prepared by
Dr. Girdhar Gopal
Faculty Advisor
Department of ECE, NIT Patna

Introduction to the Syllabus and its Significance

Introduction

Analog Electronics is a core course of the ECE programme. It builds on the basics learnt in Elements of Electronics Engineering and moves from single transistors to complete analog integrated circuits. The course explains how MOS transistors are combined to amplify signals, how these amplifiers behave at different frequencies, and how they are packaged as op amps and special-purpose ICs used in everyday electronic systems.

Brief Overview of the Units

- **Unit I – Transistor Amplifiers:** small-signal model, CS/CG/CD stages, frequency response, Miller effect, cascode and multistage amplifiers, power amplifier classes and current mirrors.
- **Unit II – Differential Amplifiers:** MOS differential pair, CMRR, offset, active load and design of the two-stage CMOS op amp.
- **Unit III – Operational Amplifier:** ideal and practical op amp, basic configurations, instrumentation amplifier, log/antilog amplifiers, multiplier and divider.
- **Unit IV – Feedback and Oscillators:** feedback topologies, stability and compensation, RC/LC oscillators, Schmitt trigger and waveform generators.
- **Unit V – Special-Purpose ICs:** 555 timer, 565 PLL and 566 VCO with their pin diagrams and applications.

The units follow a natural order: transistor → amplifier stage → differential pair → op amp → feedback circuits → complete ICs.

Significance of the Syllabus

- **The real world is analog.** Sound, temperature, light, pressure and biomedical signals are all analog. Every sensor, microphone and antenna needs analog circuits to amplify and condition its signal before a digital system can process it.
- **Foundation of analog and mixed-signal IC design.** Current mirrors, differential pairs and the two-stage CMOS op amp are the building blocks of ADCs, DACs, PLLs, voltage regulators and RF front-ends found in every chip.
- **Design thinking and trade-offs.** Students learn to balance gain, bandwidth, power, stability and noise, a skill needed in every engineering design.
- **Basis for later courses.** It supports Analog/VLSI Design, Communication Systems, Control Systems, Power Electronics, Instrumentation and Embedded Systems.
- **Practical and lab relevance.** Op amps, 555 timers, oscillators and PLLs are used directly in projects, laboratory work and products such as audio systems, signal generators, FM receivers and frequency synthesizers.
- **Career relevance.** With the growth of the semiconductor industry in India, there is strong demand for engineers in analog, mixed-signal and RF circuit design, testing and verification.

Expected Outcome

After this course, students will be able to analyse the frequency response of MOS amplifiers, understand the differential amplifier as the building block of the op amp, apply op amps and special ICs in practical circuits, and use feedback to make circuits stable and reliable.

Small Signal Analysis of BJT Amplifiers

Phase Shift in the CE Configuration

If a BJT is used in the **common emitter configuration**, a phase shift of 180° occurs between the input and output *voltage* waveforms.

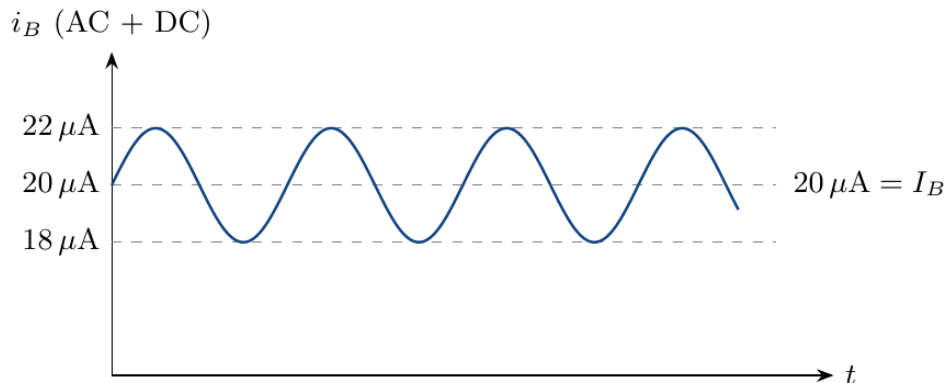
No phase shift is obtained between the input and output **currents**. The 180° shift applies to the voltage waveform only.

Notation for DC, AC and Total Quantities

Four different symbols are used for base current, and they mean different things:

Symbol	Meaning
I_B	DC value
i_b	instantaneous AC value
i_B	total value (AC + DC)
I_b	RMS AC value

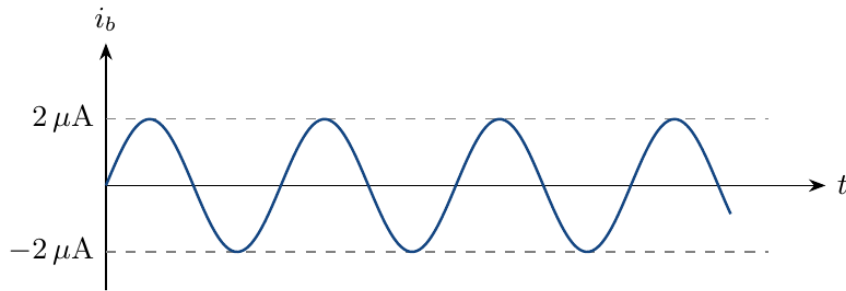
Total value i_B (AC + DC)



DC value

$$I_B = 20 \mu\text{A}$$

Instantaneous AC value i_b



RMS AC value

$$I_b = \frac{2}{\sqrt{2}} = \sqrt{2} \mu\text{A}$$

Signal Variation

Signal variation is the maximum (peak) change in the value of current or voltage when an AC input is applied.

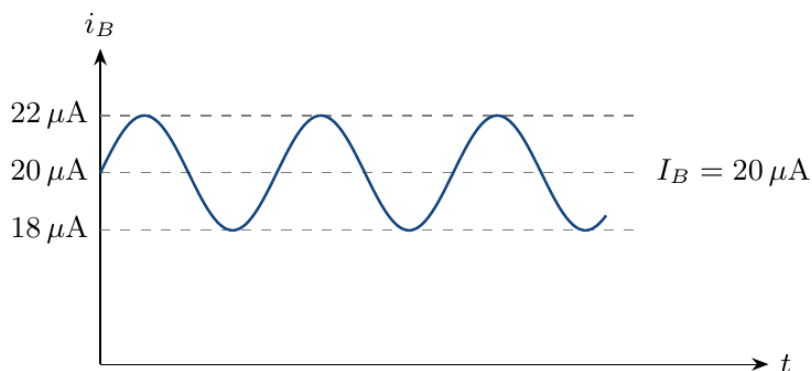
For the waveform above, the peak change in the output is only $2 \mu\text{A}$, since i_B swings between $18 \mu\text{A}$ and $22 \mu\text{A}$ about a DC level of $20 \mu\text{A}$.

According to signal variation, amplifiers are divided into:

- i. Small signal amplifiers
- ii. Large signal amplifiers

Small Signal Amplifier

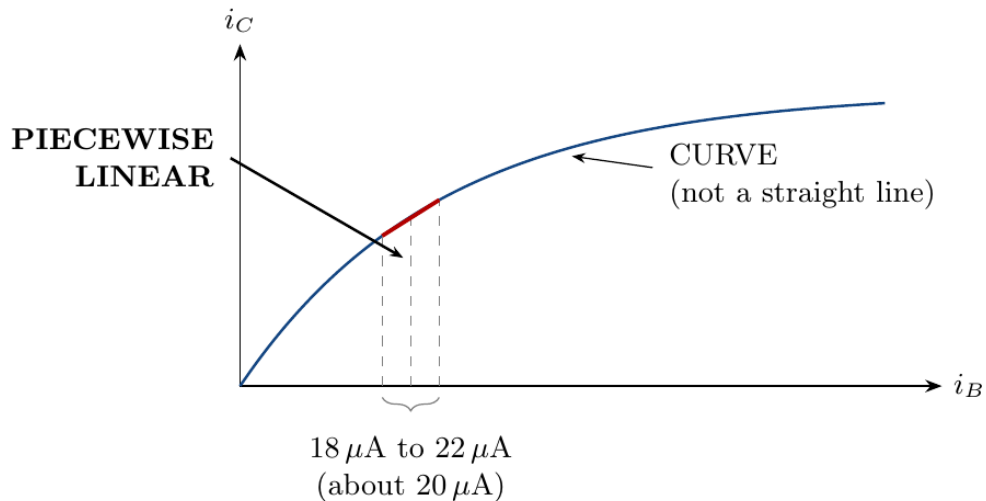
An amplifier in which the signal variation is **much smaller** than the DC value.



$$\Delta i_B = 2 \mu\text{A} \lll I_B = 20 \mu\text{A}$$

Why the BJT behaves linearly here

A BJT is a **non-linear device**, because its dynamic characteristic — the plot of i_C against i_B — is a curve, not a straight line.

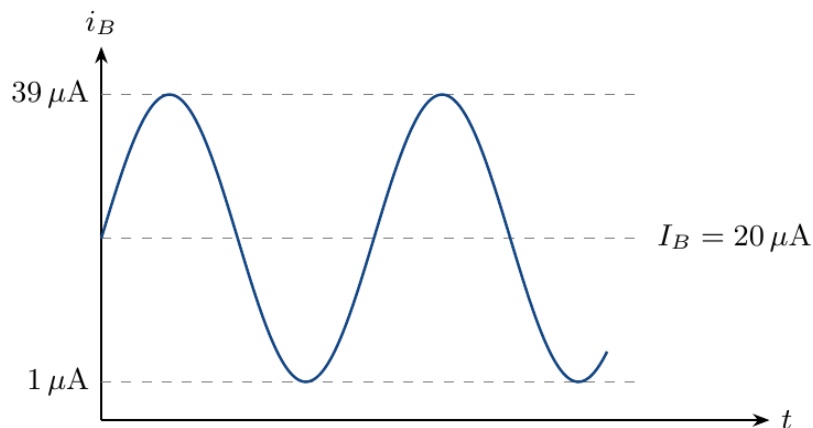


- In a small signal amplifier the BJT behaves as a **linear element**, because over such a small window the curve is effectively a straight line (piecewise linearity).
- Due to this linearity, a small signal amplifier provides **distortionless output**. It is used as a **voltage amplifier**.

SMALL SIGNAL AMPLIFIER $\equiv$ VOLTAGE AMPLIFIER

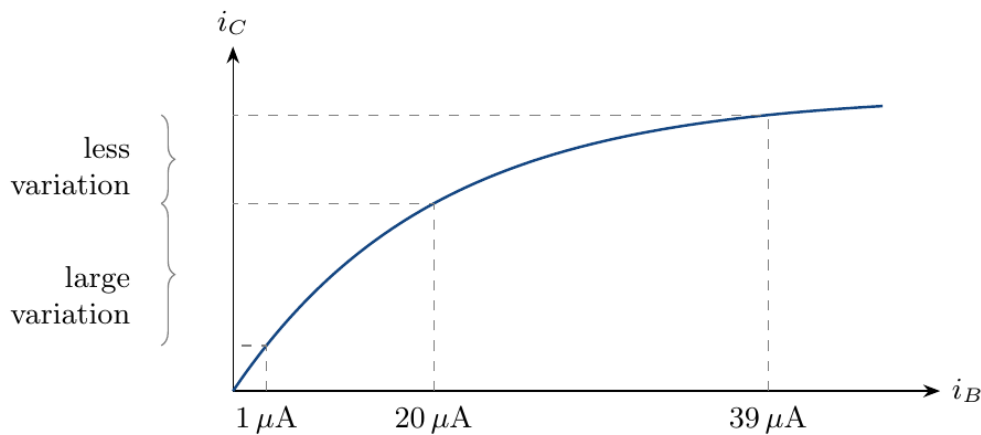
Large Signal Amplifier

An amplifier in which the signal variation is **large**, i.e. comparable to the DC value.



$$\Delta i_B = 19 \mu\text{A} \approx I_B = 20 \mu\text{A} \quad (\text{signal variation comparable})$$

Over such a wide swing the curvature of the characteristic can no longer be ignored:

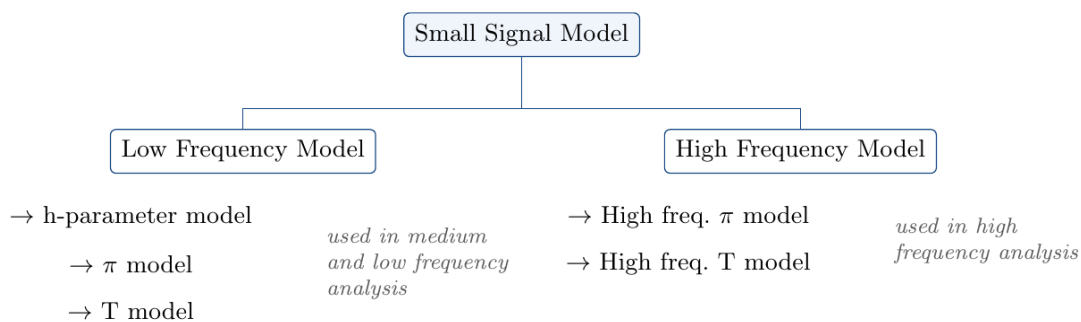


- In large signal amplifiers the BJT becomes a **non-linear device**, and **distortion appears** in the output.
- Because of the large signal variation, the AC output current and voltage are large, so **large AC output power** can be achieved.

LARGE SIGNAL AMPLIFIER $\equiv$ POWER AMPLIFIER

Small Signal Model of the BJT

The small signal model is the **equivalent circuit of the BJT**, used in the analysis of small signal BJT amplifiers.



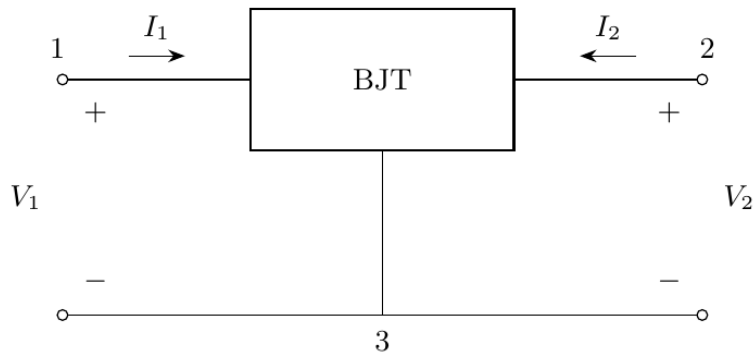
The h-Parameter Model

The model is obtained from the following two h-parameter equations:

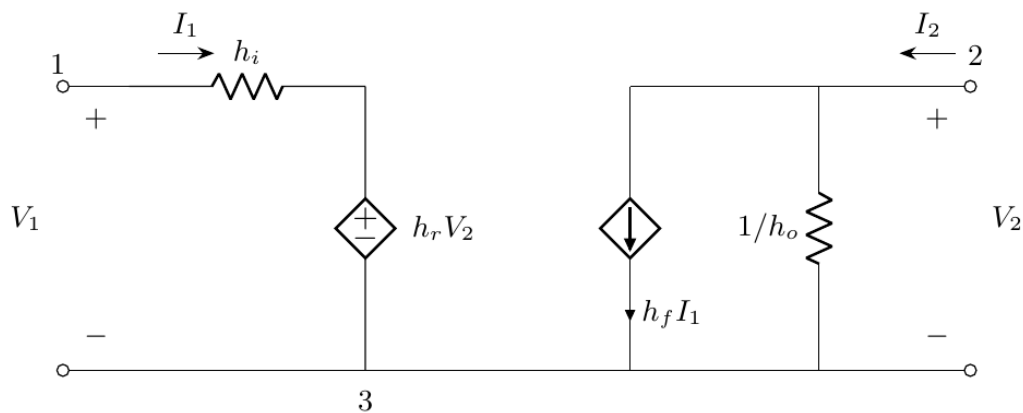
$$V_1 = h_i I_1 + h_r V_2 \quad (1)$$

$$I_2 = h_f I_1 + h_o V_2 \quad (2)$$

The BJT is treated as a two-port network with a common terminal 3:



Using equations (1) and (2), the equivalent circuit is:



Meaning of the four h-parameters

- (i) $h_i = \frac{V_1}{I_1} \Big|_{V_2=0}$ — input resistance, output node shorted to the common node
- (ii) $h_r = \frac{V_1}{V_2} \Big|_{I_1=0}$ — reverse voltage gain, input node open circuited
- (iii) $h_f = \frac{I_2}{I_1} \Big|_{V_2=0}$ — forward current gain, output node shorted to the common node
- (iv) $h_o = \frac{I_2}{V_2} \Big|_{I_1=0}$ — output conductance, input node open circuited

Consequently $1/h_o$ is the **output resistance**.

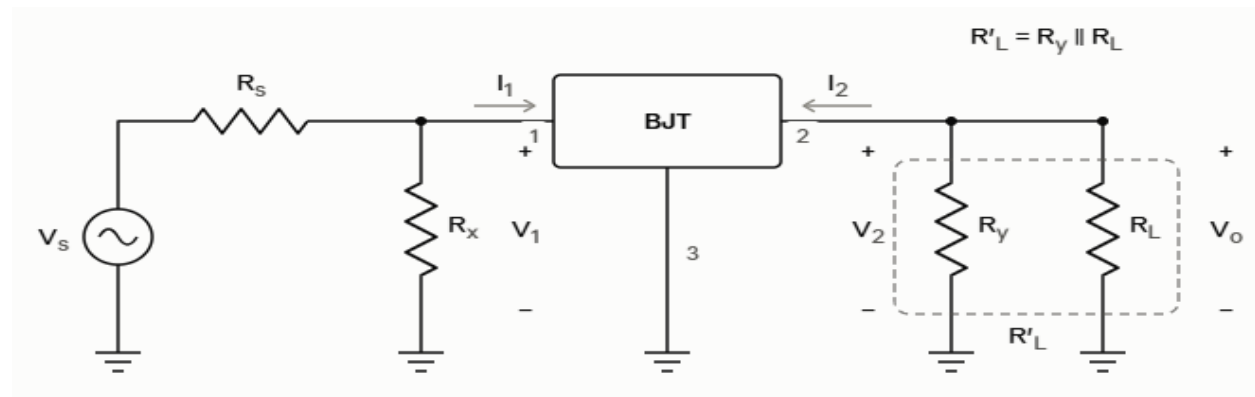
h-parameters are **low frequency parameters** and are **real numbers**. They are calculated from the input and output characteristics of the BJT.

Standard h-parameter values for the CE configuration

Parameter	Value	Parameter	Value
h_{ie}	$1.1 \text{ k}\Omega = 1100 \Omega$	h_{fe}	50
h_{re}	2.5×10^{-4}	h_{oe}	$25 \mu\text{S}$

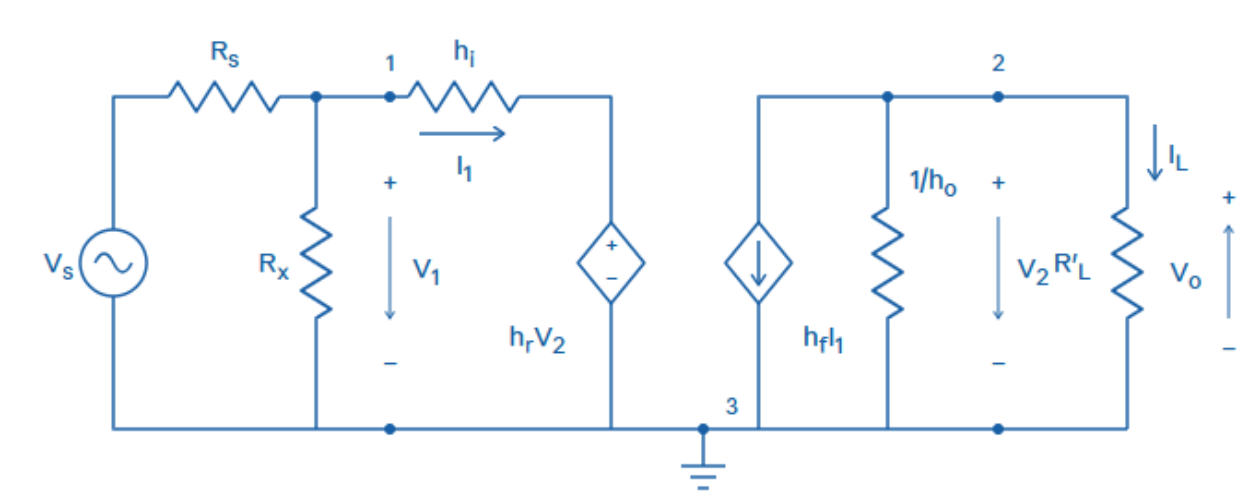
Exact Analysis of a BJT Amplifier

This analysis can be used for **all** BJT configurations (CE, CB and CC).



Symbol	Meaning
V_s	small signal AC source (mV), generally from a transducer such as a microphone
R_s	resistance of the signal source (100Ω to a few $\text{k}\Omega$)
R_x	biasing resistor connected to the input node
R_y	biasing resistor connected to the output node
R_L	external load resistance, generally a loudspeaker
R'_L	effective load resistance

Replacing the BJT with its h-parameter model gives the full small signal circuit:



Small signal analysis of an amplifier refers to the calculation of **current gain, voltage gain, input resistance** and **output resistance**.

Current Gain A_I

A_I is the ratio of load current to input current:

$$A_I = \frac{I_L}{I_1}$$

The current $(-h_f I_1)$ gets divided between $1/h_o$ and R_L' . Using the current divider rule:

$$I_L = \frac{-h_f I_1 \times \frac{1}{h_o}}{R_L' + \frac{1}{h_o}} \Rightarrow \frac{I_L}{I_1} = \frac{-h_f \times \frac{1}{h_o}}{\frac{1 + R_L' h_o}{h_o}}$$

$$A_I = \frac{I_L}{I_1} = \frac{-h_f}{1 + h_o R_L'}$$

Input Resistance R_i

R_i is the ratio of input voltage to input current, i.e. the resistance seen between the input node and ground:

$$R_i = \frac{V_1}{I_1}$$

Applying KVL in the input loop:

$$V_1 = h_i I_1 + h_r V_2$$

$$V_1 = h_i I_1 + h_r (I_L R_L')$$

$$V_1 = h_i I_1 + h_r R_L' (A_I I_1) \quad \text{since } A_I = I_L / I_1$$

$$V_1 = I_1 (h_i + h_r A_I R_L')$$

$$R_i = \frac{V_1}{I_1} = h_i + h_r A_I R_L'$$

Voltage Gain A_V

A_V is the ratio of output voltage to input voltage — the **internal voltage gain**:

$$A_V = \frac{V_2}{V_1} = \frac{V_o}{V_1}$$

Substituting $V_2 = I_L R_L'$ and $I_L = A_I I_1$:

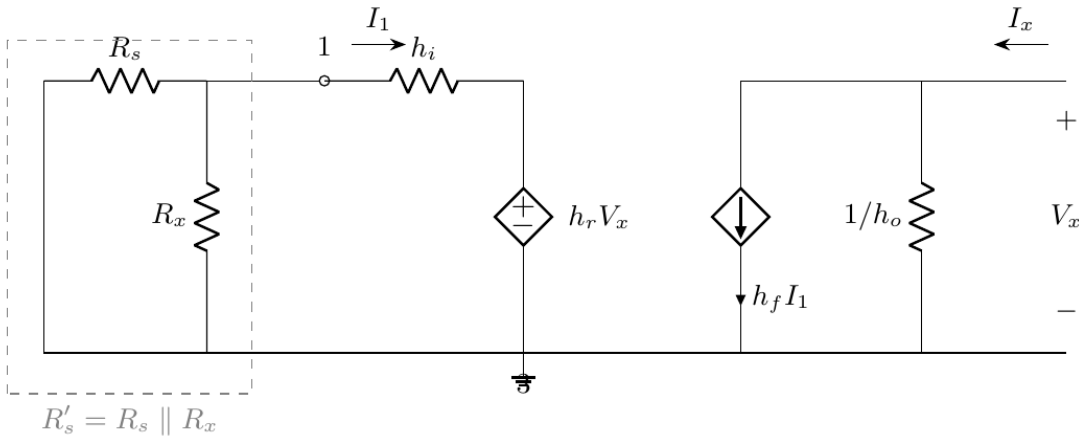
$$A_V = \frac{V_2}{V_1} = \frac{I_L R_L'}{V_1} = \frac{A_I I_1 R_L'}{V_1} \quad \text{where } R_i = V_1/I_1$$

$$A_V = \frac{A_I R_L'}{R_i}$$

Output Resistance R_o

R_o is the resistance seen between the output node and ground. The procedure is:

- Disable the external source present in the input circuit.
- Disconnect the load R_L' .
- Apply a voltage V_x across the output port, with a current I_x flowing into the output node. Then $R_o = V_x/I_x$.



KVL in the input loop:

$$I_1 R_s' + I_1 h_i + h_r V_x = 0$$

$$I_1 (R_s' + h_i) = -h_r V_x$$

$$\frac{I_1}{V_x} = \frac{-h_r}{R_s' + h_i} \quad (1)$$

KCL at node 2:

$$I_x = h_o V_x + h_f I_1 \quad \Rightarrow \quad \frac{I_x}{V_x} = h_o + \frac{h_f I_1}{V_x} = \frac{1}{R_o}$$

Substituting equation (1):

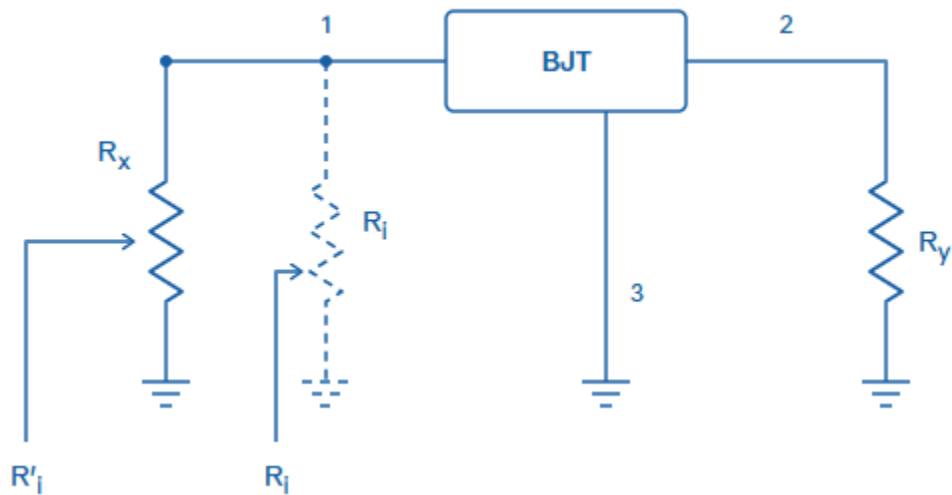
$$\frac{1}{R_o} = h_o - \frac{h_f h_r}{R_s' + h_i}, \quad R_s' = R_s \parallel R_x$$

Net Input and Output Resistances

R_i' — net input resistance

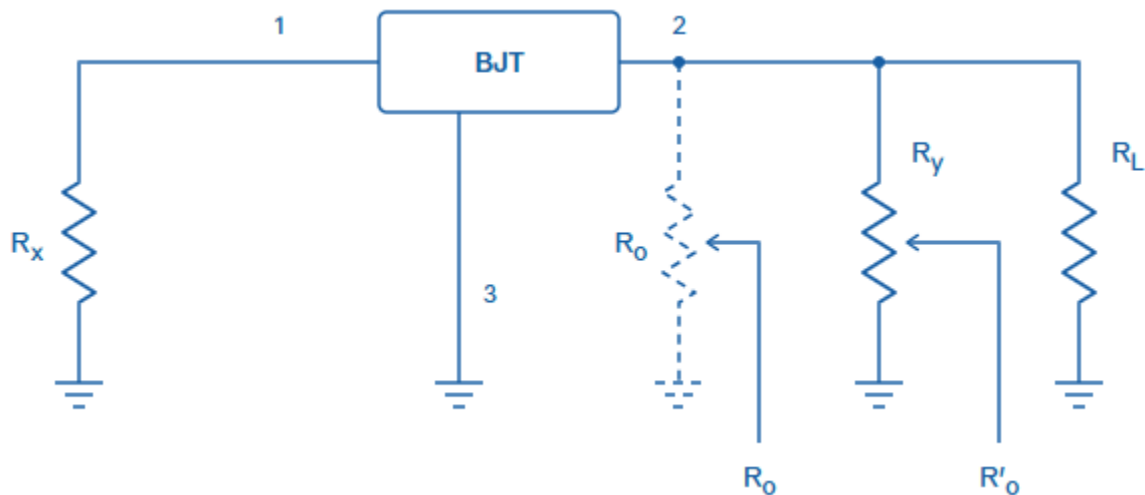
R_i' is the net input resistance calculated by **including the effect of the biasing resistor R_x** :

$$R_i' = R_i \parallel R_x$$



R_o' — net output resistance

R_o' is the net output resistance calculated by **including the effect of R_y** :



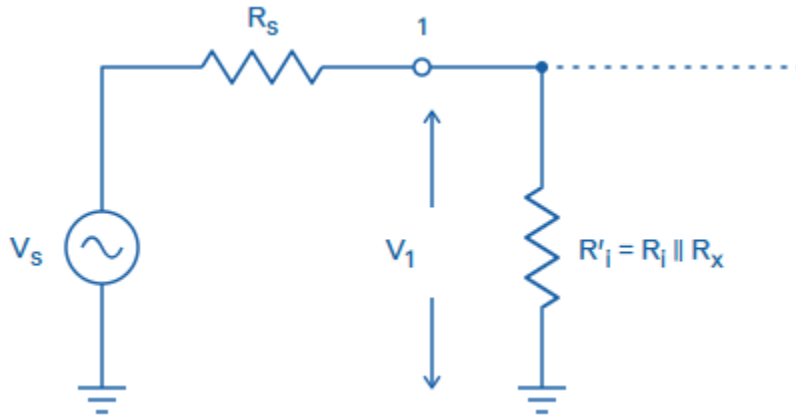
$$R_o' = R_o \parallel R_y$$

Overall Voltage Gain A_{VS}

A_{VS} is the ratio of output voltage to the *signal source* voltage — the **external voltage gain**:

$$A_{VS} = \frac{V_2}{V_s}$$

The source voltage V_s gets divided between R_s and R_i' :



Using the voltage divider rule:

$$V_1 = \frac{V_s R_i'}{R_s + R_i'} \Rightarrow \frac{V_1}{V_s} = \frac{R_i'}{R_s + R_i'} \quad (1)$$

Now,

$$A_{VS} = \frac{V_2}{V_s} = \frac{V_2}{V_1} \times \frac{V_1}{V_s}$$

$$A_{VS} = A_V \times \frac{R_i'}{R_s + R_i'}$$

If the signal source is **ideal**, i.e. $R_s = 0$, then $A_{VS} = A_V$.

Resistance seen by the signal source

$$\text{Resistance} = R_s + R_i'$$

Conversion Formulae:

CE to CC h-parameters

$$h_{fc} = -(1 + h_{fe}) \quad h_{oc} = h_{oe}$$

$$h_{ic} = h_{ie} \quad h_{rc} = 1 - h_{re} \approx 1$$

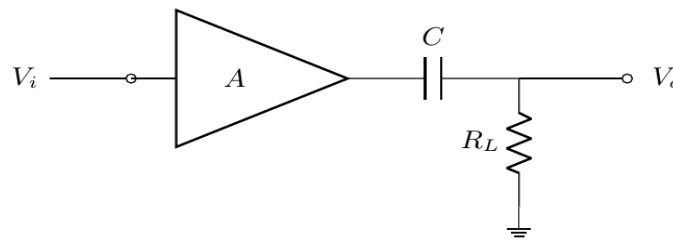
CE to CB h-parameters

$$h_{fb} = \frac{-h_{fe}}{1 + h_{fe}} \quad h_{ob} = \frac{h_{oe}}{1 + h_{fe}}$$

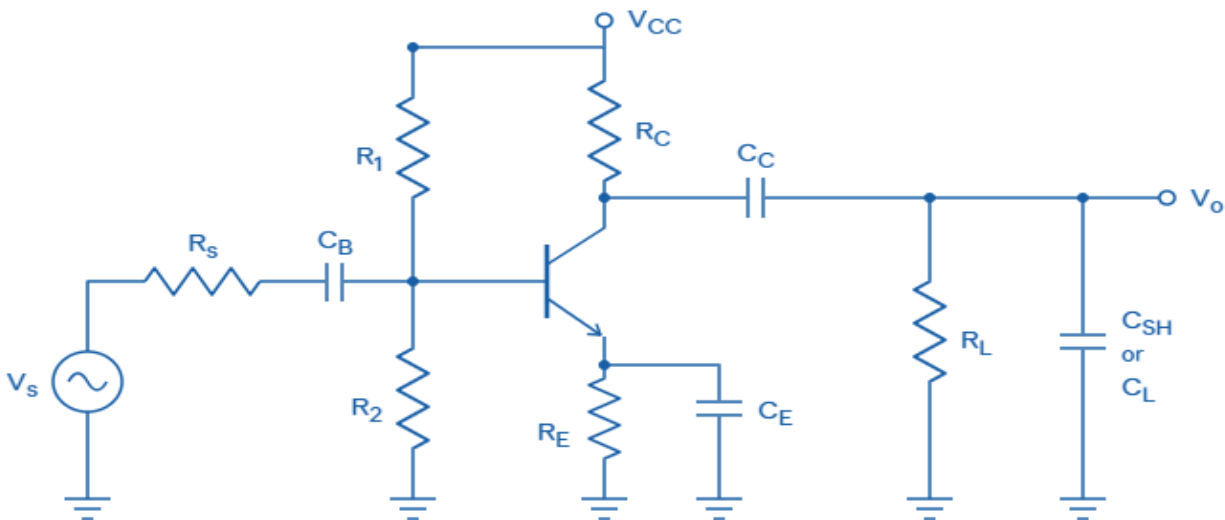
$$h_{ib} = \frac{h_{ie}}{1 + h_{fe}} \quad h_{rb} = \frac{h_{ie}h_{oe}}{1 + h_{fe}} - h_{re}$$

RC-Coupled CE Amplifier

An amplifier in which the load is a **resistor** and a **capacitor** is used to interconnect the amplifier and the load is called an **RC-coupled CE amplifier**.



- The capacitor provides **DC isolation** between the amplifier and the load.
- It is an **audio frequency voltage amplifier**.
- Operating signal frequency is **20 Hz to 20 kHz**.
- It is a **small signal amplifier** (mV range).
- A **single stage** RC-coupled CE amplifier introduces a phase shift of **180°**.
- A **two stage** RC-coupled CE amplifier introduces a phase shift of **360°**.



Role of each element

- **Self-bias circuit** (R_1, R_2, R_E) is used to operate the BJT in the **active region** and to keep the operating point stable.
- **Emitter bypass capacitor** C_E is connected to prevent a decrease in voltage gain. If C_E is not present, R_E will reduce the voltage gain.
- **Blocking capacitor** C_B and **coupling capacitor** C_C provide DC isolation to the biasing circuit, so that the external elements V_s, R_s and R_L do not disturb the operating point.
- If C_C is not present, R_L will draw DC current and the operating point gets disturbed.
- If C_B is not present, V_s and R_s can disturb the operating point.
- C_B, C_E and C_C are **large capacitors** (μF), because they should offer negligible reactance at the frequency of the AC signal.
- **Shunt capacitor** C_{SH} is connected to control the bandwidth of the amplifier.

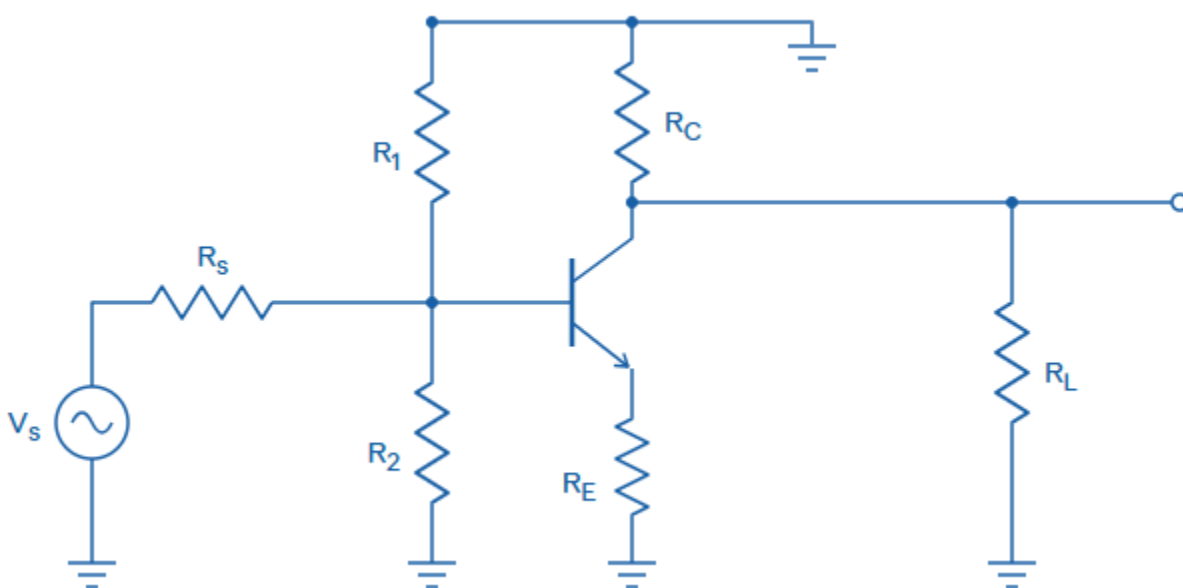
$$\text{Bandwidth} \propto \frac{1}{C_{SH}}$$

C_{SH} is a **small capacitor** (pF), because it should offer high reactance to the AC signal.

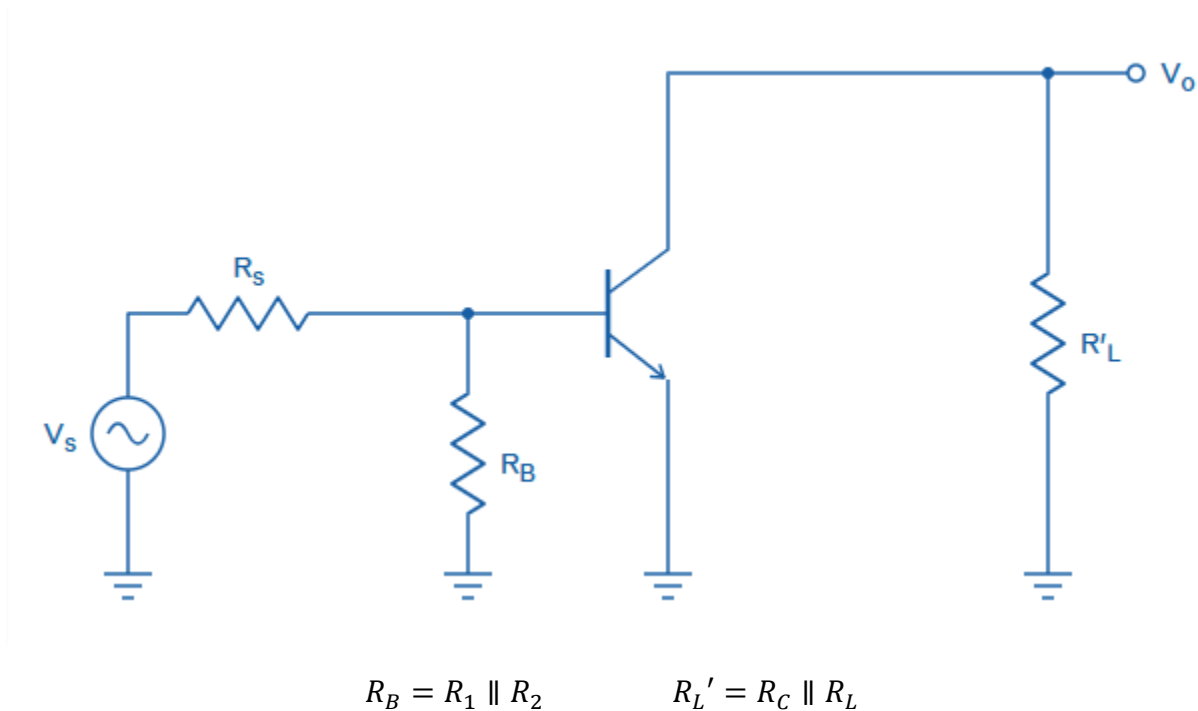
Procedure to Analyse an Amplifier

- Identify the BJT configuration (CE, CB or CC) in the given amplifier.
- Draw the **medium frequency AC equivalent circuit** by replacing large capacitors with a short circuit, small capacitors with an open circuit, and by connecting V_{CC} to ground. (The DC supply V_{CC} is disabled in AC or small signal analysis.)
- Replace the BJT with the appropriate small signal model and calculate the desired values.

Step (ii) applied to the RC-coupled CE amplifier



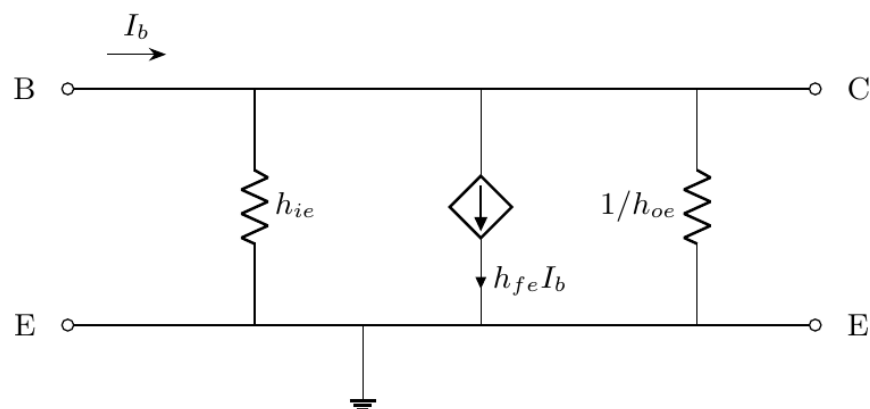
Combining the parallel resistors gives the compact form — the **medium frequency AC equivalent**:



Approximate Analysis of BJT Amplifiers

In approximate analysis the BJT is replaced with the **approximate CE h-parameter model**, which is obtained by **neglecting** h_{re} :

$$h_{re} \approx 2.5 \times 10^{-4} \approx 0 \quad \text{hence } h_{re} \text{ is neglected}$$



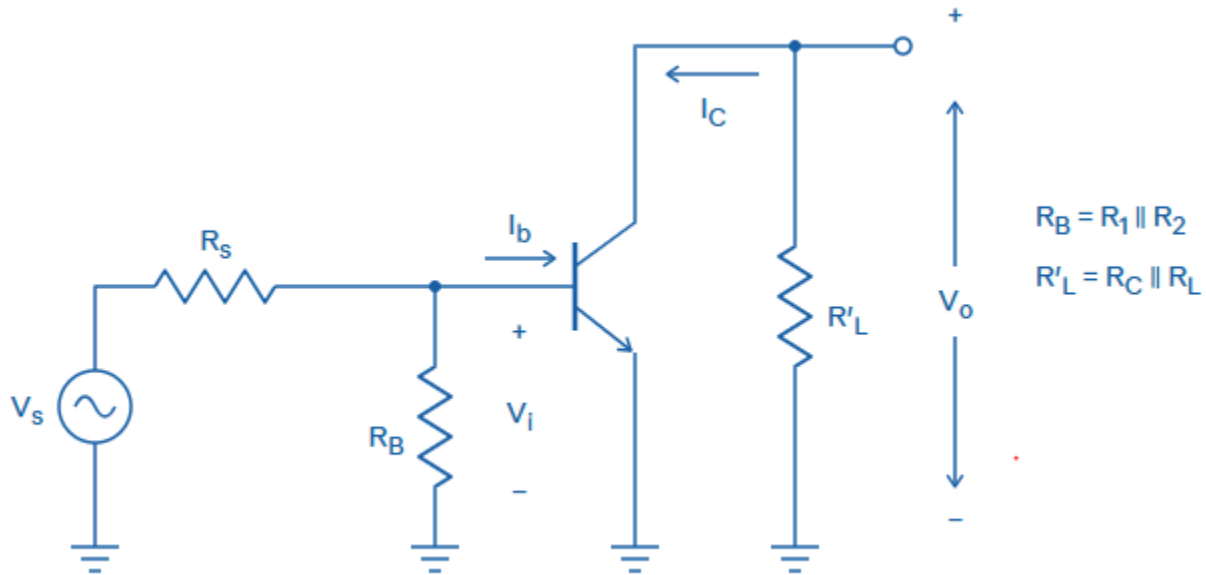
If h_{oe} is not provided, or if h_{oe} is stated to be negligible, then $1/h_{oe}$ is replaced with an **open circuit**.

Approximate analysis becomes valid if:

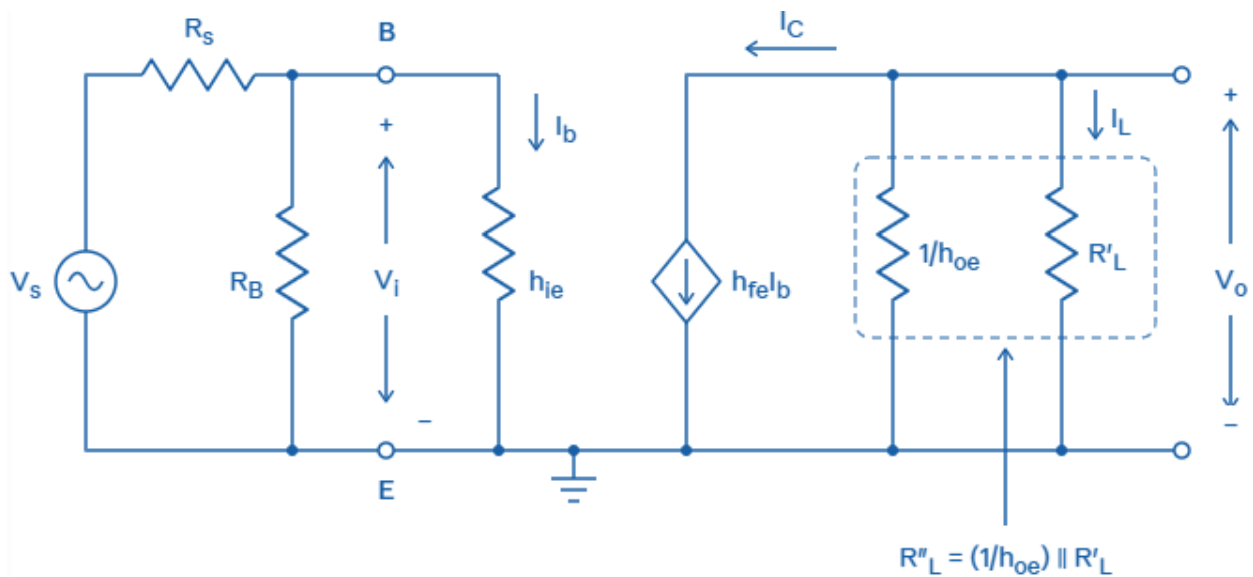
$$h_{oe}R_L' \leq 0.1$$

CE Amplifier with Bypass Capacitor C_E

The AC equivalent circuit of the BJT amplifier is:



Replacing the BJT with the approximate CE h-parameter model:



(I) Current gain A_I

$$A_I = \frac{I_L}{I_b}$$

By the current divider rule:

$$I_L = \frac{-h_{fe}I_b \times \frac{1}{h_{oe}}}{\frac{1}{h_{oe}} + R_L'}$$
$$\frac{I_L}{I_b} = A_I = \frac{-h_{fe}}{1 + h_{oe}R_L'}$$

If $h_{oe}R_L' \leq 0.1$, that term can be neglected, so:

$$\frac{I_L}{I_b} = -h_{fe} = A_I$$

The negative sign appears because the load current I_L and the output current I_c are in **opposite directions**: $I_L/I_b = -h_{fe}$ while $I_c/I_b = h_{fe}$.

(II) Input resistance R_i

$$R_i = \frac{V_i}{I_b}$$

From the input circuit, $V_i = h_{ie}I_b$, hence:

$$\frac{V_i}{I_b} = R_i = h_{ie}$$

(III) Voltage gain A_V

$$A_V = \frac{V_o}{V_i} = \frac{-h_{fe}I_b \cdot R_L''}{I_b \cdot h_{ie}}$$
$$A_V = \frac{-h_{fe}}{h_{ie}} \times R_L'' \quad \text{where} \quad R_L'' = R_L' \parallel \frac{1}{h_{oe}}$$

If $h_{oe} \approx 0$ (neglected), then $R_L'' = R_L'$, so:

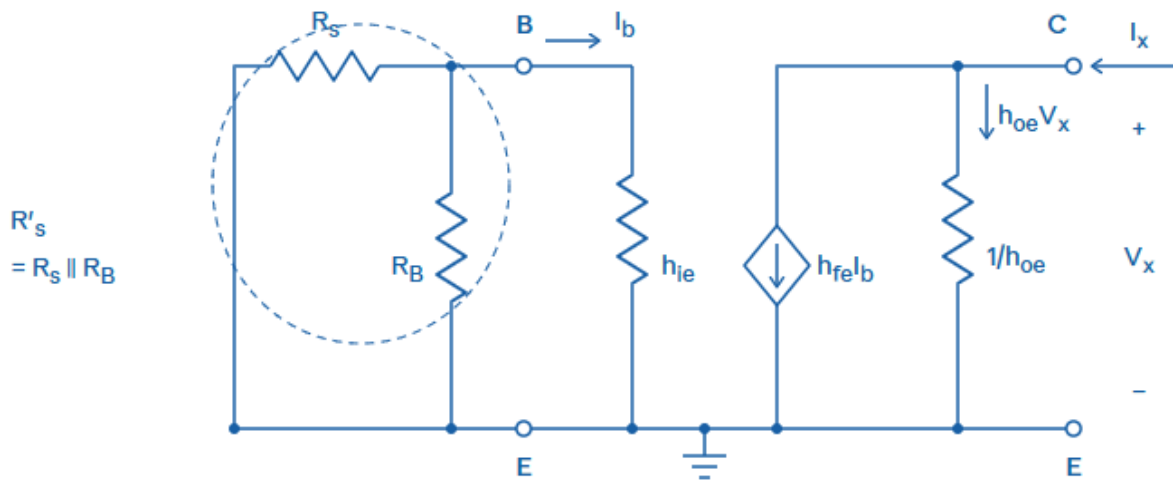
$$A_V = \frac{-h_{fe}}{h_{ie}} \times R_L'$$

The negative sign in A_V indicates a phase shift of **180°** between V_o and V_i .

(IV) Output resistance R_o

To calculate R_o :

- Replace V_s with a short circuit.
- Disconnect the load R_L' .
- Apply a voltage V_x across the output port.



KVL in the input loop:

$$I_b R'_s + h_{ie} I_b = 0 \quad \Rightarrow \quad I_b = 0$$

KCL at the collector:

$$I_x = h_{oe} V_x + h_{fe} I_b = h_{oe} V_x$$

$$\frac{V_x}{I_x} = R_o = \frac{1}{h_{oe}}$$

(V) Net resistances and overall gain

R'_i , R'_o and A_{VS} are obtained in the same way as in the exact analysis.

$$R'_i = R_i \parallel R_x = R_i \parallel R_B = h_{ie} \parallel R_B$$

$$R'_o = R_o \parallel R_y = \frac{1}{h_{oe}} \parallel R_C$$

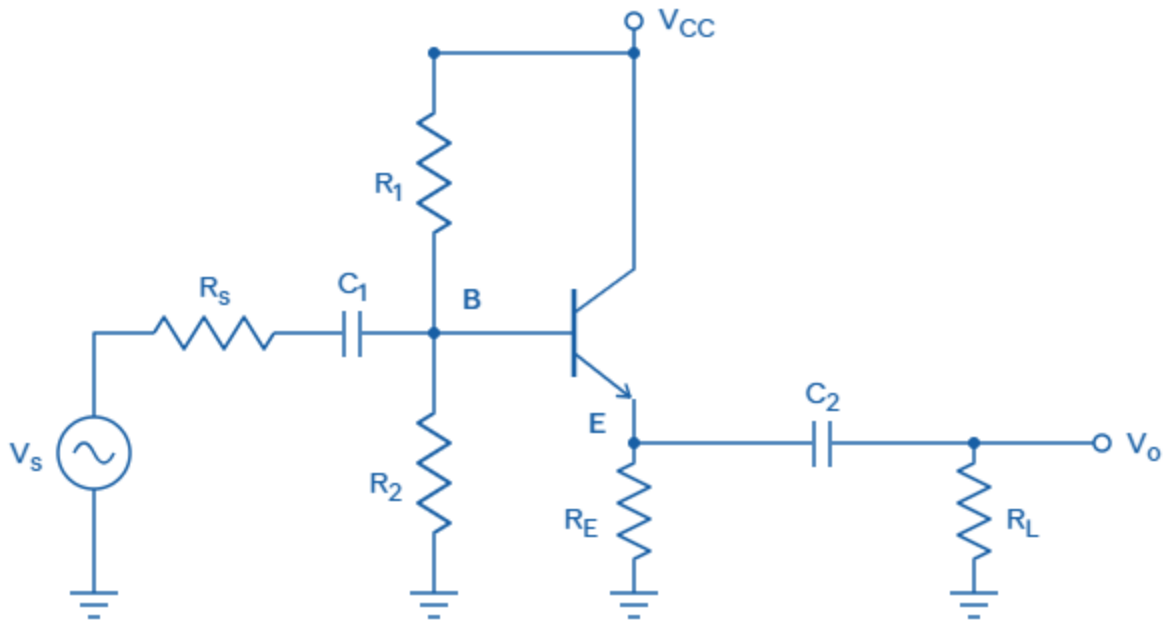
$$A_{VS} = \frac{V_o}{V_s} = \frac{A_V \cdot R'_i}{R'_i + R_s}$$

$$\text{Resistance seen by } V_s = R_s + R'_i$$

Here x denotes the input node and y the output node.

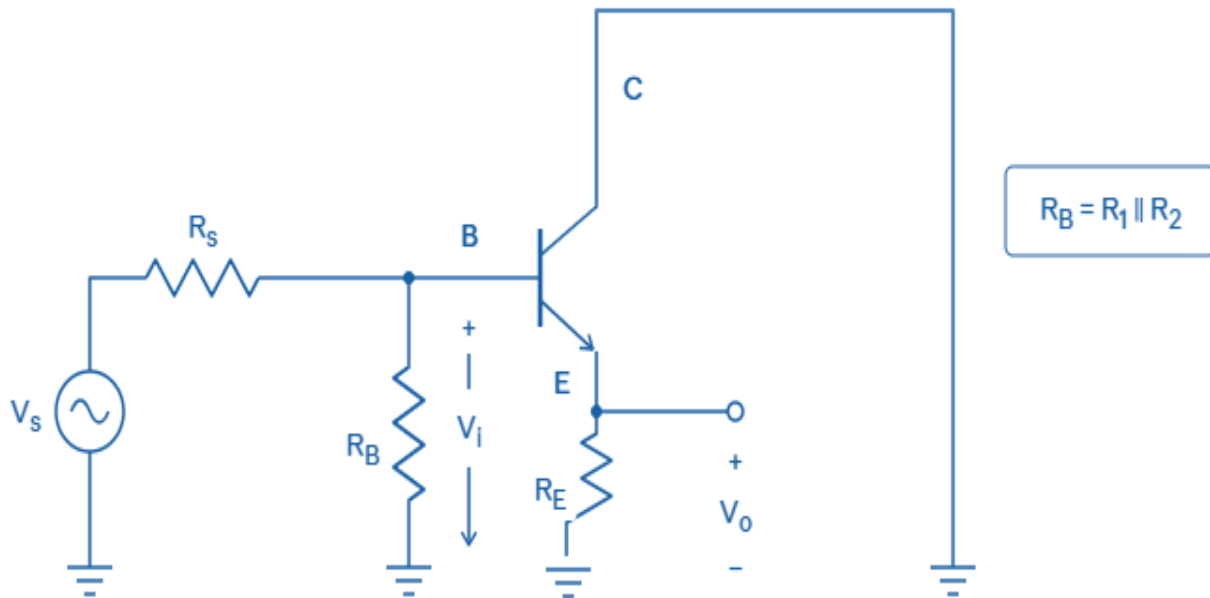
Common Collector Amplifier

Input node: Base • Output node: Emitter



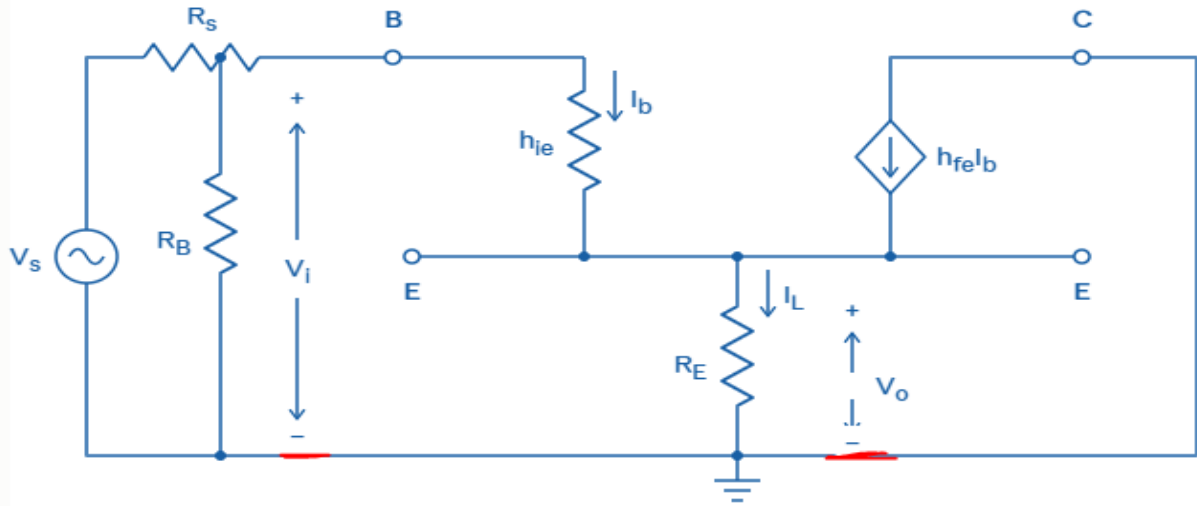
- In a CE amplifier, R_E acts as the load.
- If an external load R_L is also present, then $(R_E \parallel R_L)$ becomes the effective load.

AC equivalent circuit



with $R_B = R_1 \parallel R_2$.

With the approximate CE h-parameter model (h_{oe} neglected)



(a) Current gain A_I

$$A_I = \frac{I_L}{I_b}$$

KCL at the emitter node:

$$I_L = I_b + h_{fe} I_b \quad \Rightarrow \quad I_L = I_b (1 + h_{fe})$$

$$\frac{I_L}{I_b} = A_I = (1 + h_{fe}) \quad \text{— } A_I \text{ is large, since } h_{fe} \text{ is large}$$

The exact formula for A_I is:

$$A_I = \frac{-h_{fc}}{1 + h_{oc} R_E} \quad \text{with} \quad h_{fc} = -(1 + h_{fe}), \quad h_{oc} = h_{oe}$$

$$A_I = \frac{1 + h_{fe}}{1 + h_{oe} R_E}$$

If $h_{oe} R_E \leq 0.1$, this reduces to $A_I = (1 + h_{fe})$.

(b) Input resistance R_i

$$R_i = \frac{V_i}{I_b}$$

Applying KVL in the input loop:

$$V_i = h_{ie} I_b + I_L R_E = h_{ie} I_b + (1 + h_{fe}) R_E \times I_b$$

$$\frac{V_i}{I_b} = R_i = h_{ie} + (1 + h_{fe})R_E$$

The CC amplifier has a **high input resistance**.

(c) Voltage gain A_V

$$A_V = \frac{V_o}{V_i} = \frac{I_L R_E}{V_i} = \frac{I_b (1 + h_{fe}) R_E}{I_b \{h_{ie} + (1 + h_{fe}) R_E\}}$$

$$A_V = \frac{V_o}{V_i} = \frac{(1 + h_{fe}) R_E}{h_{ie} + (1 + h_{fe}) R_E}$$

Since $(1 + h_{fe})R_E \gg h_{ie}$:

$$A_V = \frac{V_o}{V_i} \approx \frac{(1 + h_{fe}) R_E}{(1 + h_{fe}) R_E} \approx 1$$

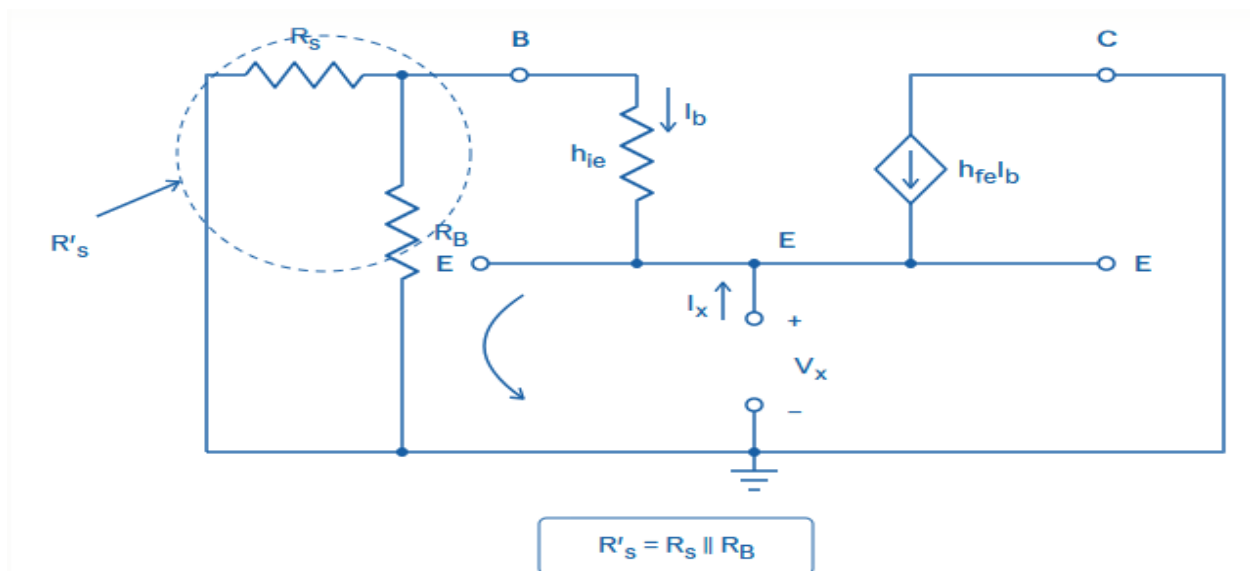
The common collector amplifier has **unity voltage gain**, $A_V \approx 1$.

The AC output voltage is the same as the AC input voltage. The CC amplifier is therefore also called an **emitter follower**, because the emitter voltage (output voltage) follows the input voltage.

(d) Output resistance R_o

To calculate R_o :

- Replace V_s with a short circuit.
- Disconnect the load R_E .
- Apply a voltage V_x across the output port.



KVL in the input loop:

$$I_b R_s' + I_b h_{ie} + V_x = 0 \quad \Rightarrow \quad V_x = -I_b (h_{ie} + R_s') \quad (1)$$

KCL at the emitter:

$$I_x + I_b + h_{fe} I_b = 0 \quad \Rightarrow \quad I_x = -I_b (1 + h_{fe}) \quad (2)$$

Dividing (1) by (2):

$$\frac{V_x}{I_x} = R_o = \frac{h_{ie} + R_s'}{1 + h_{fe}} \quad \text{--- } R_o \text{ is small}$$

If $R_s = 0$, then:

$$R_o = \frac{h_{ie}}{1 + h_{fe}}$$

(e) Net resistances and overall gain

$$R_i' = R_i \parallel R_x = R_i \parallel R_B$$

$$R_o' = R_o \parallel R_y = R_o \parallel R_E$$

$$A_{VS} = \frac{V_o}{V_s} = A_V \times \frac{R_i'}{R_i' + R_s}$$

$$\text{Resistance seen by } V_s = R_s + R_i'$$

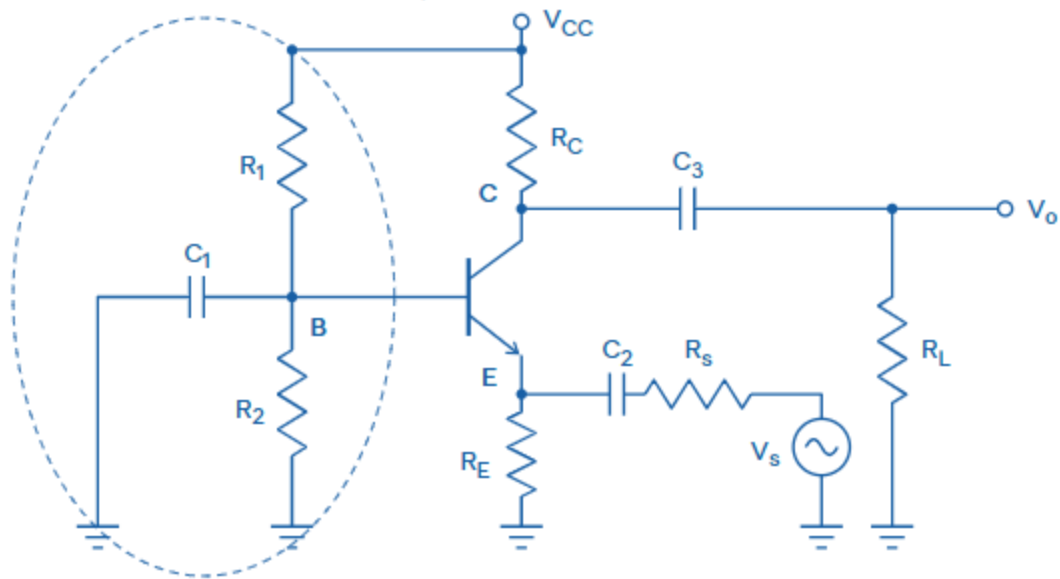
6. Summary — Common Collector Amplifier

- Unity voltage gain** ($A_V \approx 1$) — slightly less than 1.
- Large current gain.**
- Large input resistance** (hundreds of $k\Omega$).
- Small output resistance** (tens of Ω).

II) COMMON BASE (CB) AMPLIFIER

- Input node** : Emitter
- Output node** : Collector

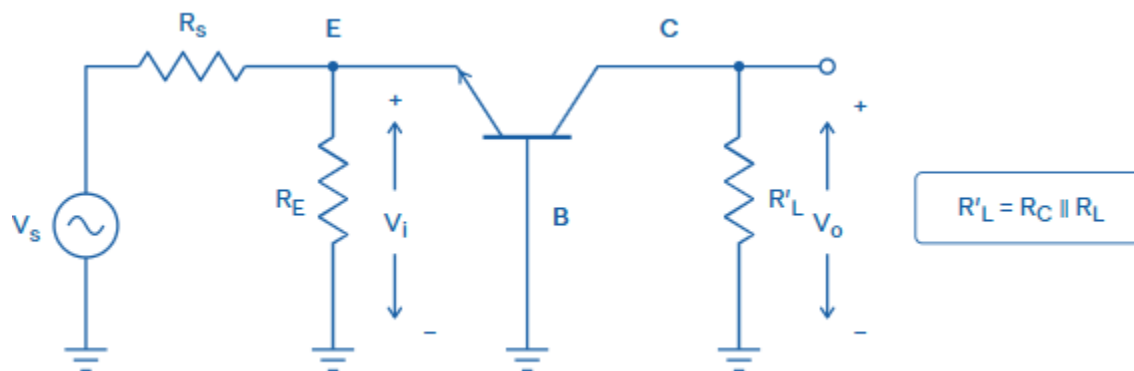
For AC analysis, all in parallel, hence $R_{eq} = 0$



NOTE

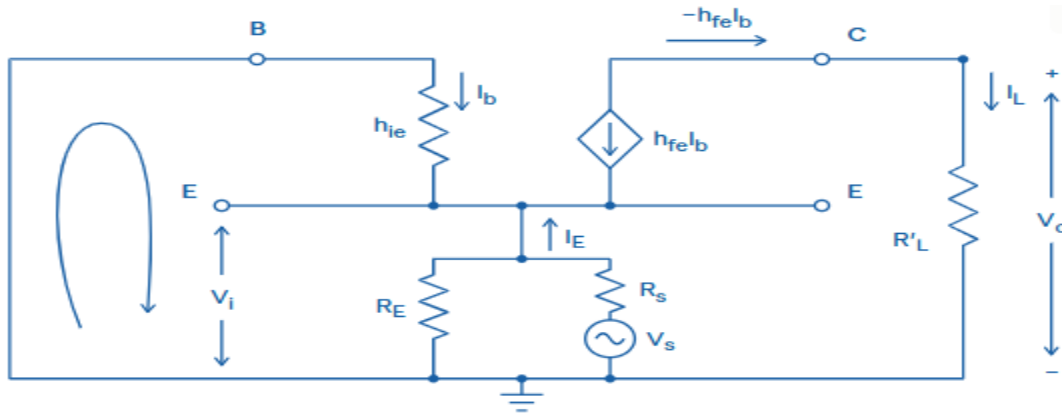
R_1 , R_2 and C_1 are all in parallel for AC analysis, hence $R_{eq} = 0$.

★ Redrawing the **AC equivalent circuit** (medium frequency):



★ R_1 and R_2 do not appear in the AC equivalent circuit because they are in parallel with capacitor C_1 , which is acting as a short circuit.

★ Replace the BJT with the approximate CE h-parameter model and neglect h_{oe} :



a) Current Gain (A_I)

Mathematically:

$$A_I = \frac{I_L}{I_e}$$

From the output circuit:

$$I_L = -h_{fe} I_b \dots (1)$$

★ KCL at the emitter, we get:

$$I_e + I_b + h_{fe} I_b = 0 \quad I_e = -I_b(1 + h_{fe}) \dots (2)$$

So,

$$\frac{I_L}{I_e} = \frac{-h_{fe} I_b}{-(1 + h_{fe}) I_b}$$

$$\frac{I_L}{I_e} = A_I = \frac{h_{fe}}{(1 + h_{fe})}$$

★ As $h_{fe} \gg 1 \Rightarrow 1 + h_{fe} \approx h_{fe}$:

$$A_I \approx 1$$

CB amplifier has **unity current gain** (slightly less than 1).

b) Input Resistance (R_i)

Mathematically:

$$R_i = \frac{V_i}{I_e}$$

★ KVL in the input loop, we get:

$$I_b h_{ie} + V_i = 0 \quad V_i = -I_b h_{ie} \dots (3)$$

So,

$$\frac{V_i}{I_e} = \frac{-I_b h_{ie}}{-I_b(1 + h_{fe})}$$

$$R_i = \frac{V_i}{I_e} = \frac{h_{ie}}{(1 + h_{fe})}$$

Smaller R_i for the CB amplifier ($\approx$ tens of Ω).

c) Voltage Gain (A_V)

Mathematically:

$$A_V = \frac{V_o}{V_i} = \frac{I_L R_L'}{V_i}$$

$$A_V = \frac{-h_{fe} I_b R_L'}{-I_b h_{ie}}$$

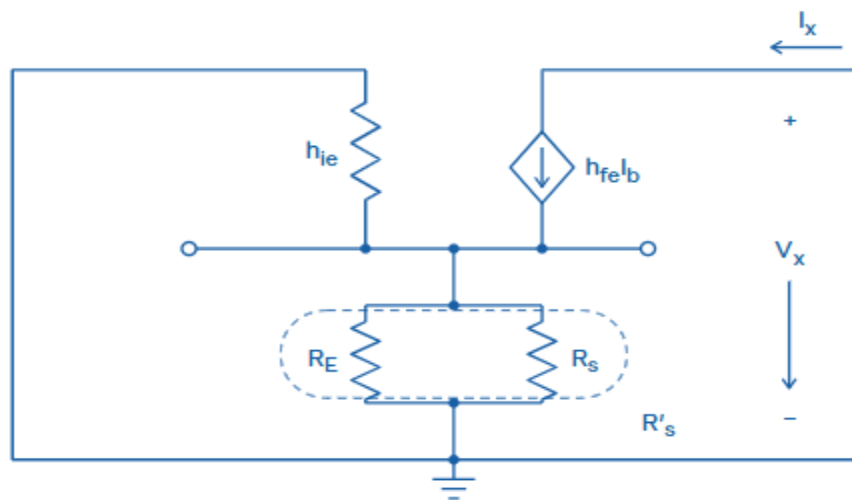
$$A_V = \frac{h_{fe} R_L'}{h_{ie}}$$

Large value of A_V , i.e. large voltage gain.

NOTE

Voltage gain formula is similar for CE and CB. But A_V is **negative** in the common emitter and **positive** in the common base amplifier.

d) Output Resistance (R_o)



$$R_o = \frac{V_x}{I_x} = \frac{V_x}{0} = \infty$$

$$R_s' = R_s \parallel R_E$$

★ If h_{oe} is considered, then:

$$R_o \approx \frac{1 + h_{fe}}{h_{oe}}$$

Large value of R_o .

Note

$$(i) R_i' = R_i \parallel R_x = R_i \parallel R_E$$

$$(ii) R_o' = R_o \parallel R_y = R_o \parallel R_C$$

with $R_o \rightarrow \infty$, $R_o' = R_C$

$$(iii) A_{VS} = \frac{A_V \times R_i'}{R_s + R_i'}$$

$$(iv) \text{ Resistance seen by the voltage source} = R_s + R_i'$$

SUMMARY

Common base amplifier has:

- unity current gain (slightly less than 1)
- large voltage gain
- smaller input resistance ($\approx$ tens of Ω)
- large output resistance ($M\Omega$ or ∞)

★ CB has the opposite values to those of the CC amplifier.

COMPARISON — CE, CC AND CB AMPLIFIER

PARAMETERS	CE AMP.	CC AMP.	CB AMP.
i) A_V	LARGE	$A_V \approx 1$	LARGE
ii) A_I	LARGE	LARGE	$A_I \approx 1$
iii) R_i	MEDIUM	LARGE	SMALL
iv) R_o	MEDIUM	SMALL	LARGE
v) Phase shift b/w V_o and V_i	180° phase shift	0° phase shift	0° phase shift
vi) Application	Audio frequency amplifier (20 Hz < f < 20 kHz)	Voltage buffer	i) Current buffer ii) High frequency amplifier

NOTE

- i) Common emitter configuration provides both current and voltage amplification. Hence CE is the best configuration in terms of amplification.
- ii) CE configuration has the highest power gain ($A_P = A_V \times A_I$).

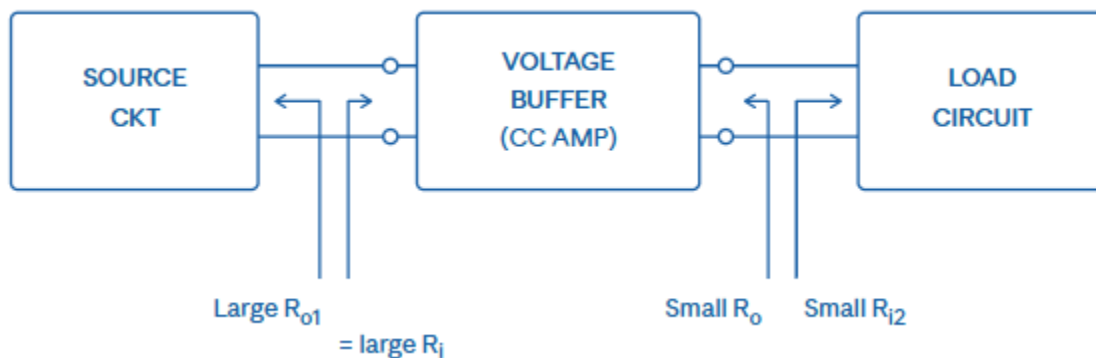
NOTE

The performance of the CE amplifier will suffer or degrade at high frequencies due to the **MILLER EFFECT**.

The frequency response of the CE transistor amplifier is inferior to that of the CB and CC transistor amplifiers (due to the Miller effect only).

BUFFER

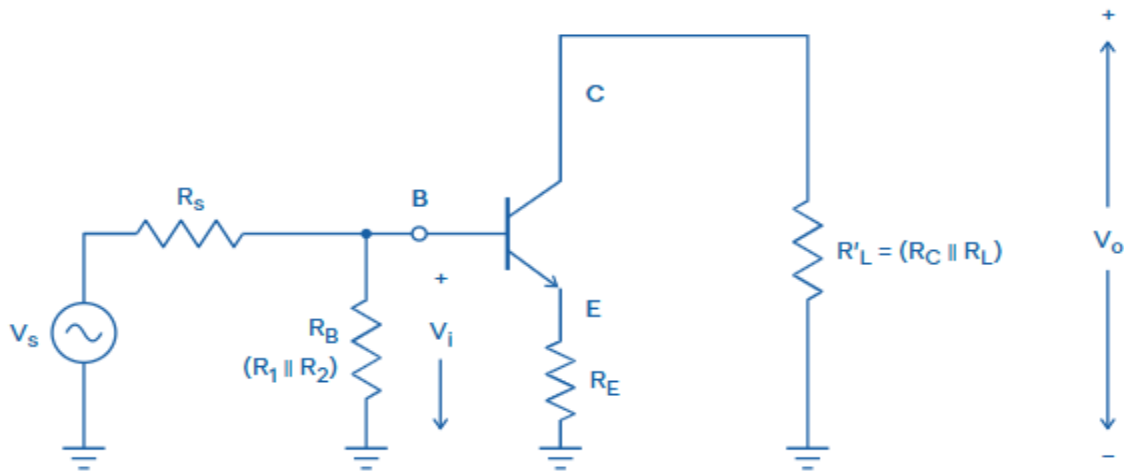
- ★ It is a circuit used for **impedance matching** between source and load, so that maximum power is transferred from source to load.
- ★ If the source circuit has a large output resistance and the load circuit has a smaller input resistance, then a **voltage buffer** such as a common collector amplifier is used for impedance matching.



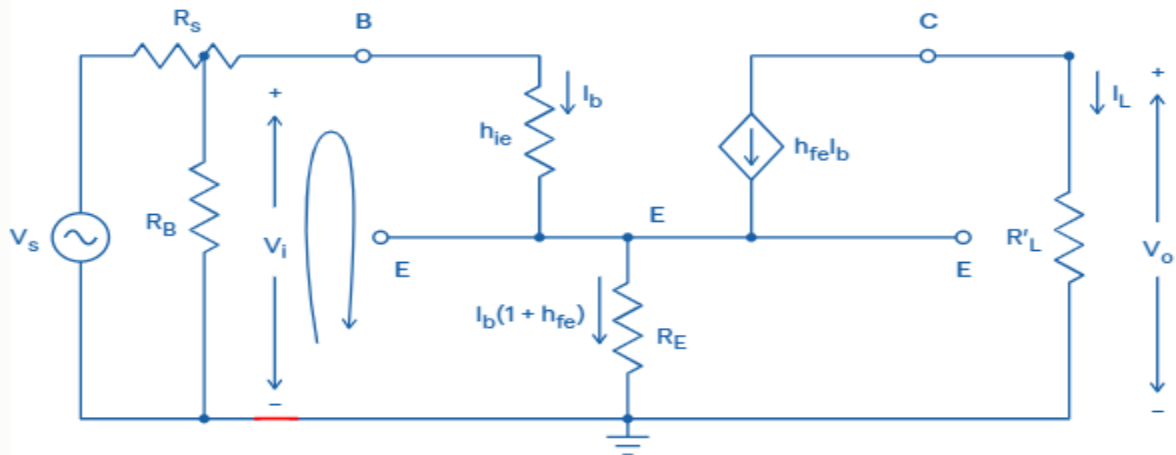
- ★ A **VOLTAGE BUFFER** has large R_i , smaller R_o and unity voltage gain.
- ★ If the source circuit has smaller R_o and the load circuit has large R_i , then a **CURRENT BUFFER**, i.e. a CB amplifier, is used for impedance matching.
- ★ A **CURRENT BUFFER** has smaller R_i , large R_o and unity current gain.

IV) CE AMPLIFIER WITHOUT BYPASS CAPACITOR

- ★ If the bypass capacitor is not present, then the effect of R_E should be considered in the analysis.



★ Replace the BJT with the approximate CE h-parameter model and neglect h_{oe} :



a) Current Gain (A_I)

$$A_I = \frac{I_L}{I_b}$$

$$I_L = -h_{fe} I_b$$

$$\frac{I_L}{I_b} = A_I = -h_{fe}$$

b) Input Resistance (R_i)

$$R_i = \frac{V_i}{I_b}$$

★ KVL in the input loop, we get:

$$V_i = h_{ie}I_b + I_b(1 + h_{fe})R_E$$

$$\frac{V_i}{I_b} = R_i = h_{ie} + (1 + h_{fe})R_E$$

NOTE

- The value of R_i **increases** when the bypass capacitor is not used.
- CC amplifier and CE amplifier without bypass capacitor have equal R_i .

c) Voltage Gain (A_V)

$$A_V = \frac{V_o}{V_i} = \frac{I_L R_L'}{V_i}$$

$$A_V = \frac{-h_{fe} I_b \times R_L'}{I_b [h_{ie} + (1 + h_{fe})R_E]}$$

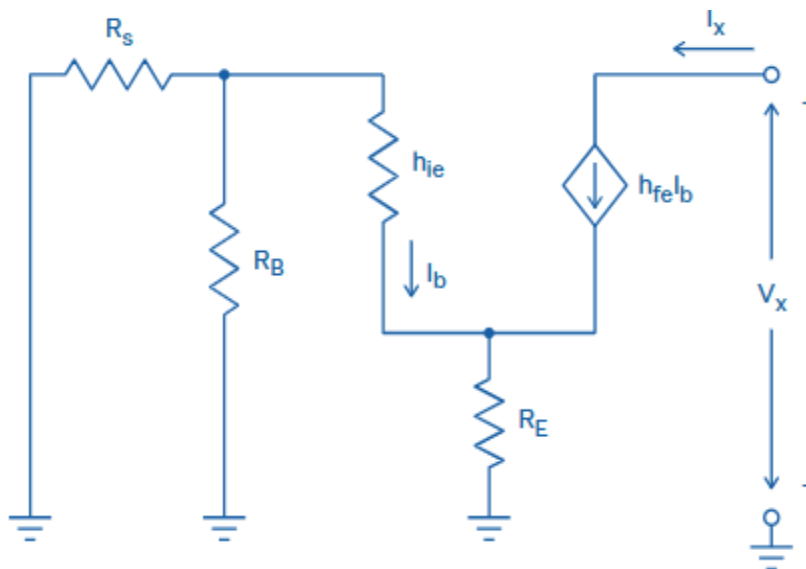
$$A_V = \frac{-h_{fe} R_L'}{h_{ie} + (1 + h_{fe})R_E} \Rightarrow |A_V| \text{ decreases}$$

NOTE

- When the bypass capacitor is not connected, resistor R_E will reduce the voltage gain.
- Hence to prevent this reduction in A_V , a bypass capacitor should be connected.
- If R_E is made zero, then A_V will be high but the **Q point will be unstable**.

d) Output Resistance (R_o)

$$R_o = \frac{V_x}{I_x} = \frac{V_x}{0} = \infty$$



★ If h_{oe} is considered, then:

$$R_o \approx \frac{1 + h_{fe}}{h_{oe}} \leftarrow R_o \text{ increases}$$

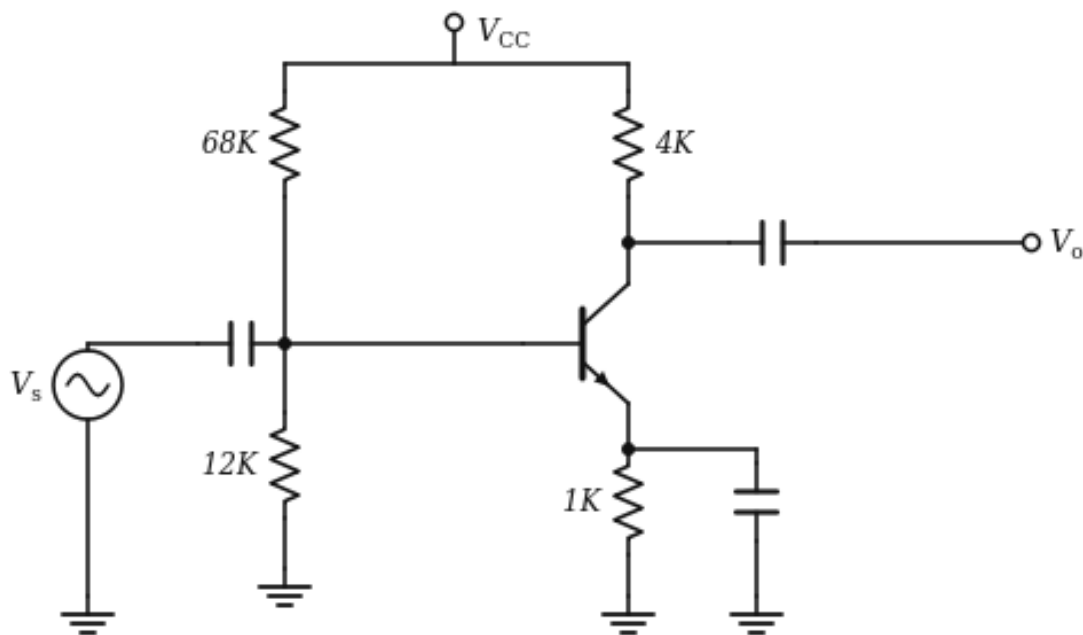
SUMMARY

If the bypass capacitor is disconnected in a CE amplifier, then:

- $|A_V|$ decreases
- R_i increases
- R_o increases

PROBLEM 1

In the amplifier shown, the transistor has $h_{fe} = 75$, $h_{ie} = 2.5 \text{ k}\Omega$ and $h_{oe} = 20 \mu\text{S}$. Calculate the voltage gain, input and output resistances.



SOLUTION

★ The above configuration is a **CE transistor amplifier with bypass capacitor**.

i) Voltage gain

$$\text{Voltage gain} = \frac{V_o}{V_s} = \frac{V_o}{V_i} \Rightarrow A_V = \frac{-h_{fe} R_L''}{h_{ie}} \quad \text{As } \{R_L'' = R_L' \parallel \frac{1}{h_{oe}}\} \text{ So, } R_L'' = R_L' \parallel \frac{1}{h_{oe}} = 4\text{k}\Omega \parallel 50\text{k}\Omega = 3.7\text{k}\Omega \Rightarrow A_V = \frac{-75 \times 3.7\text{k}\Omega}{2.5\text{k}\Omega}$$

$$\Rightarrow A_V = -111$$

ii) Input resistance

$$R_i = h_{ie} = 2.5k\Omega$$

$$R_i' = h_{ie} \parallel R_B = h_{ie} \parallel R_x = 2.5k\Omega \parallel (68k\Omega \parallel 12k\Omega)$$

$$\Rightarrow R_i' = 2k\Omega$$

iii) Output resistance

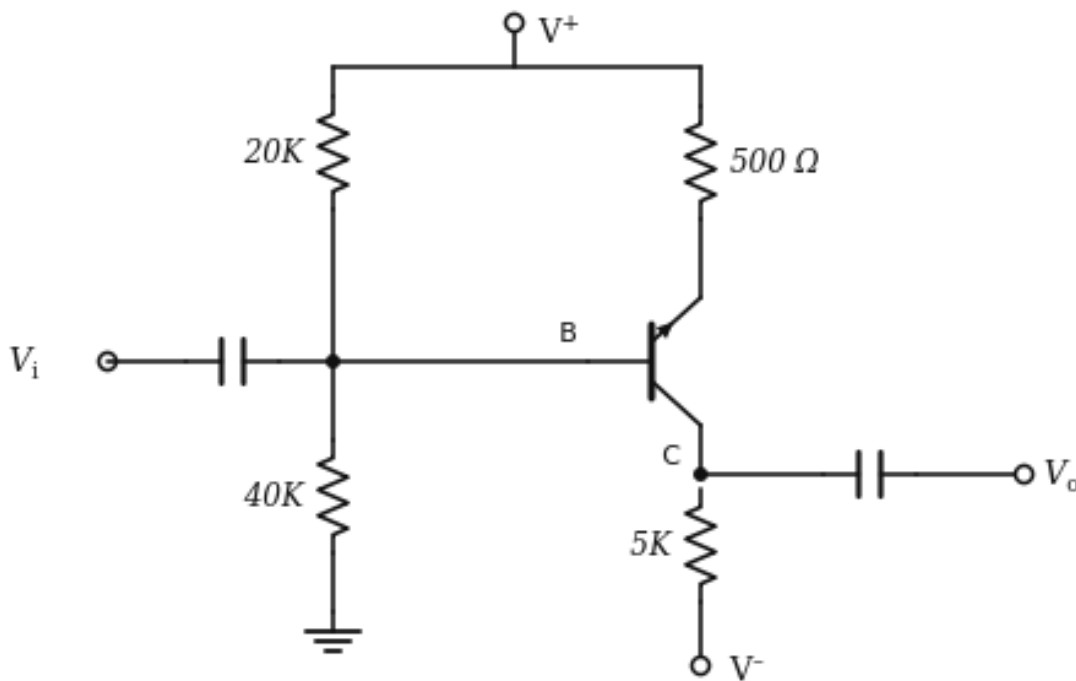
$$R_o = \frac{1}{h_{oe}} = 50k\Omega$$

$$R_o' = R_o \parallel R_C = 50k\Omega \parallel 4k\Omega$$

$$\Rightarrow R_o' = 3.7k\Omega$$

PROBLEM 2

Calculate the voltage gain and input resistance in the amplifier shown, if the transistor has $h_{fe} = 60$ and $h_{ie} = 2\text{ k}\Omega$.



SOLUTION

★ The above circuit is a **CE configuration without bypass capacitor**.

i) Voltage gain

$$A_V = \frac{-h_{fe} R_L'}{h_{ie} + (1 + h_{fe})R_E} = \frac{-60 \times 5k\Omega}{2k\Omega + (1 + 60) \times 0.5k\Omega}$$

$$A_V = -9.23$$

ii) Input resistance

$$R_i = h_{ie} + (1 + h_{fe})R_E = 2k\Omega + (1 + 60) \times 0.5k\Omega$$

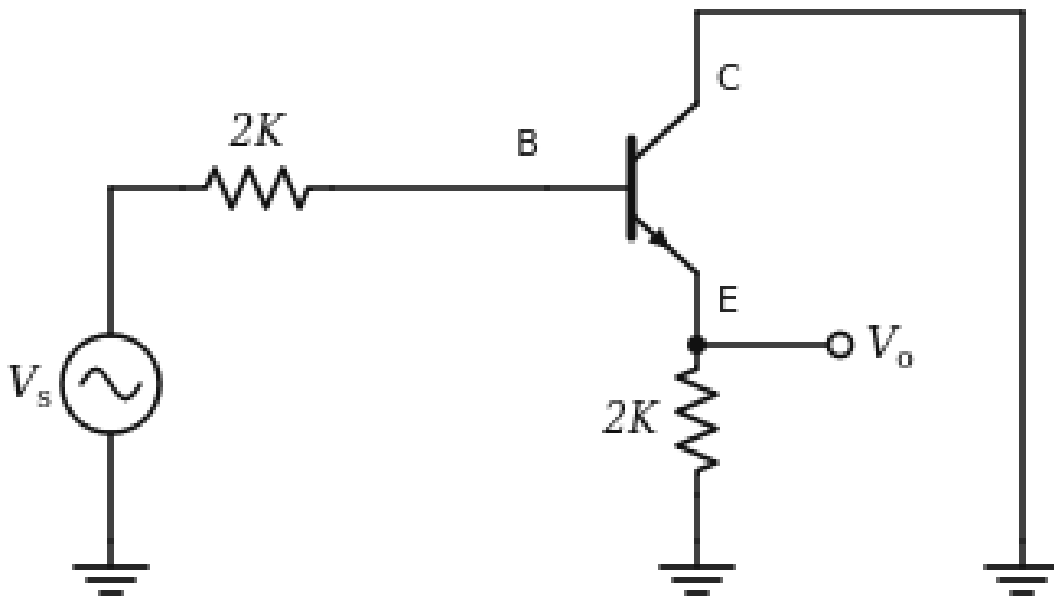
$$\Rightarrow R_i = 32.5k\Omega$$

$$R_i' = R_i \parallel R_x = 32.5k\Omega \parallel 20k\Omega \parallel 40k\Omega$$

$$\Rightarrow R_i' = 9.4545k\Omega$$

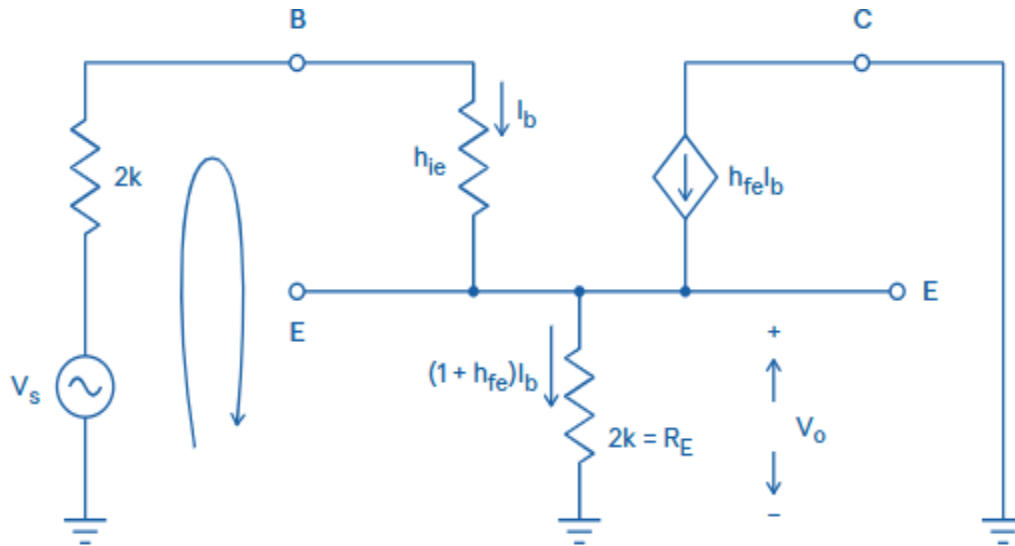
PROBLEM 3

For the amplifier shown, calculate the voltage gain if the transistor has $h_{fe} = 50$ and $h_{ie} = 1\text{ k}\Omega$.



SOLUTION

- ★ The circuit shown is a **CC configuration**.
- ★ Replace the BJT with the approximate h-parameter model:



$$\text{So, } V_o = I_b(1 + h_{fe}) \times 2k\Omega$$

$$\& V_s = (2k\Omega + h_{ie})I_b + 2k\Omega(1 + h_{fe})I_b$$

$$= I_b [2k\Omega + h_{ie} + 2k\Omega + 2k\Omega h_{fe}]$$

So,

$$\frac{V_o}{V_s} = A_V = \frac{(1 + h_{fe})R_E}{(1 + h_{fe})R_E + R_s + h_{ie}} = \frac{51 \times 2k\Omega}{2k\Omega + 50 \times 2k\Omega + 2k\Omega + 1k\Omega}$$

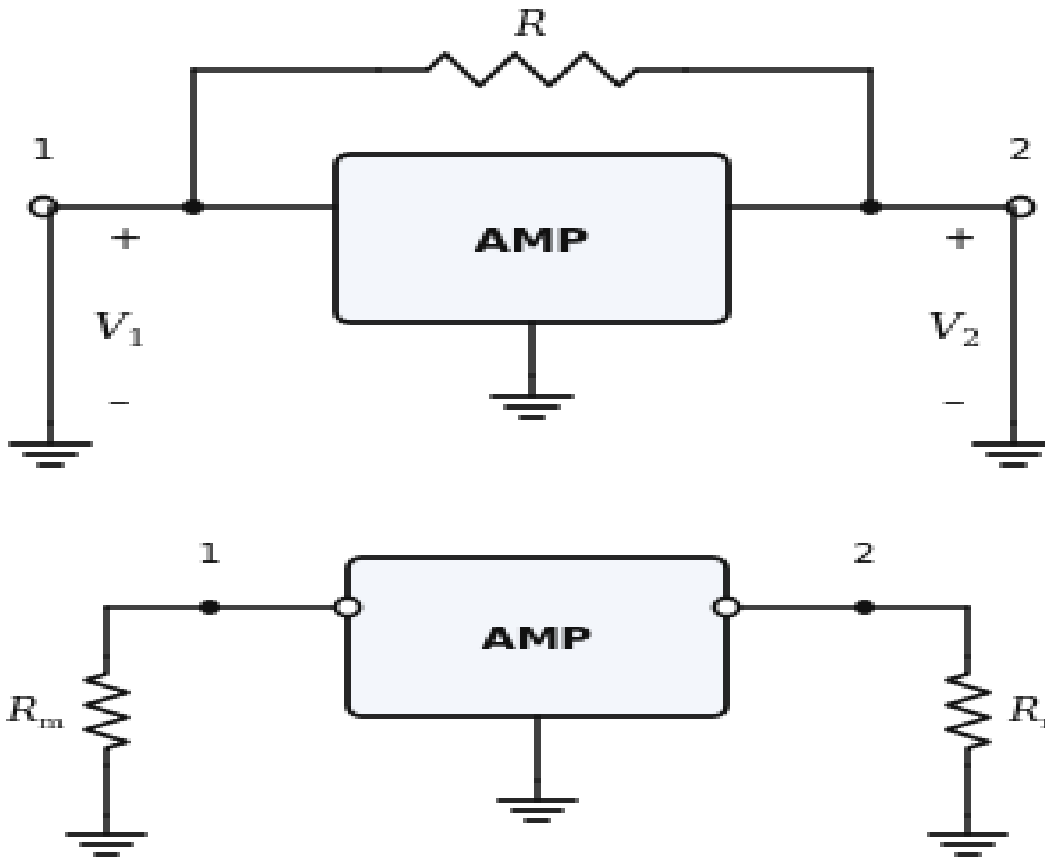
$$\frac{V_o}{V_s} = A_V = 0.9714$$

MILLER'S THEOREM

★ It is used in SMALL SIGNAL ANALYSIS or AC ANALYSIS of amplifiers **if a resistor is present between the output node and the input node.**

Statement

★ A resistor R present between node 1 and node 2 can be replaced with two resistances R_m and R_n , such that R_m is connected between node 1 and ground and R_n is connected between node 2 and ground.



Mathematically:

$$R_m = \frac{R}{(1 - K)}$$

$$R_n = \frac{R}{1 - \frac{1}{K}}$$

$$K = \frac{V_2}{V_1} = \frac{V_o}{V_i} = \text{internal voltage gain}$$

NOTE

K represents the internal voltage gain A_v . If A_v is large, then $\frac{1}{A_v}$ can be neglected; then:

$$R_n = \frac{R}{1 - \frac{1}{A_v}} \approx R$$

NOTE

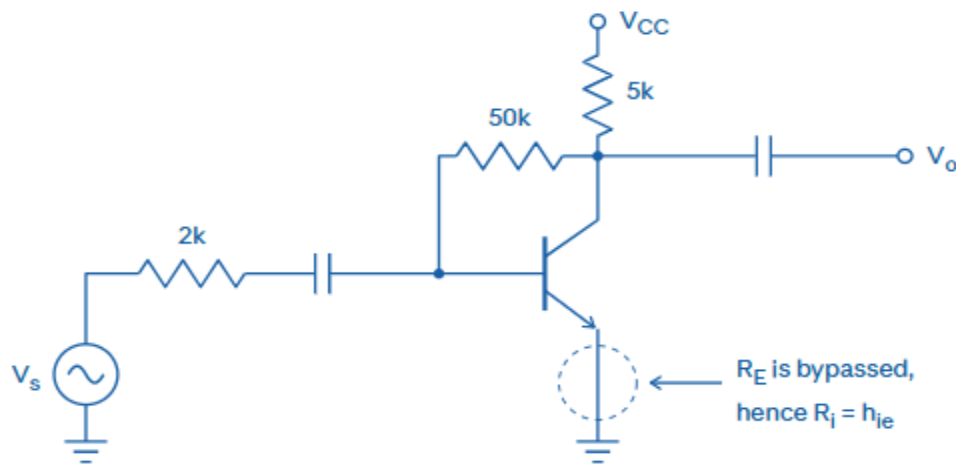
If Miller's theorem is applied to a capacitor, then:

$$C_m = C(1 - K)$$

$$C_n = C \left(1 - \frac{1}{K} \right)$$

PROBLEM 4

Calculate the voltage gain for the amplifier shown, if the transistor has $h_{fe} = 50$ and $h_{ie} = 2 \text{ k}\Omega$.

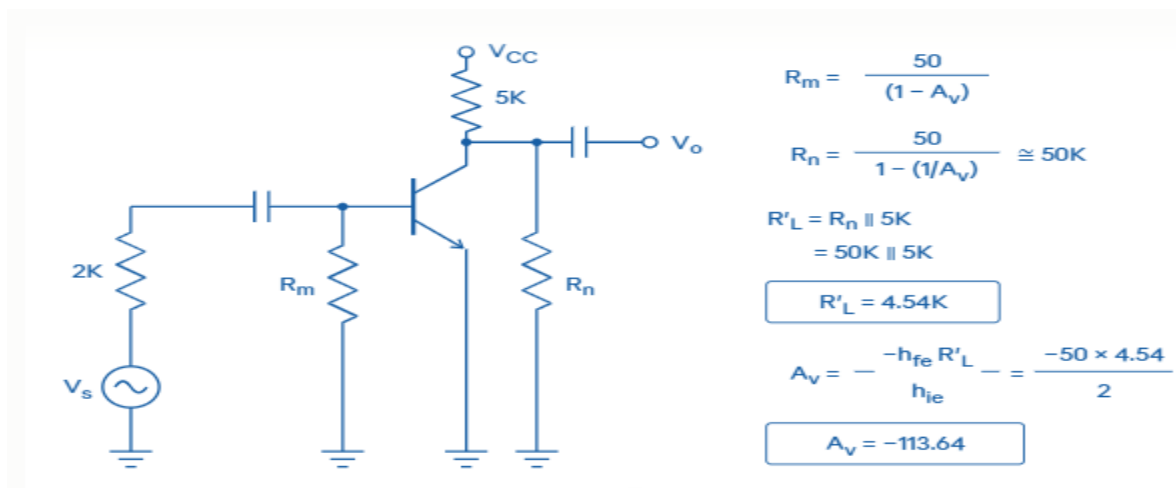
**SOLUTION****NOTE**

The given circuit is a CE amplifier in which a $50 \text{ k}\Omega$ resistor is present between the output node (collector) and the input node (base). Hence the $50 \text{ k}\Omega$ resistor should be replaced with R_m and R_n using Miller's theorem.

★ CE configuration $\Rightarrow$ emitter is directly grounded; hence a bypass capacitor is used.

$$R_m = \frac{50 \text{ k}\Omega}{(1 - A_v)} = \frac{50 \text{ k}\Omega}{1 + 113.65}$$

$$R_m = 0.436 \text{ k}\Omega$$



So, $R_i' = h_{ie} \parallel R_x = h_{ie} \parallel R_B = h_{ie} \parallel R_m = 2k\Omega \parallel 0.436k\Omega$

$$R_i' = 0.358k\Omega$$

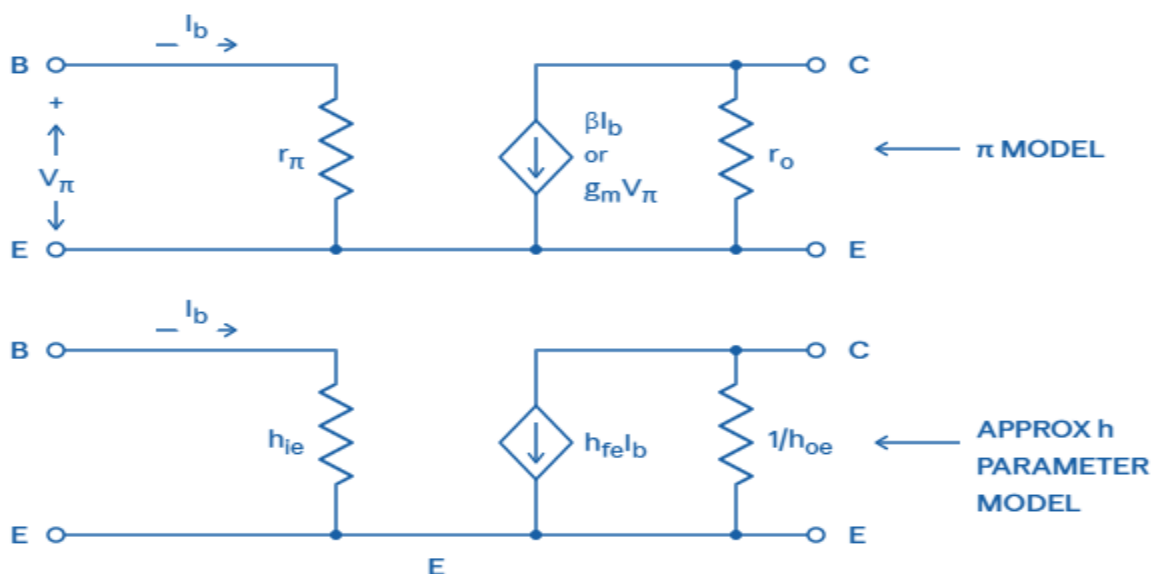
$$\frac{V_o}{V_s} = A_{VS} = \frac{A_v R_i'}{R_s + R_i'} = \frac{-113.5 \times 0.358}{2 + 0.358}$$

$$A_{VS} = -17.23$$

Final answer, as the source is given.

π – MODEL OF BJT

★ It is also called the **COMMON EMITTER MODEL**.



★ π MODEL is identical to the approximate "h parameter model".

★ Through comparison of the π model with the h-parameter model we get:

$$r_{\pi} = h_{ie}$$

$$\beta = h_{fe}$$

$$r_o = \frac{1}{h_{oe}}$$

Calculation of π parameters

★ β , g_m , r_{π} and r_o are known as the **π PARAMETERS**.

i) Common emitter current gain (β)

$$\beta = \frac{\partial I_C}{\partial I_B}$$

ii) Transconductance (g_m)

★ g_m = transconductance of the BJT.

★ It is the rate of change of collector current I_C with respect to V_{BE} .

$$g_m = \frac{\partial I_C}{\partial V_{BE}}$$

$$\text{As } I_C = I_S e^{V_{BE}/V_T} \frac{\partial I_C}{\partial V_{BE}} = I_C \times e^{V_{BE}/V_T} \times \frac{1}{V_T}$$

$$\frac{\partial I_C}{\partial V_{BE}} = \frac{I_C}{V_T} = g_m$$

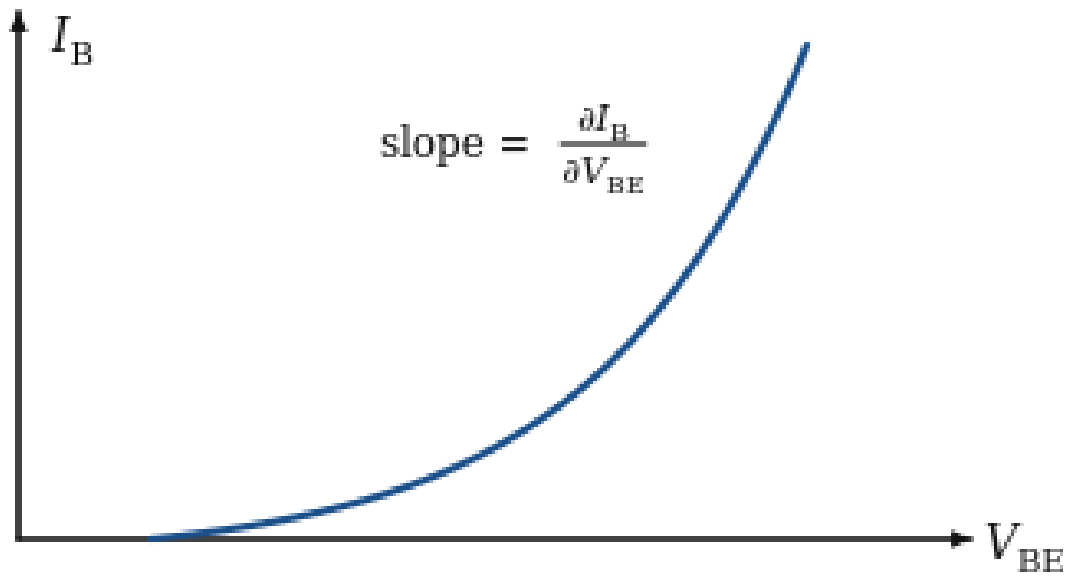
I_C : DC collector current

NOTE

BJT has greater transconductance than MOSFET and JFET, as in a BJT I_C has an exponential dependence on V_{BE} , whereas the drain current of an FET has a quadratic dependence on V_{GS} .

iii) CE input resistance (r_{π})

★ It is the reciprocal of the slope of the CE input characteristics.



$$r_\pi = \frac{1}{\text{slope}} = \frac{\partial V_{BE}}{\partial I_B}$$

Also,

$$r_\pi = \frac{\partial V_{BE}}{\partial I_B} = \frac{\partial V_{BE}}{\partial I_C} \times \frac{\partial I_C}{\partial I_B}$$

$$r_\pi = \frac{1}{g_m} \times \beta$$

Also,

$$r_\pi = \frac{\beta}{g_m} = \frac{\beta}{\frac{I_C}{V_T}} = \frac{\beta V_T}{I_C} = \frac{\beta V_T}{\beta I_B}$$

$$r_\pi = \frac{V_T}{I_B}$$

I_B : DC base current

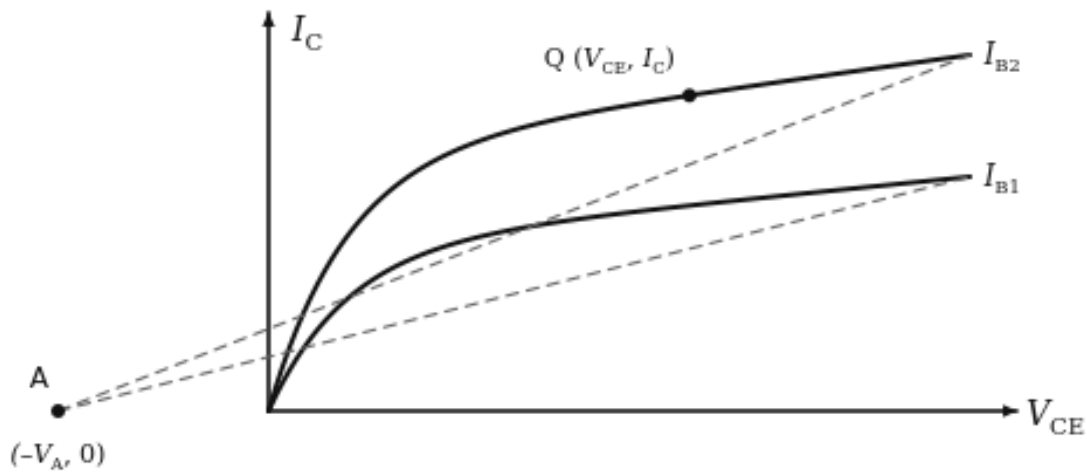
NOTE

At a constant I_C (maintained constant using stabilisation and compensation techniques), if temperature increases then $V_T \uparrow$, $g_m \downarrow$ and $r_\pi \uparrow$.

With $T \uparrow \Rightarrow \beta \uparrow$ and $r_\pi \uparrow$ (similar variation).

iv) CE output resistance (r_o)

★ It is the reciprocal of the slope of the CE output characteristics.



$$r_o = \frac{1}{\text{slope}} = \frac{\partial V_{CE}}{\partial I_C}$$

NOTE

If the CE output characteristics are projected in the backward direction, they cut the voltage axis at a common point $(-V_A, 0)$, where V_A is called the **EARLY VOLTAGE**.

★ Considering the straight line between A and Q we get:

$$\text{slope} = \frac{I_C - 0}{V_{CE} - (-V_A)} = \frac{I_C}{V_A + V_{CE}}$$
$$r_o = \frac{1}{\text{slope}} = \frac{V_A + V_{CE}}{I_C}$$

V_A = Early voltage (≈ 100 V) and $V_{CE} \approx 4$ V or 5 V

★ As $V_A \gg V_{CE}$, so:

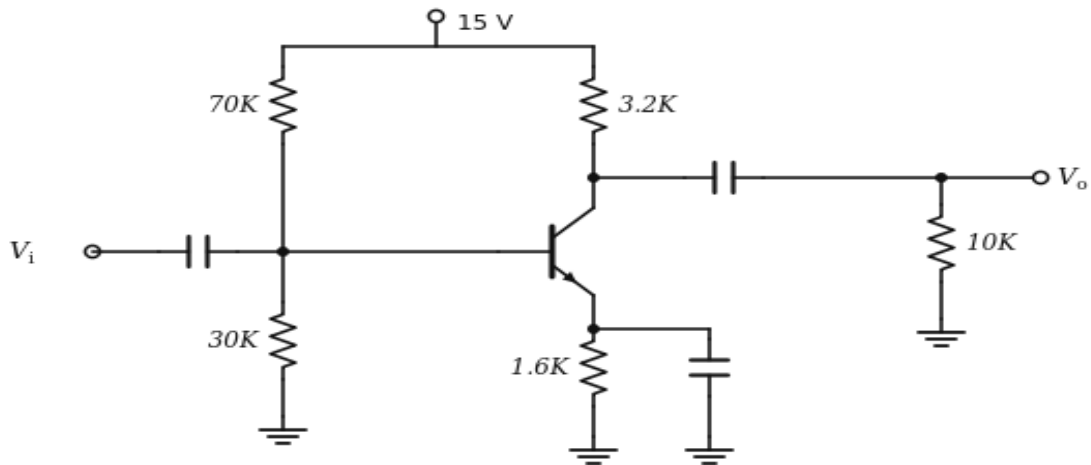
$$r_o = \frac{V_A}{I_C}$$

NOTE

If the BJT does not have the EARLY EFFECT, then the output characteristics remain flat or horizontal. Hence slope = 0, $r_o = \infty$ and $V_A = \infty$.

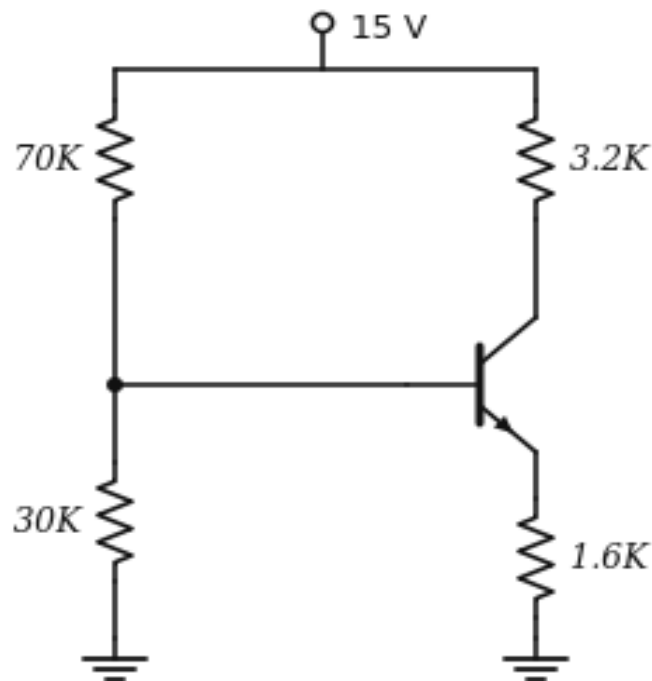
PROBLEM 5

In the circuit shown, the BJT has $\beta = 100$, $V_{BE} = 0.7\text{ V}$ and $V_A = \infty$. Calculate: i) DC operating point, ii) π parameters, iii) small signal voltage gain (V_o/V_i).



SOLUTION

i) In calculation of the operating point, all capacitors are considered as open circuit. Hence redrawing the circuit we get:



$$V_{th} = \frac{15 \times 30k\Omega}{100k\Omega} = 4.5V ; R_{th} = 70k\Omega \parallel 30k\Omega = 21k\Omega ; \text{ Now, } I_B = \frac{V_{th} - V_{BE}}{R_B + (1 + \beta)R_E} = \frac{4.5 - 0.7}{21 + (1 + 100) \times 1.6}$$

$$\Rightarrow I_B = 0.0208\text{mA}$$

$$\text{Now, } I_C = \beta I_B = 2.08\text{mA} ; V_{CE} = V_{CC} - I_C(R_C + R_E) = 15 - 2.08(4.8)$$

$$\Rightarrow V_{CE} = 5.016\text{V}$$

Hence the operating point is (V_{CE} , I_C):

$$Q(5\text{V}, 2.08\text{mA})$$

ii) π parameters

$$a) g_m = \frac{I_C}{V_T} = \frac{2.08\text{mA}}{26\text{mV}} = 0.082127 = 80\text{mA/V} \quad b) r_\pi = \frac{\beta}{g_m} = \frac{100}{80\text{mA/V}} = 1.25\text{k}\Omega ; c) r_o = \infty \{ \because V_A = \infty \}$$

iii) Small signal voltage gain

★ Given circuit is a CE amplifier with bypass capacitor. Hence:

$$\frac{V_o}{V_i} = \frac{-h_{fe} R_L'}{h_{ie}}$$

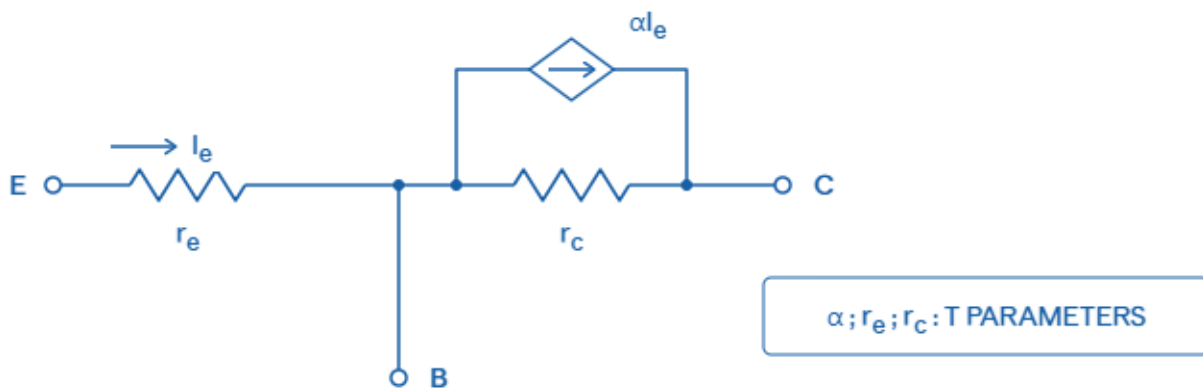
$$A_V = \frac{-\beta \cdot R_L'}{r_\pi} = -g_m R_L'$$

$$\text{Now, } A_V = -80\text{mA/V} \times (3.2\text{k}\Omega \parallel 10\text{k}\Omega) = -80 \times 10^{-3} \times 2.4\text{k}\Omega$$

$$A_V = -194$$

T – MODEL OF BJT

★ It is also called the **COMMON BASE MODEL** (r_e model).



i) Common base current gain (α)

$$\alpha = \frac{\beta}{1 + \beta}$$

slightly less than 1

ii) Common base input resistance (r_e)

★ It is the reciprocal of the slope of the common base input characteristics, **OR** it is the dynamic resistance of the emitter junction.

$$r_e = \frac{V_T}{I_E}$$

I_E : DC EMITTER CURRENT

Also,

$$r_e = \frac{V_T}{I_C} \times \frac{I_C}{I_E} \left\{ \frac{V_T}{I_C} = \frac{1}{g_m} \text{ and } \frac{I_C}{I_E} = \alpha \right\}$$

$$r_e = \frac{\alpha}{g_m} \Rightarrow r_e \propto \frac{1}{g_m} \Rightarrow r_e \approx \frac{1}{g_m} \text{ \{since } \alpha \approx 1 \}}$$

Also,

$$r_e = \frac{\alpha}{g_m} = \frac{\beta}{(1 + \beta)g_m}$$

$$r_e = \frac{r_\pi}{(1 + \beta)}$$

iii) Common base output resistance (r_c)

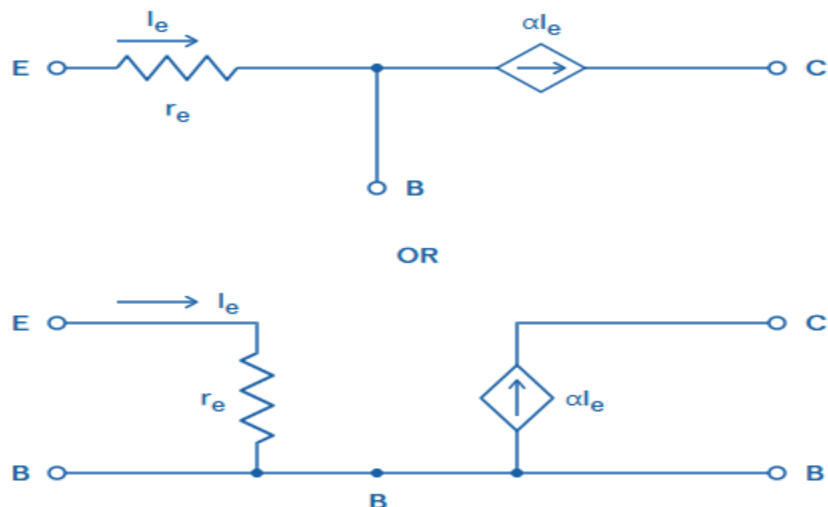
★ It is the reciprocal of the slope of the CB output characteristics.

$$r_c = (1 + \beta)r_o$$

Range of r_c is in $M\Omega$

NOTE

As r_c is very high ($M\Omega$), it is replaced with an open circuit in the analysis. Hence redrawing the T model we get:



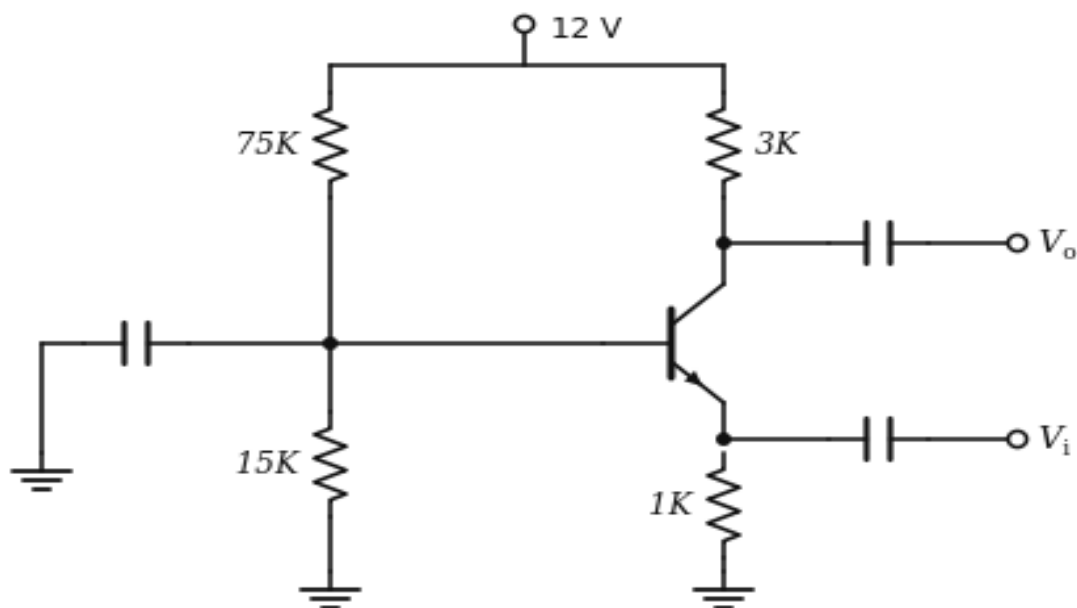
NOTE

T-parameter model of the BJT should be used while analysing a CB transistor amplifier, as the analysis becomes simpler than using h parameters for the same analysis.

Also, T parameters are used in the analysis of the **differential amplifier**.

PROBLEM 6

For the amplifier shown, the BJT has $\beta = 99$, $V_{BE} = 0.7 \text{ V}$ and $V_A = \infty$. Calculate: i) T parameters, ii) small signal voltage gain, iii) input resistance.



SOLUTION

i) T PARAMETERS

★ To obtain the T parameters we need to calculate the DC emitter current by replacing the capacitors with open circuit.

$$V_{th} = \frac{12 \times 15}{75 + 15} = 2V ; R_{th} = 75k\Omega \parallel 15k\Omega = 12.5k\Omega ; I_B = \frac{V_{th} - V_{BE}}{R_{th} + (1 + \beta)R_E} = \frac{2 - 0.7}{12.5 + (1 + 99) \times 1} = 0.0115mA$$

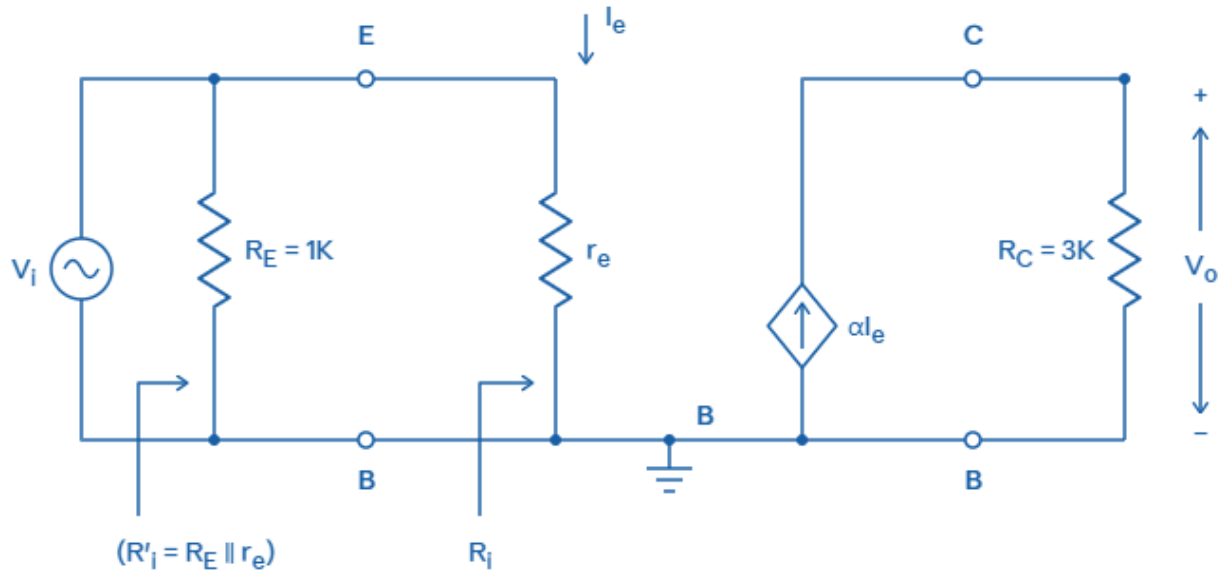
So, $I_E = (1 + \beta)I_B = 100I_B = 1.15mA$; $r_e = \frac{V_T}{I_E} = \frac{26mV}{1.15mA}$

$$r_e = 22.6\Omega$$

$$r_c = \infty ; V_A = \infty ; \alpha = \frac{\beta}{(1 + \beta)} = 0.99$$

ii) Calculation of gain

★ The given circuit is a CB amplifier. It can be analysed by replacing the BJT with the T model:



$$V_o = \alpha I_E \times R_C ; V_i = I_E r_e ; \text{ So, } A_V = \frac{V_o}{V_i} = \frac{\alpha I_E R_C}{I_E r_e}$$

$$A_V = \frac{\alpha R_C}{r_e} = \frac{0.99 \times 3k\Omega}{22.6\Omega} = 131.416$$

iii) Input resistance

$$R_i = r_e = 22.6\Omega ; R'_i = R_i \parallel 1k\Omega = 22.1\Omega$$

PROBLEM 7

A BJT in CB configuration is used to amplify a signal received by an antenna of impedance $50\ \Omega$. If $V_T = 25\text{ mV}$, then calculate the DC collector current of the BJT such that the antenna impedance is matched with the input impedance of the CB amplifier.

SOLUTION

For a CB amplifier:

$$R_i = r_e = \frac{V_T}{I_E} \approx \frac{V_T}{I_C} = 50\ \Omega \quad I_C = \frac{V_T}{50} = \frac{25\text{mV}}{50}$$

$$I_C = 0.5\text{mA}$$

SUMMARY OF VOLTAGE GAIN EXPRESSIONS

★ Voltage gain of CE amplifier **with** bypass capacitor is:

$$\frac{V_o}{V_i} = A_V = \frac{-h_{fe} R_L'}{h_{ie}} = -g_m R_L'$$

★ Voltage gain of CE amplifier **without** bypass capacitor is:

$$\frac{V_o}{V_i} = A_V = \frac{-h_{fe} R_L'}{h_{ie} + (1+h_{fe})R_E} = \frac{-\beta R_L'}{r_\pi + (1+\beta)R_E} \quad \text{So, } A_V = \frac{-R_L'}{\frac{r_\pi}{\beta} + \frac{(1+\beta)R_E}{\beta}} = \frac{-R_L'}{\frac{1}{g_m} + R_E} \quad \{(1+\beta) \approx \beta\}$$

$$A_V = \frac{-g_m R_L'}{1 + g_m R_E}$$

NOTE

If $g_m R_E \gg 1$, then $A_V = \frac{-g_m R_L'}{g_m R_E} = \frac{-R_L'}{R_E} \quad \{ R_L' = R_L \parallel R_C \}$

If $R_L = \infty$, then $A_V \approx \frac{-R_C}{R_E}$.

★ Voltage gain of CC amplifier is given as:

$$\frac{V_o}{V_i} = A_V = \frac{(1+h_{fe})R_E}{h_{ie} + (1+h_{fe})R_E} = \frac{(1+\beta)R_E}{r_\pi + (1+\beta)R_E} = \frac{R_E}{\frac{r_\pi}{(1+\beta)} + R_E}$$

$$A_V = \frac{R_E}{r_e + R_E} \quad \left\{ \frac{r_\pi}{1+\beta} = r_e \right\}$$

also, $r_e \approx \frac{1}{g_m}$, so:

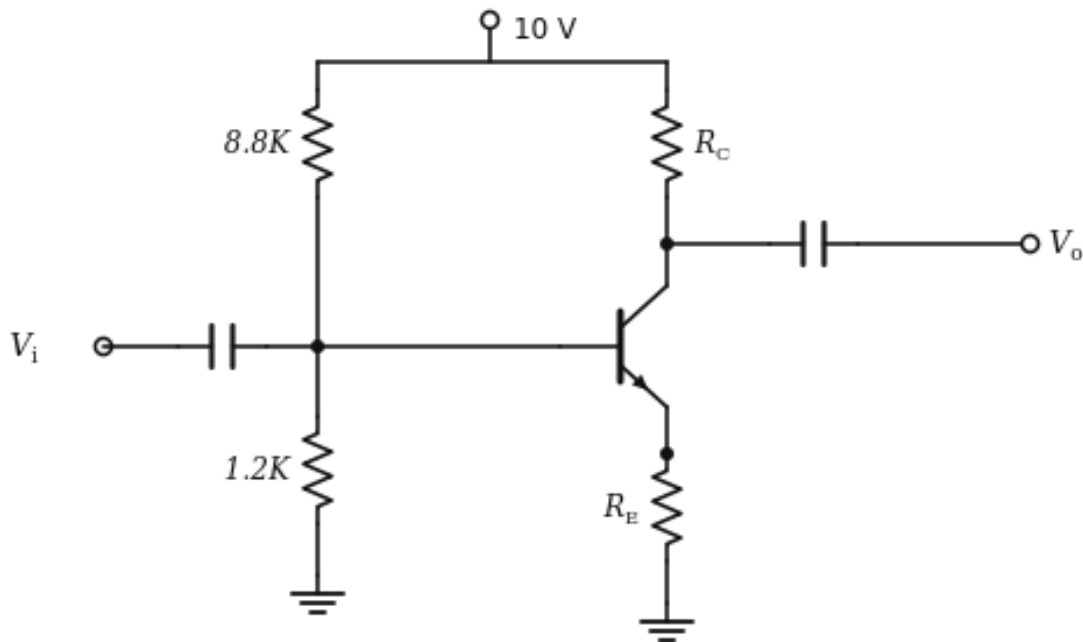
$$A_V = \frac{R_E}{\frac{1}{g_m} + R_E}$$

★ Voltage gain of CB amplifier is:

$$\frac{V_o}{V_i} = A_V = \frac{h_{fe} R_L'}{h_{ie}} = g_m R_L'$$

PROBLEM 8

Calculate the small signal voltage gain (V_o/V_i) in the amplifier shown, if the transistor has large β , $V_{BE} = 0.7$ V and $V_{CE} = V_{CC}/2$.



SOLUTION

★ Given amplifier is CE **without** bypass capacitor.

★ As g_m could not be found because R_E is not given, hence we have to go to:

$$A_V = \frac{-R_C}{R_E}$$

★ For the self bias circuit:

$$V_{th} = \frac{1.2k\Omega \times 10V}{10k\Omega} = 1.2V; I_C = \frac{V_{th} - V_{BE}}{R_E} = \frac{1.2 - 0.7}{R_E} = \frac{0.5}{R_E} \dots (i)$$

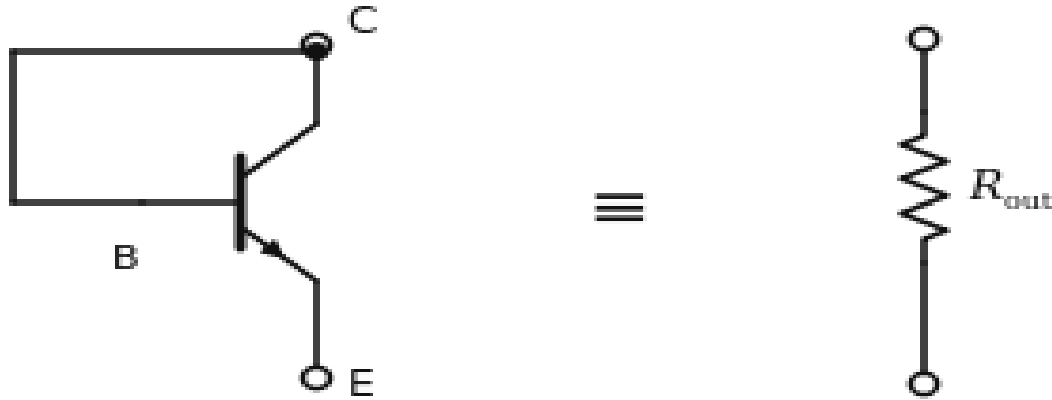
$$\text{Also, } V_{CE} = V_{CC} - I_C(R_C + R_E) \quad 5 = 10 - I_C(R_C + R_E) \Rightarrow I_C(R_C + R_E) = 5 \quad \frac{0.5}{R_E}(R_C + R_E) = 5$$

$$\{ \text{from equation (i)} \} \frac{R_C + R_E}{R_E} = 10 \Rightarrow \frac{R_C}{R_E} + 1 = 10$$

$$\frac{R_C}{R_E} = 9 = -A_V \Rightarrow A_V = -9$$

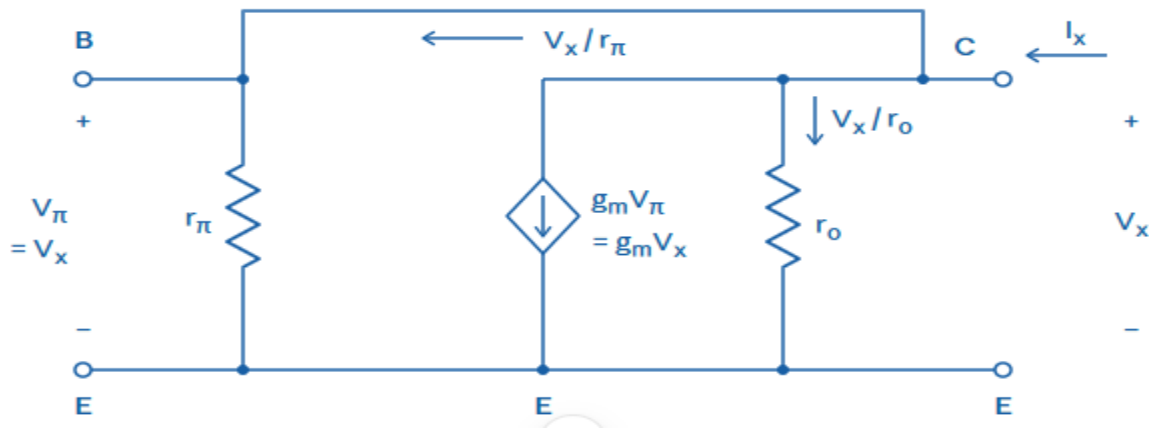
BJT AS AN ACTIVE LOAD

★ A BJT with the **collector shorted to the base** can be used as an **ACTIVE LOAD** in IC amplifiers. Its output resistance will act as load.



R_{out} = small signal resistance seen between C and E

★ R_{out} can be calculated by replacing the BJT with the π MODEL. Redrawing the circuit we get:



NOTE

To calculate R_o , connect V_x at the output; I_x flows through it.

★ KCL at the collector, we get:

$$I_x = \frac{V_x}{r_\pi} + g_m V_x + \frac{V_x}{r_o} I_x = V_x \left[\frac{1}{r_\pi} + g_m + \frac{1}{r_o} \right] \frac{I_x}{V_x} = \left[\frac{1}{r_\pi} + \frac{1}{r_o} + \frac{1}{\frac{1}{g_m}} \right] \frac{1}{R_{out}} = \frac{1}{r_\pi} + \frac{1}{r_o} + \frac{1}{\frac{1}{g_m}}$$

$$\text{So, } R_{out} = \frac{1}{g_m} \parallel r_\pi \parallel r_o$$

★ If NO EARLY EFFECT: $V_A = \infty \Rightarrow r_o = \infty$:

$$R_{\text{out}} = \frac{1}{g_m} \parallel r_{\pi}$$

★ As $\frac{1}{g_m} \ll r_{\pi}$:

$$R_{\text{out}} \approx \frac{1}{g_m}$$

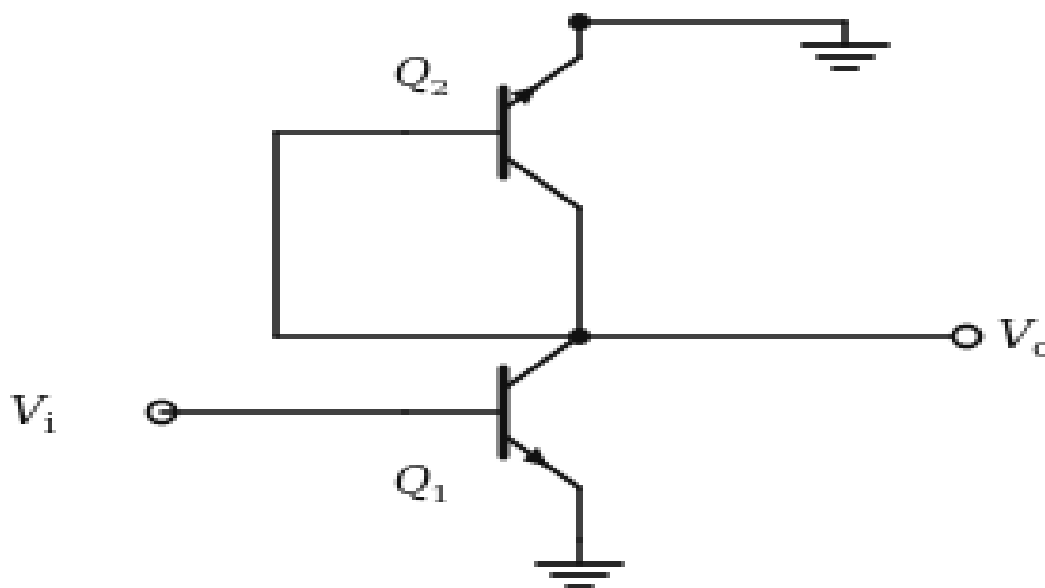
NOTE

A BJT with collector shorted to base can be replaced with $\frac{1}{g_m}$ in small signal or AC ANALYSIS.

In DC analysis the BJT should be treated as it is.

PROBLEM 9

In the AC equivalent circuit shown, the two BJTs are biased in the active region and have identical parameters with $\beta \gg 1$. The open circuit small signal voltage gain is approximately ...

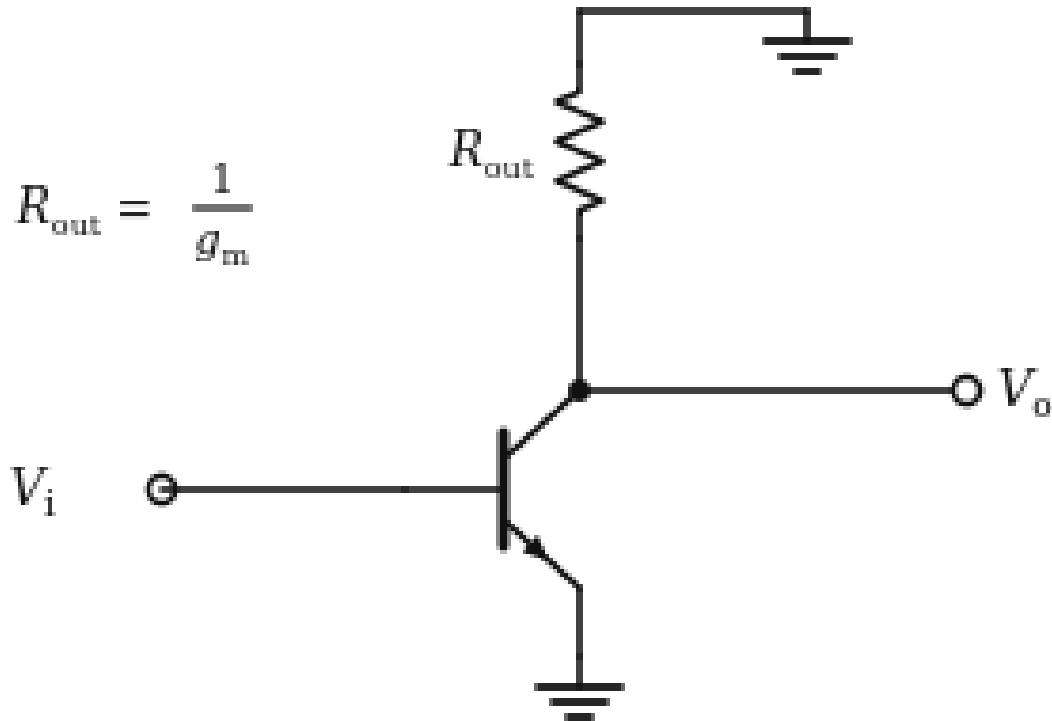


SOLUTION

NOTE

Q_1 is a CE amplifier.

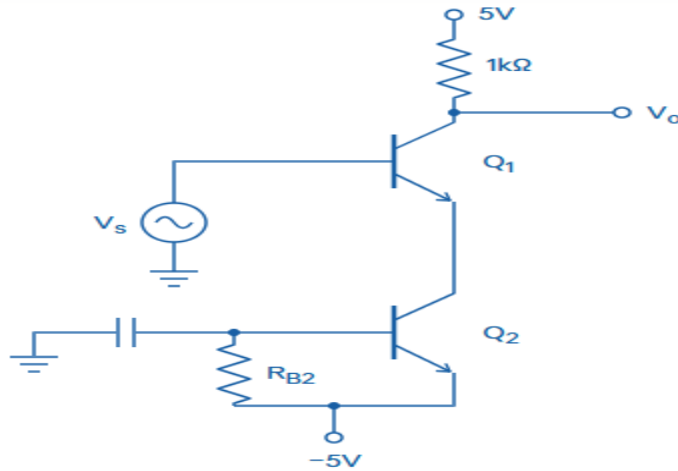
Q_2 , with collector shorted to base, is an ACTIVE LOAD for Q_1 . Hence Q_2 should be replaced with its R_{out} in small signal analysis.



$$\text{So, Gain} = \frac{V_o}{V_i} = -g_m R_L' = -g_m \times \frac{1}{g_m}$$
$$\frac{V_o}{V_i} = -1$$

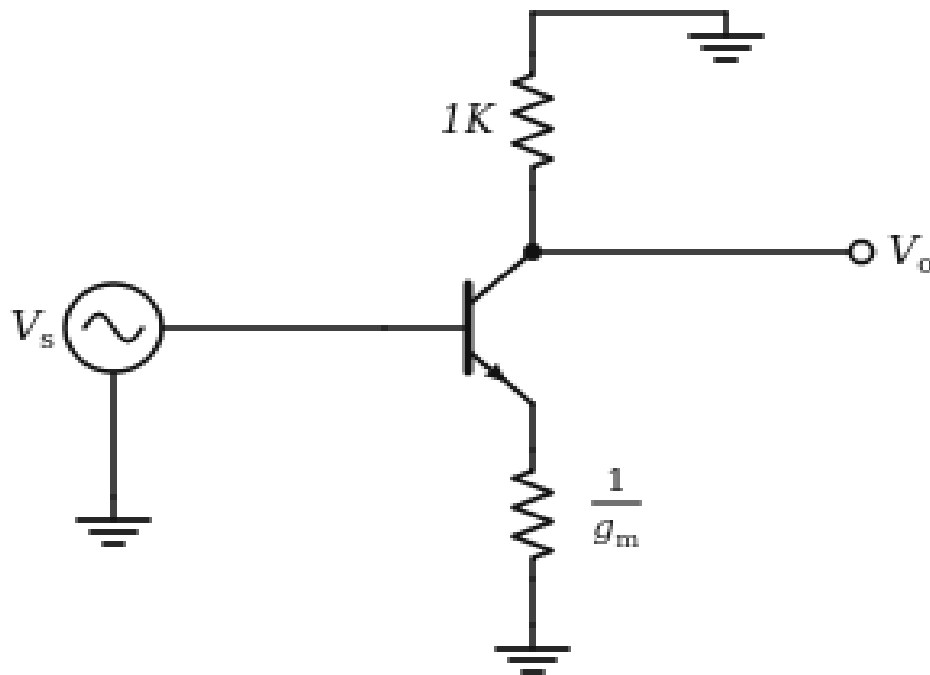
PROBLEM 10

In the circuit shown, transistors Q_1 and Q_2 are biased at a collector current of 2.6 mA. Assuming that the transistor current gains are sufficiently large to assume collector current equal to emitter current, and thermal voltage of 26 mV, the magnitude of voltage gain V_o/V_s in the mid band frequency range is ... (up to second decimal place).



SOLUTION

★ In AC analysis, capacitors are shorted and source voltages are grounded. Then the collector and base of Q₂ get shorted. Hence Q₂ should be replaced by its $R_{out} = \frac{1}{g_m}$.



CE amplifier without bypass capacitor

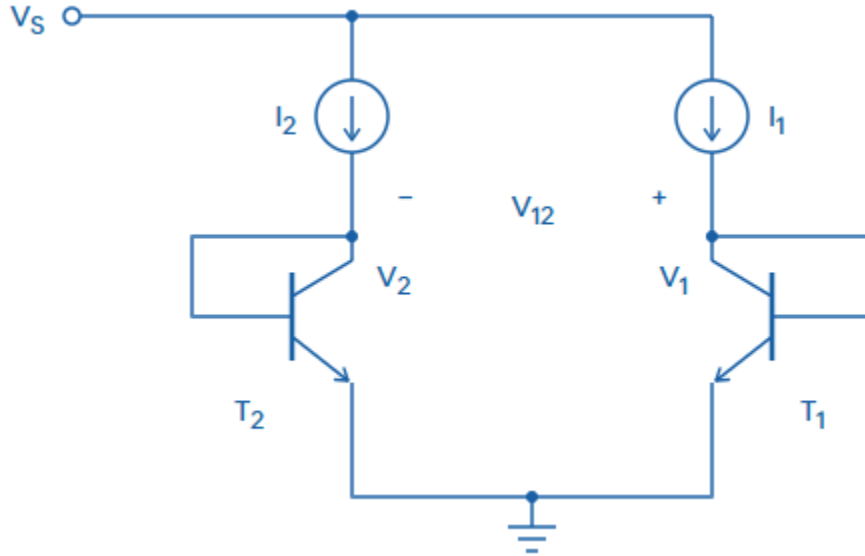
$$\text{So, } \frac{V_o}{V_s} = \frac{V_o}{V_i} = \frac{-g_m R_L'}{1 + g_m R_E}; g_m = \frac{I_C}{V_T} = \frac{2.6\text{Ma}}{26\text{Mv}} = 100\text{mA/V}; R_L' = R_C = 1\text{k}\Omega;$$

$$\frac{V_o}{V_i} = \frac{-100 \times 10^{-3} \times 1 \times 10^3}{1 + (100 \times 10^{-3}) \times \frac{1}{g_m}} = \frac{-100}{2}$$

$$\frac{V_o}{V_i} = -50$$

PROBLEM 11

In the circuit shown, $I_1 = 80 \text{ mA}$ and $I_2 = 4 \text{ mA}$. Transistors T_1 and T_2 are identical. Assume that the thermal voltage V_T is 26 mV at 27°C . At 50°C the value of the voltage $V_{12} = V_1 - V_2$ (in mV) is ...

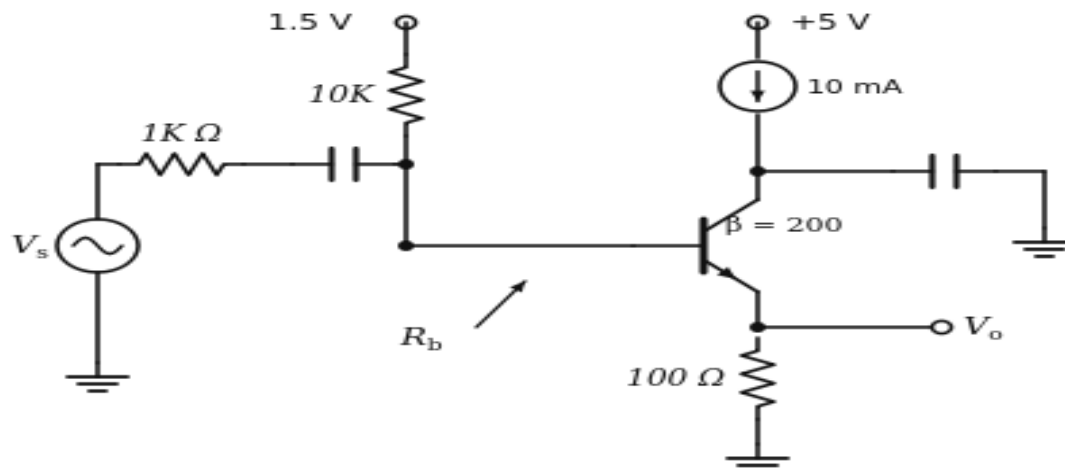


SOLUTION

$$\begin{aligned} V_{12} &= V_1 - V_2 = V_{BE1} - V_{BE2} \\ &= V_T \ln\left(\frac{I_{E1}}{I_{ES}}\right) - V_T \ln\left(\frac{I_{E2}}{I_{ES}}\right) \\ &= V_T \ln\left(\frac{I_{E1}}{I_{E2}}\right) = \frac{T}{11600} \ln\left(\frac{80}{4}\right) = \frac{50+273}{11600} \ln(20) \\ V_{12} &= 83.415 \text{ mV} \end{aligned}$$

PROBLEM 12

In the given circuit, the impedance R_b seen through the base of the transistor is ... Ω .



CE with bypass

SOLUTION

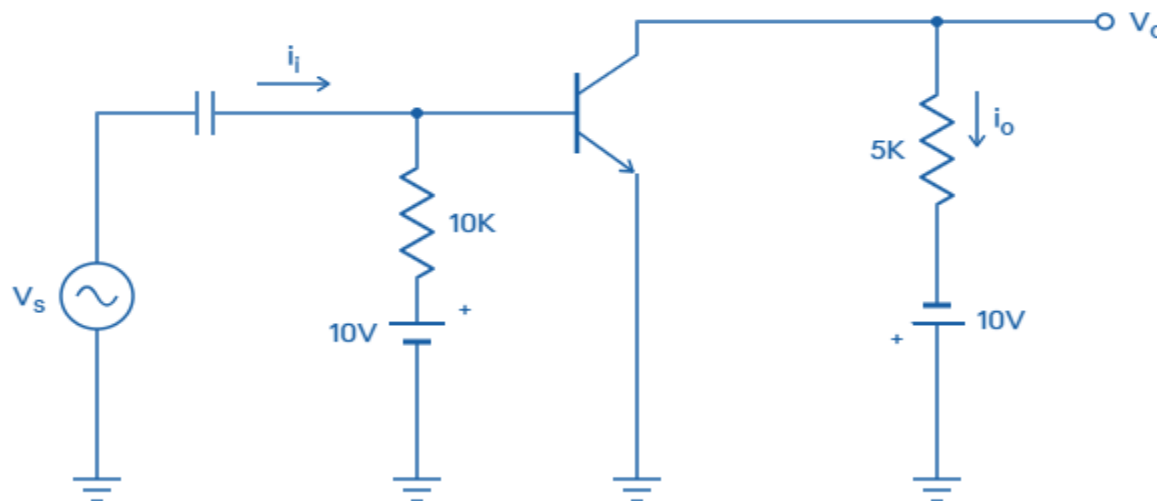
★ Resistance seen through the base is the input resistance of the amplifier.

$$R_i = h_{ie} = r_\pi = \frac{V_T}{I_B}; I_B = \frac{I_E}{1+\beta} = \frac{10mA}{1+200} = 0.04975mA; R_i = r_\pi = \frac{26mV}{0.04975mA}$$

$$R_i = r_\pi = 522.6\Omega$$

PROBLEM 13

Consider the transistor amplifier circuit shown below. The transistor parameters are $\beta = 100$, $V_{BE(on)} = -0.7V$ and $V_A = \infty$. What is the current gain $A_i = i_o/i_1$?



SOLUTION

★ Given circuit is a common base amplifier. Hence current gain can be calculated by replacing the BJT with the T MODEL.

By KVL we get:

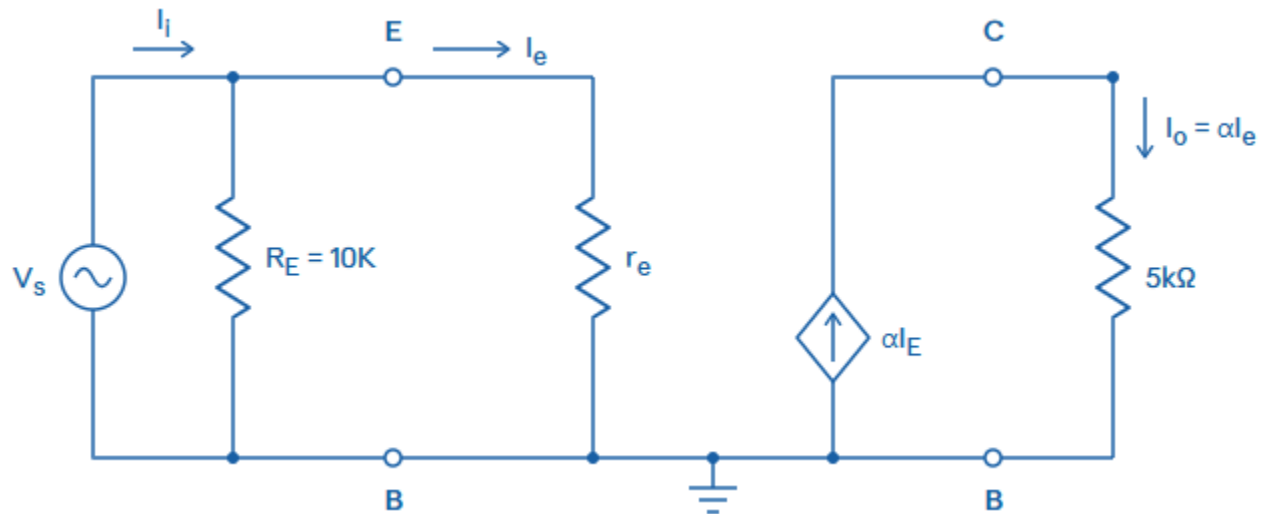
$$-10 + 10 I_E - V_{BE} = 0 \Rightarrow I_E = \frac{10 + V_{BE}}{10}$$

$$I_E = \frac{10 - 0.7}{10} = \frac{9.3}{10} = 0.93\text{mA}$$

$$\text{So, } r_e = \frac{V_T}{I_E} = \frac{26\text{mV}}{0.93\text{mA}} \approx 28\Omega$$

$$r_c = \infty; V_A = \infty; \alpha = \frac{\beta}{1 + \beta} = \frac{100}{101} = 0.99$$

★ Replace the BJT with the T MODEL, we get:



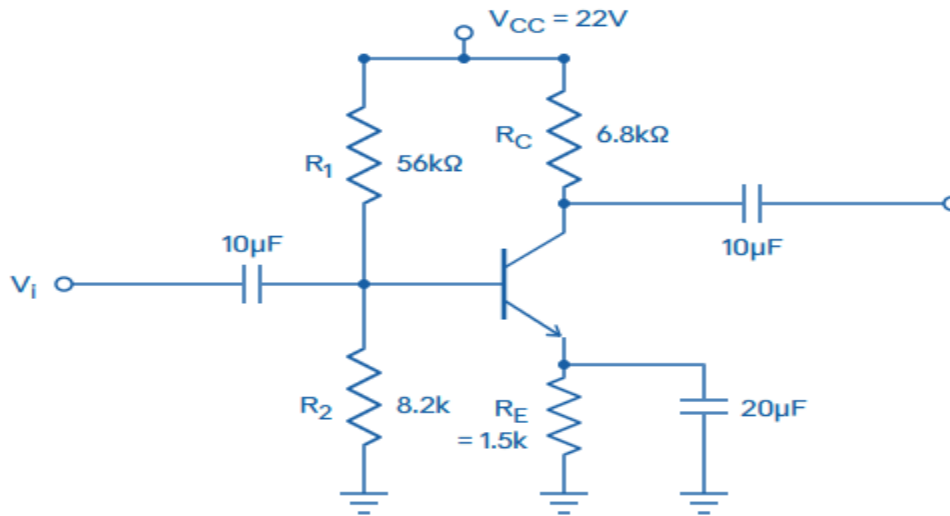
Now, by current divider we get:

$$I_E = \frac{I_1 \times 10k\Omega}{10k\Omega + 28\Omega} = \frac{I_1 \times R_E}{R_E + r_e} \frac{I_o}{I_E} = \frac{I_o}{I_E} \times \frac{I_E}{I_1} = \alpha \times \frac{R_E}{R_E + r_e} = 0.99 \times \frac{10k\Omega}{10k\Omega + 28\Omega}$$

$$\frac{I_o}{I_1} = A_i = 0.98723$$

PROBLEM 14

For the circuit given below, $V_{\text{thermal}} = 26 \text{ mV}$. What is the value of the current gain $A_i = i_o/i_1$? (Take $\beta = 90$.)



SOLUTION

★ Given circuit is a CE amplifier with bypassed emitter.

★ Current gain can be calculated by replacing the BJT with the π MODEL.

★ For the self bias circuit:

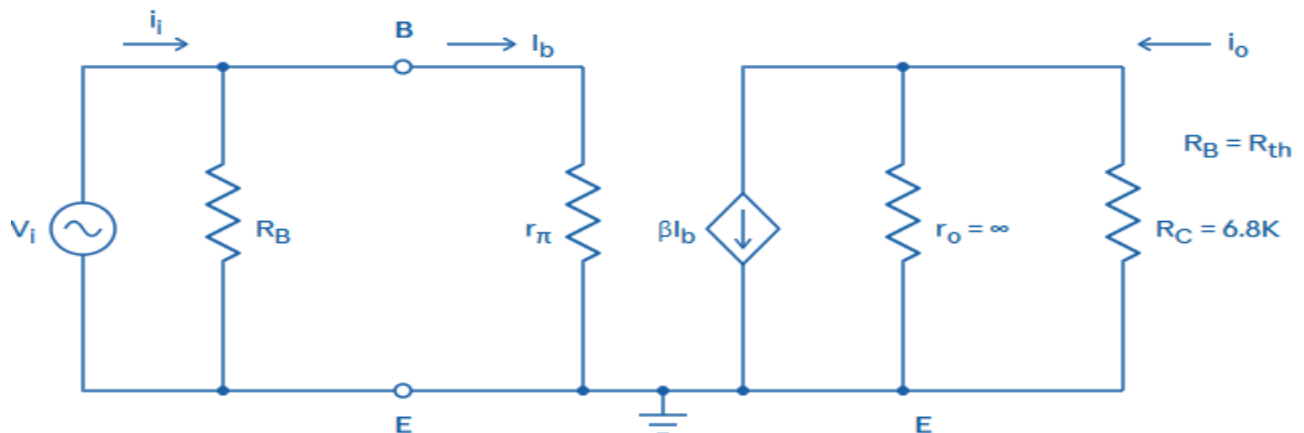
$$V_{th} = \frac{22 \times 8.2}{8.2 + 56} = 2.80V ; R_{th} = 56 \parallel 8.2 = 7.15k\Omega ;$$

$$\text{So, } I_B = \frac{V_{th} - V_{BE}}{R_B + (1 + \beta)R_E} = \frac{2.8 - 0.7}{7.15 + (1 + 90) \times 1.5} = 0.0146\text{mA}$$

$$\text{So, } r_{\pi} = \frac{V_T}{I_B} = \frac{26\text{mV}}{0.0146\text{mA}}$$

$$r_{\pi} = 1.78k\Omega ; r_o = \infty ; \beta = 90$$

★ Replace the BJT with the π MODEL, we get:



$$R_B = R_{th}, R_C = 6.8 k\Omega, r_o = \infty$$

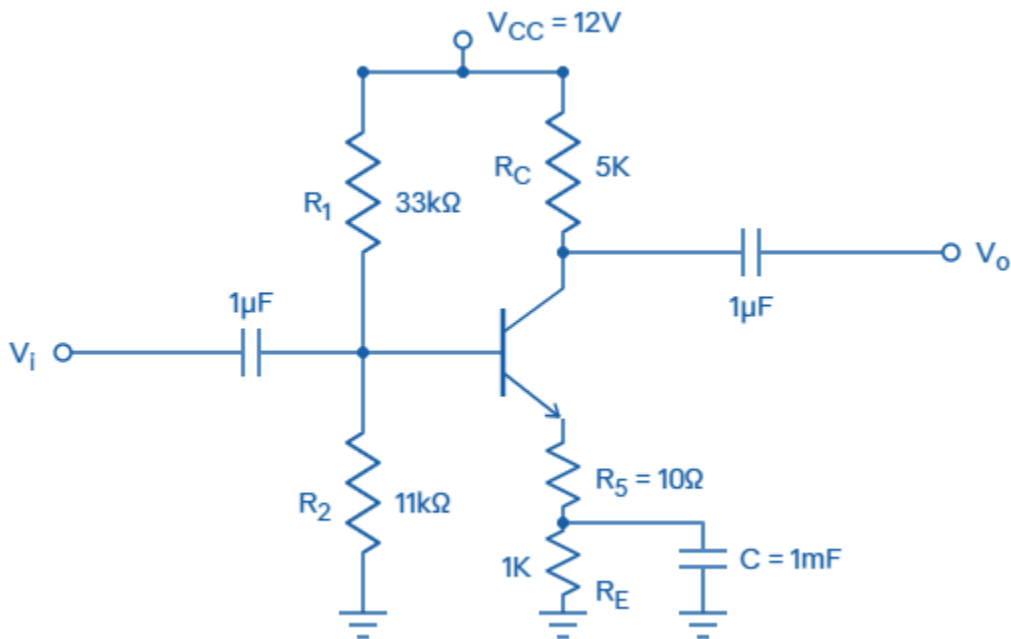
$$i_o = \beta I_B$$

$$i_o = \beta \times \frac{i_1 \times R_B}{R_B + r_\pi}$$

$$\frac{i_o}{i_1} = A_i = \frac{\beta \times R_B}{R_B + r_\pi} = 72.0604$$

PROBLEM 15

For the amplifier shown in the figure, the BJT parameters are $V_{BE} = 0.7$ V, $\beta = 200$ and thermal voltage $V_T = 25$ mV. The voltage gain (V_o/V_i) of the amplifier is ...



SOLUTION

NOTE

Given circuit is a CE amplifier in which 1 kΩ is bypassed and 10 Ω is unbypassed.

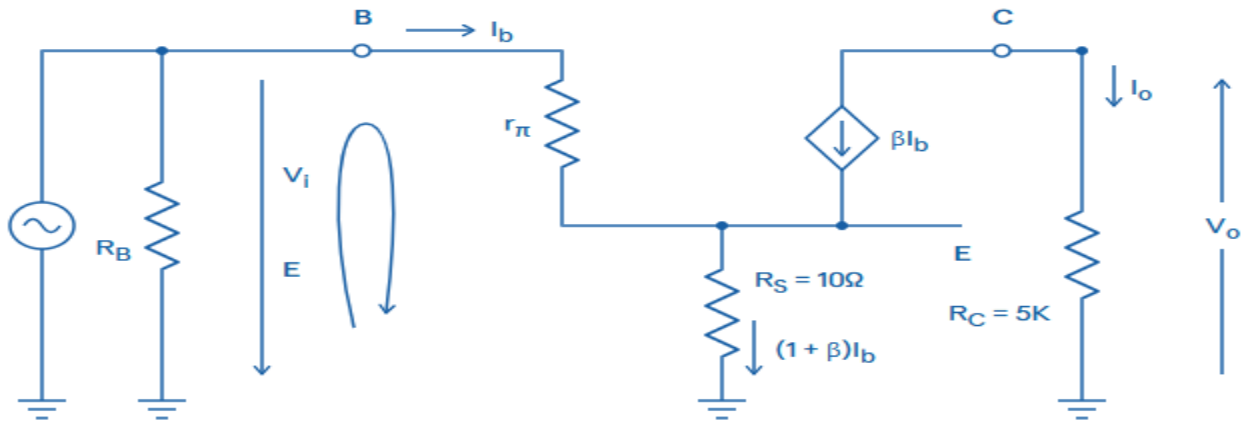
$$V_{th} = \frac{12 \times 11k\Omega}{11k\Omega + 33k\Omega} = \frac{12 \times 11}{44} = 3V$$

$$R_{th} = 33k\Omega \parallel 11k\Omega = 8.25k\Omega$$

$$\text{Now, } I_B = \frac{V_{th} - V_{BE}}{R_B + (1 + \beta)R_E'} = \frac{3 - 0.7}{8.25 + (201) \times 1010} = 0.0109\text{mA}$$

$$I_C = \beta I_B = 2.19\text{mA}$$

$$\text{Now, } r_{\pi} = \frac{V_T}{I_B} = \frac{25\text{mV}}{0.0109\text{mA}} = 2.293\text{k}\Omega$$



$$\text{So, } V_o = 5\text{k}\Omega \times \beta I_B = 5000 \times 200 \times 0.0109 \times 10^{-3} = 10.9 \text{ V}_i = I_B r_{\pi} + 0.01 \times (1 + \beta) I_B = 0.0469$$

$$A_V = \frac{V_o}{V_i} = \frac{-10.9}{0.0469} = -232.4$$

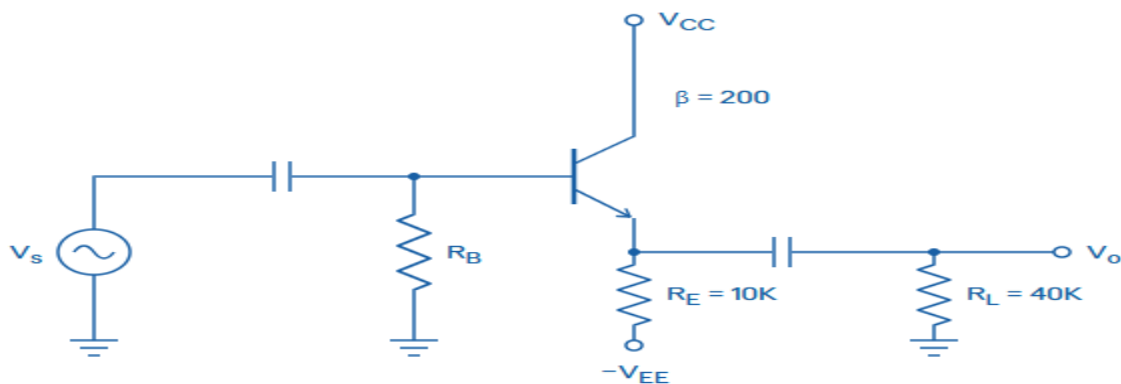
Also,

$$A_V = \frac{-\beta R_L'}{r_{\pi} + (1 + \beta) R_E} = \frac{-200 \times 5}{2.29 + 201 \times 0.01}$$

$$A_V = -232$$

PROBLEM 16

With reference to the CC circuit shown in the figure below, the input resistance looking into the base is ...



SOLUTION

★ Given circuit is a CC amplifier. Resistance looking into the base is the input resistance.

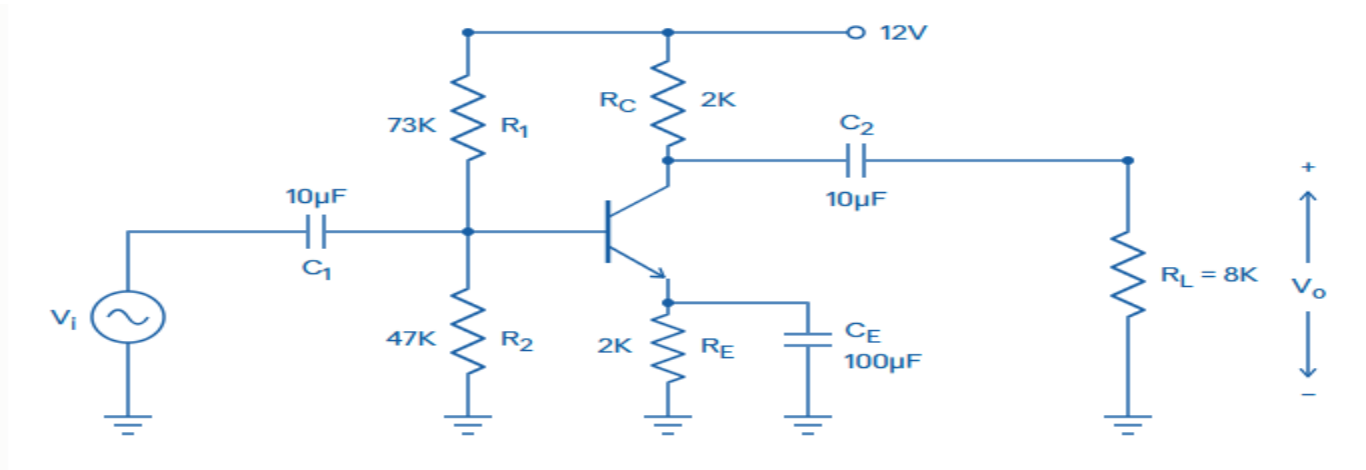
$$R_i = h_{ie} + (h_{fe} + 1)R_E = r_{\pi} + (1 + \beta)R_E \approx (1 + \beta)R_E \text{ \{ } r_{\pi} \text{ neglected as data insufficient} \}}$$

$$R_i = 201 \times (10k\Omega \parallel 40k\Omega)$$

$$R_i = 1.608M\Omega$$

PROBLEM 17

For the DC analysis of the CE amplifier shown, neglect the base current and assume that the emitter and collector currents are equal. Given that $V_T = 25 \text{ mV}$, $V_{BE} = 0.7 \text{ V}$ and the BJT output resistance r_o is practically ∞ . Under these conditions the midband voltage gain magnitude $A = |V_o/V_i|$ is ...



SOLUTION

★ Given circuit is CE with bypass capacitor.

★ For the self bias circuit:

$$R_{th} = 47 \parallel 73 = 28.59k\Omega$$

$$V_{th} = \frac{12 \times 47}{47 + 73} = 4.7V$$

$$\text{Now, } I_C = \frac{V_{th} - V_{BE}}{R_E} = \frac{4.7 - 0.7}{2k\Omega} = 2\text{mA}$$

$$g_m = \frac{I_C}{V_T} = \frac{2\text{mA}}{25\text{mV}} \Rightarrow g_m = 80\text{mA/V}$$

$$\text{So, } \frac{V_o}{V_i} = -g_m R_L' = -80 \times 10^{-3} \times (2 \parallel 8)$$

$$\left| \frac{V_o}{V_i} \right| = 128$$

Low Frequency Analysis of Amplifiers

1. Why Low Frequency Analysis Matters

At low frequencies, the **large capacitors** in an amplifier offer appreciable or medium reactance, since $X_C = 1/(2\pi fC)$ grows as f falls. They can therefore no longer be treated as short circuits, and they load the signal path.

An RC-coupled CE amplifier contains three large capacitors:

- **Coupling capacitor** C_C — couples the output to the load
- **Blocking capacitor** C_B — blocks DC at the input
- **Bypass capacitor** C_E — bypasses the emitter resistor

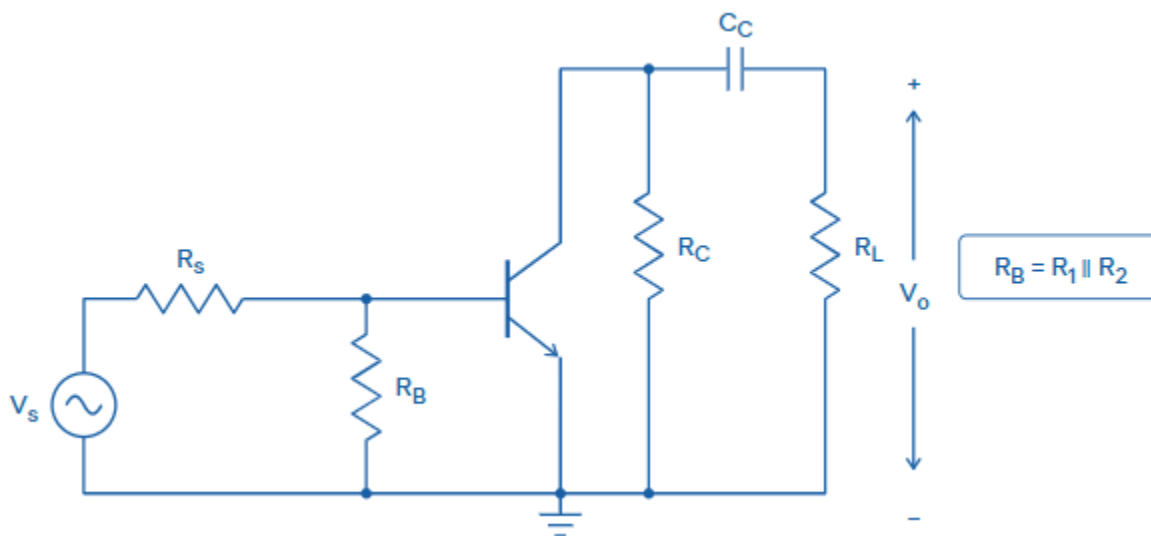
The effect of all three is included in low frequency analysis. The small internal junction capacitances are negligible here (they behave as open circuits at low f), which is why they only appear in *high* frequency analysis.

Method. To keep the algebra manageable, the effect of **one capacitor at a time** is considered. Each capacitor contributes its own lower cutoff frequency, and these are combined at the end.

2. Effect of the Coupling Capacitor

Assume C_E and C_B are short circuits, so only C_C is active.

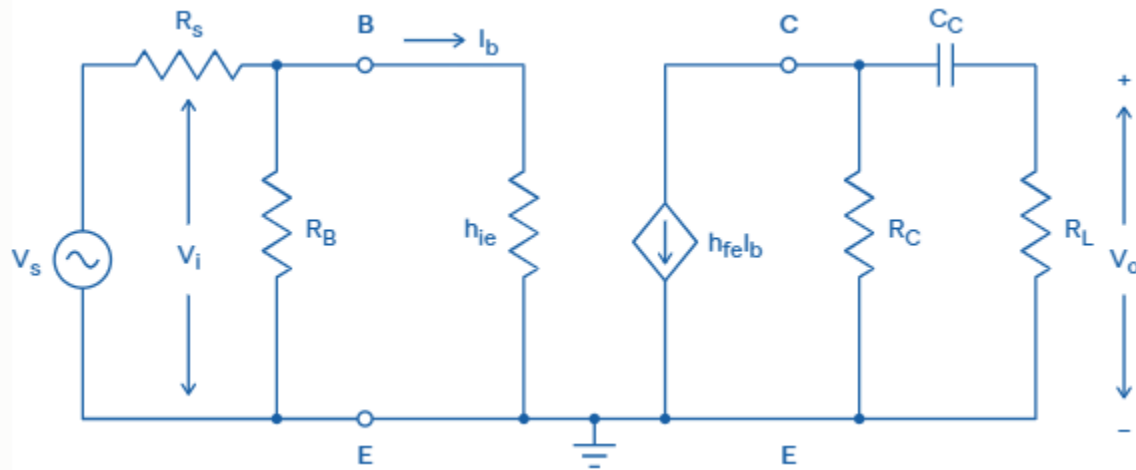
Low frequency AC equivalent circuit



where $R_B = R_1 \parallel R_2$ is the parallel combination of the bias divider resistors.

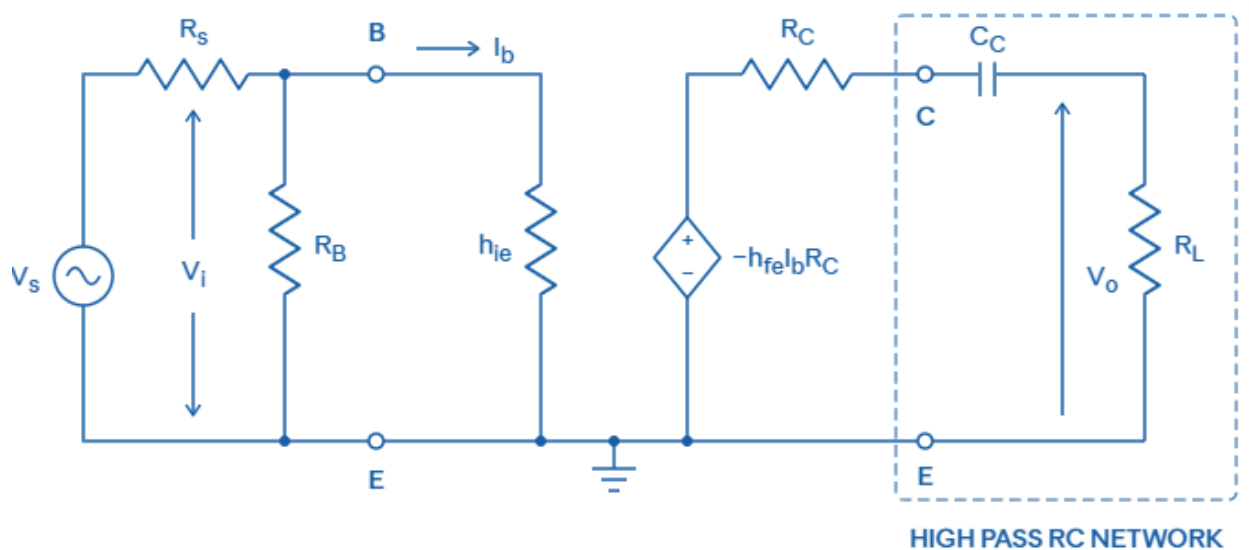
Replacing the BJT by its h-parameter model

Replace the transistor by the approximate h-parameter model, neglecting h_{oe} :



Thevenin form: a high-pass RC network

Converting the current source $h_{fe}I_b$ in parallel with R_C into a voltage source $-h_{fe}I_b R_C$ in series with R_C makes the structure obvious:

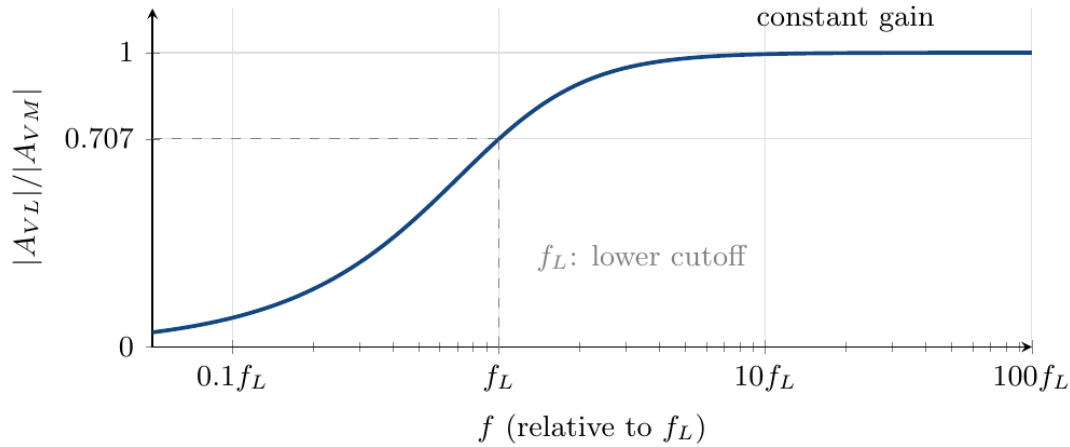


The coupling capacitor and R_L together form a **high-pass RC network**. As frequency falls:

$$f \downarrow \Rightarrow X_{C_C} \uparrow \Rightarrow V_{C_C} \uparrow \Rightarrow V_{R_L} \downarrow \Rightarrow \frac{V_o}{V_i} \downarrow$$

so the voltage gain of the RC-coupled amplifier **falls at low frequencies** because of the coupling capacitor.

Frequency response



3. Deriving the Low Frequency Voltage Gain

If the voltage gain is calculated *including* the effect of the coupling capacitor, it is called the **low frequency voltage gain**:

$$A_{VL} = \frac{V_o}{V_i} \quad (1)$$

Output circuit. By voltage division across R_L :

$$V_o = \frac{-h_{fe}I_b R_C \times R_L}{R_C + R_L + \frac{1}{j\omega C_C}}$$

Taking $(R_C + R_L)$ out of the denominator:

$$V_o = \frac{-h_{fe}I_b R_C R_L}{(R_C + R_L) \left\{ 1 + \frac{1}{j\omega C_C (R_C + R_L)} \right\}}$$

Writing $R_L' = \frac{R_C R_L}{R_C + R_L} = R_C \parallel R_L$:

$$V_o = \frac{-h_{fe}I_b R_L'}{1 + \frac{1}{j\omega C_C (R_C + R_L)}} \quad \Rightarrow \quad V_o = \frac{-h_{fe}I_b R_L'}{1 - \frac{j}{\omega C_C (R_C + R_L)}} \quad (2)$$

Input circuit. From the h-parameter model,

$$V_i = h_{ie}I_b \quad (3)$$

Dividing (2) by (3):

$$\frac{V_o}{V_i} = \frac{-h_{fe}I_b R_L'}{\left[1 - \frac{j}{\omega C_C(R_C + R_L)}\right] h_{ie}I_b} = \left(\frac{-h_{fe}R_L'}{h_{ie}}\right) \times \frac{1}{1 - \frac{j}{\omega} \cdot \frac{1}{C_C(R_C + R_L)}}$$

where the first bracket is the medium frequency gain A_{VM} and $\omega_L = \frac{1}{C_C(R_C + R_L)}$.

low frequency voltage gain:

$$A_{VL} = \frac{V_o}{V_i} = \frac{A_{VM}}{1 - j\frac{\omega_L}{\omega}} = \frac{A_{VM}}{1 - j\frac{f_L}{f}}$$

Magnitude and phase:

$$|A_{VL}| = \frac{|A_{VM}|}{\sqrt{1 + (f_L/f)^2}} \quad \angle A_{VL} = 180^\circ + \tan^{-1}\left(\frac{f_L}{f}\right)$$

- The 180° arises because A_{VM} is negative (phase inversion of the CE stage).
- The $\tan^{-1}(f_L/f)$ term arises from the complex denominator.

At $f = f_L$, $\tan^{-1}(1) = 45^\circ$, so the RC-coupled CE amplifier provides a total phase shift of 225° .

4. Cutoff Frequency

Setting $f = f_L$ in the magnitude expression:

$$|A_{VL}| = \frac{|A_{VM}|}{\sqrt{1 + 1}} = \frac{|A_{VM}|}{\sqrt{2}}$$

The frequency at which the voltage gain falls to $1/\sqrt{2}$ times the maximum gain is the **cutoff frequency** or **3 dB frequency**. For the low end this is the **lower cutoff frequency** f_L of the RC-coupled CE amplifier.

From $\omega_L = 1/[C_C(R_C + R_L)]$, the coupling capacitor contributes:

$$f_{L1} = \frac{1}{2\pi C_C(R_C + R_L)}$$

This is a special case of the general rule:

$$f_L = \frac{1}{2\pi RC}$$

where R is the **resistance seen by that capacitor** when the source V_s is disabled (shorted).

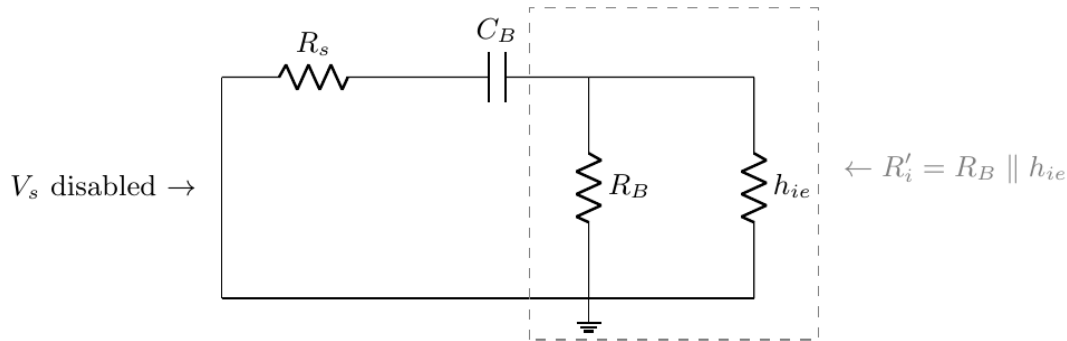
5. Contributions of the Other Two Capacitors

Bypass capacitor C_E

$$f_{L2} = \frac{1}{2\pi R C_E}, \quad R = \left(\frac{R_s' + h_{ie}}{1 + h_{fe}} \right) \parallel R_E$$

Here R is the resistance seen by the bypass capacitor, i.e. the resistance looking between emitter and ground. The $(1 + h_{fe})$ division is the usual impedance reflection from base to emitter.

Blocking capacitor C_B



With the source disabled, the blocking capacitor sees R_s in series with $R_i' = R_B \parallel h_{ie}$:

$$f_{L3} = \frac{1}{2\pi R C_B}, \quad R = R_s + R_i'$$

6. Combining the Three Cutoff Frequencies

Dominant-pole case. If one of the three frequencies is at least **4 times greater** than the other two, the net f_L is simply that highest frequency.

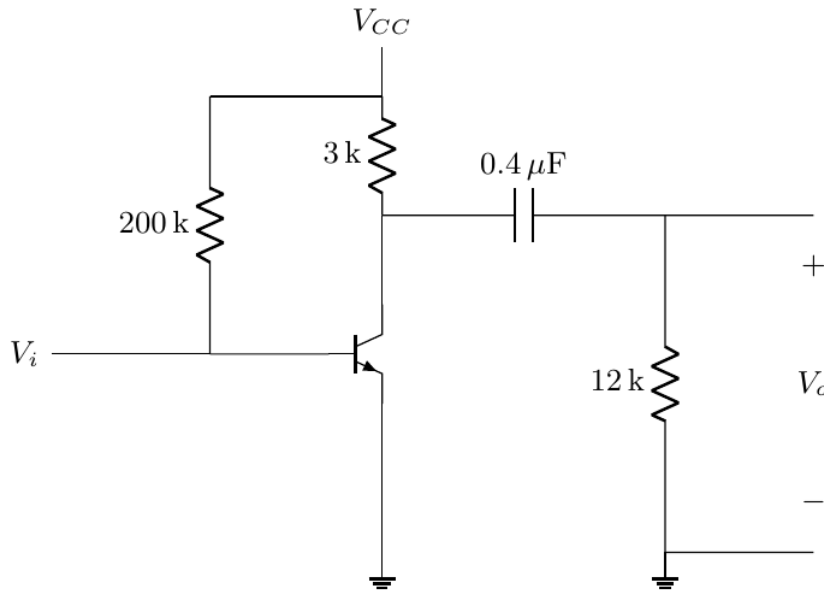
General case. If no frequency is 4 times greater than the others:

$$f_L = 1.1 \times \sqrt{f_{L1}^2 + f_{L2}^2 + f_{L3}^2}$$

7. Worked Problem

Q. For the amplifier shown, the BJT has $h_{fe} = 50$ and $h_{ie} = 1 \text{ k}\Omega$. Calculate:

- medium frequency voltage gain in dB
- lower 3 dB frequency
- voltage gain at $f = 10 \text{ Hz}$



(i) Medium frequency voltage gain

$$R_L' = R_C \parallel R_L = 3 \text{ k} \parallel 12 \text{ k} = \frac{3 \times 12}{15} \text{ k} = 2.4 \text{ k}\Omega$$

$$A_{VM} = \frac{V_o}{V_i} = \frac{-h_{fe} R_L'}{h_{ie}} = \frac{-50 \times 2.4 \text{ k}}{1 \text{ k}} = -120$$

In decibels:

$$(A_{VM})_{\text{dB}} = 20 \log |A_{VM}| = 20 \log(120) = 41.58 \text{ dB}$$

(ii) Lower 3 dB frequency

$$f_L = \frac{1}{2\pi C_C (R_C + R_L)} = \frac{1}{2\pi \times 0.4 \times 10^{-6} \times (15 \times 10^3)} = 26.526 \text{ Hz}$$

(iii) Voltage gain at $f = 10 \text{ Hz}$

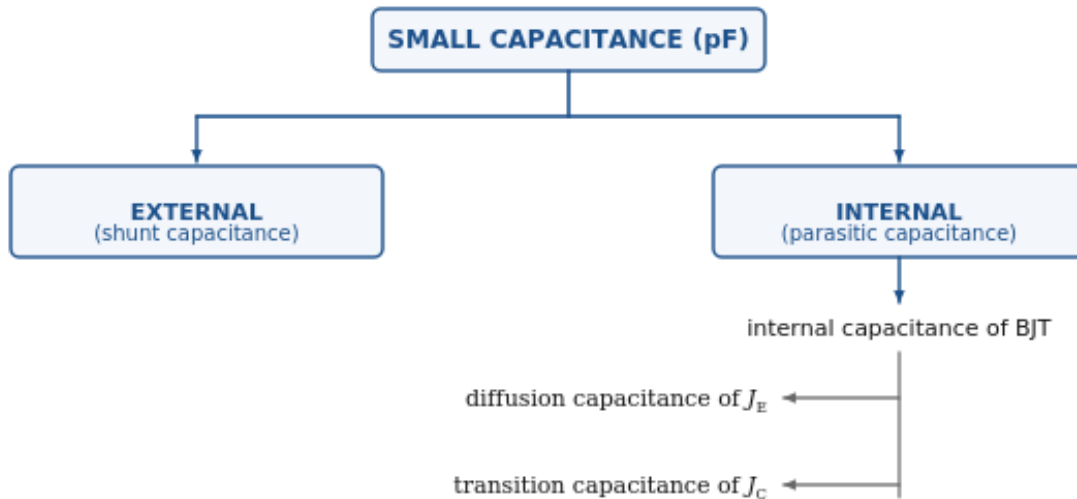
$f = 10 \text{ Hz}$ is a low frequency — if the signal frequency is less than f_L , or near f_L , it counts as low frequency. So the low frequency expression applies:

$$|A_{VL}| = \frac{|A_{VM}|}{\sqrt{1 + (f_L/f)^2}} = \frac{120}{\sqrt{1 + \left(\frac{26.526}{10}\right)^2}} = \frac{120}{\sqrt{1 + 7.036}} = \frac{120}{2.835} = 42.33$$

The gain has dropped from 120 to about 42 — confirming that in an RC-coupled CE amplifier the gain **decreases at low frequencies because of the large capacitors**.

HIGH FREQUENCY ANALYSIS OF AMPLIFIERS

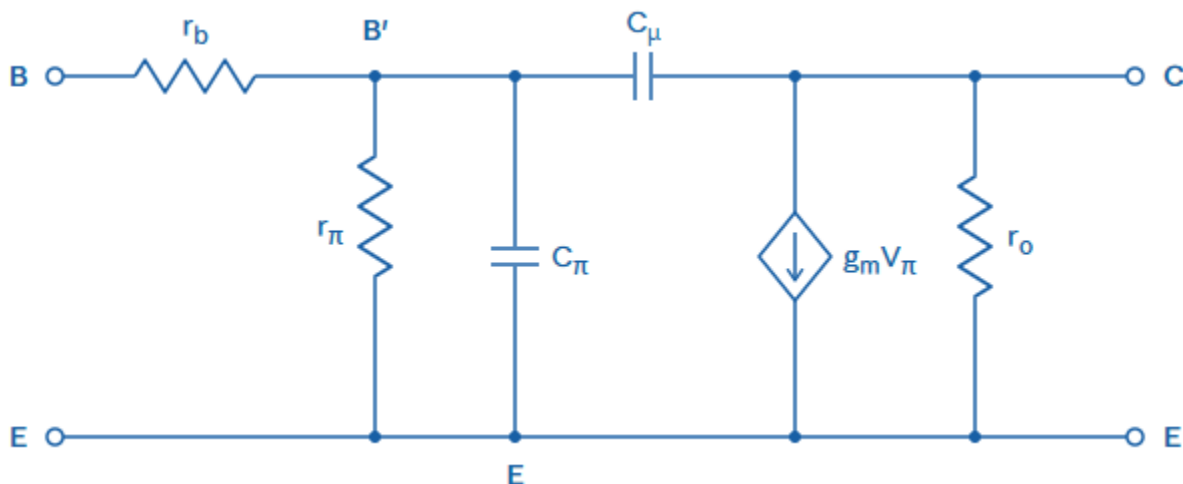
★ At high frequencies, small capacitors offer **medium reactance**. Hence the effect of small capacitors is considered in high frequency analysis.



NOTE

- Internal BJT capacitances are not included in the h-parameter model, and hence it is not used in HIGH FREQUENCY ANALYSIS.
- BJT amplifiers can be analysed at high frequencies by using the **high frequency π model** or the **high frequency T model**.

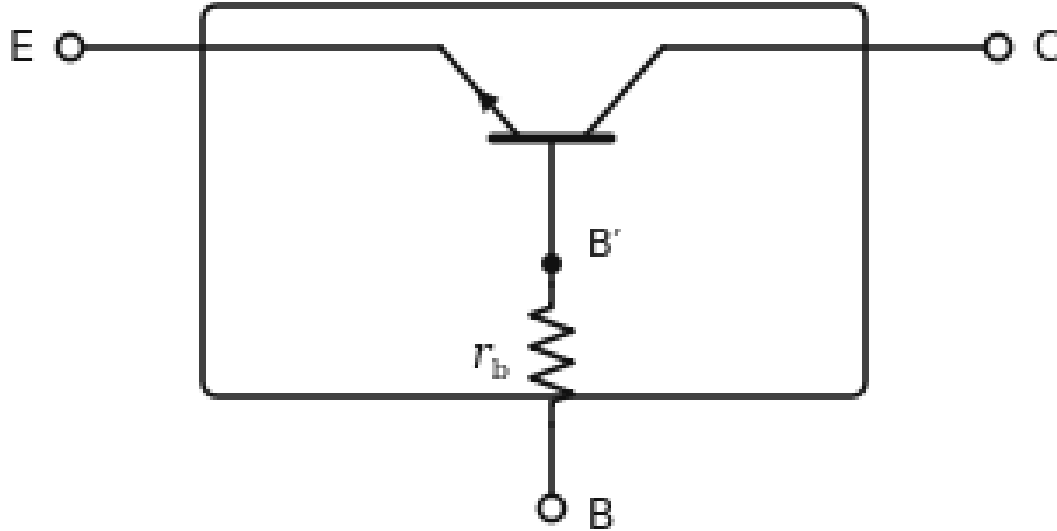
HIGH FREQUENCY π MODEL



Note

- B' : a node inside the base region.

r_b : base spreading resistance — it represents the internal resistance of the **base region**.
 $r_b = 100\Omega$ standard value (but generally ignored)



- $C_\pi (C_{b'e})$: diffusion capacitance of J_E
- $C_\mu (C_{b'c})$: transition capacitance of J_C

NOTE

If the BJT is used in CE configuration, then C_μ will appear between the input node **BASE** and the output node **COLLECTOR**.

Hence C_μ can be replaced with C_m and C_n by using **MILLER'S THEOREM**.

$$C_m = C_\mu (1 - A_V)$$

$$C_n = C_\mu \left(1 - \frac{1}{A_V}\right) \approx C_n = C_\mu$$

where $A_V = -g_m R_L'$ is high

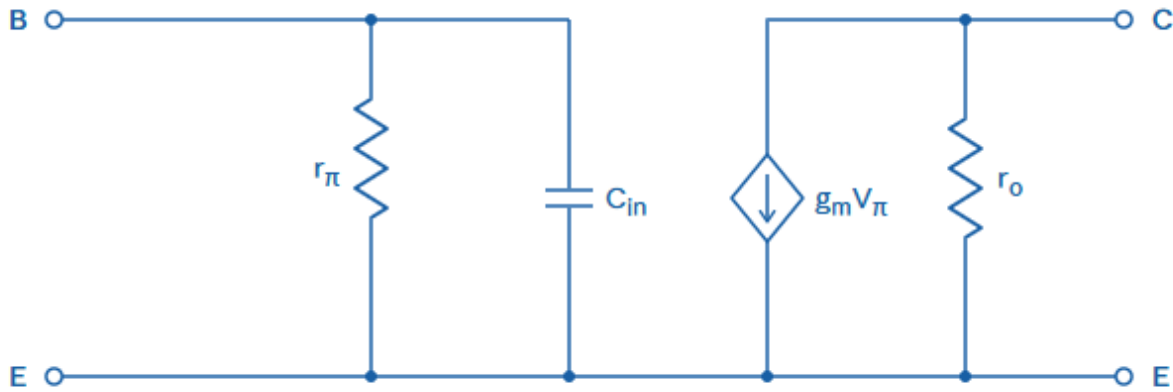
★ C_m appears in parallel to C_π . Hence the net capacitance between base and emitter, i.e. the net input capacitance of the CE configuration, will be:

$$C_{in} = C_\pi + C_m = C_\pi + C_\mu (1 - A_V)$$

★ C_n appears between collector and emitter. Hence the output capacitance becomes:

$$C_{out} = C_n \approx C_\mu$$

★ Since $C_{out} \ll C_{in}$, the effect of C_{out} is neglected. So, redrawing the circuit we get:



HIGH FREQUENCY π MODEL FOR CE AMPLIFIER

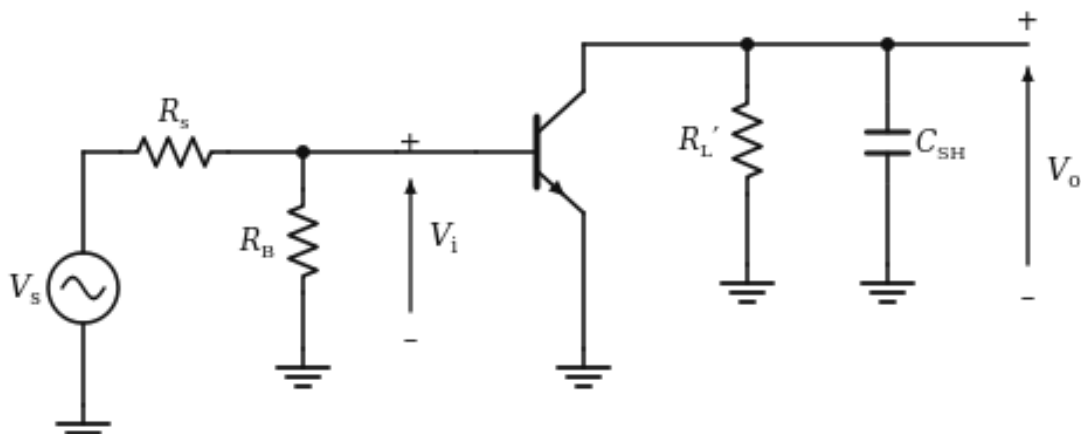
HIGH FREQUENCY VOLTAGE GAIN OF RC COUPLED CE AMPLIFIER

★ A_{VH} is calculated by considering the effect of the shunt capacitor C_{SH} and the input capacitance C_{in} .

★ To simplify the analysis, the effect of any one capacitor is considered at a time.

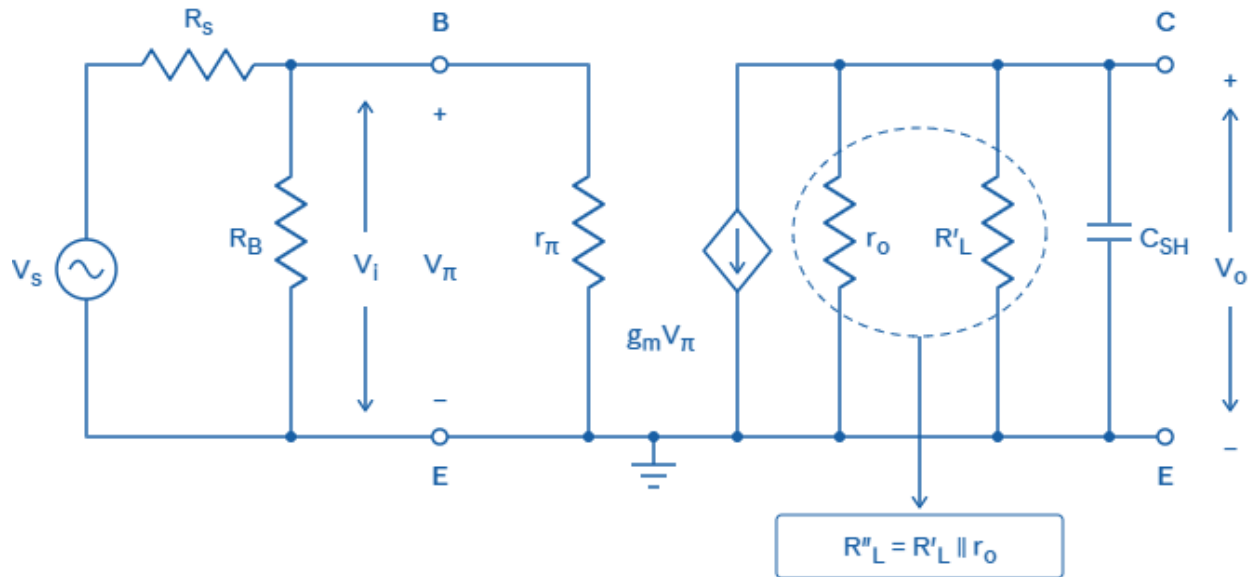
i) Effect of C_{SH}

★ Consider the effect of C_{SH} and ignore C_{in} .

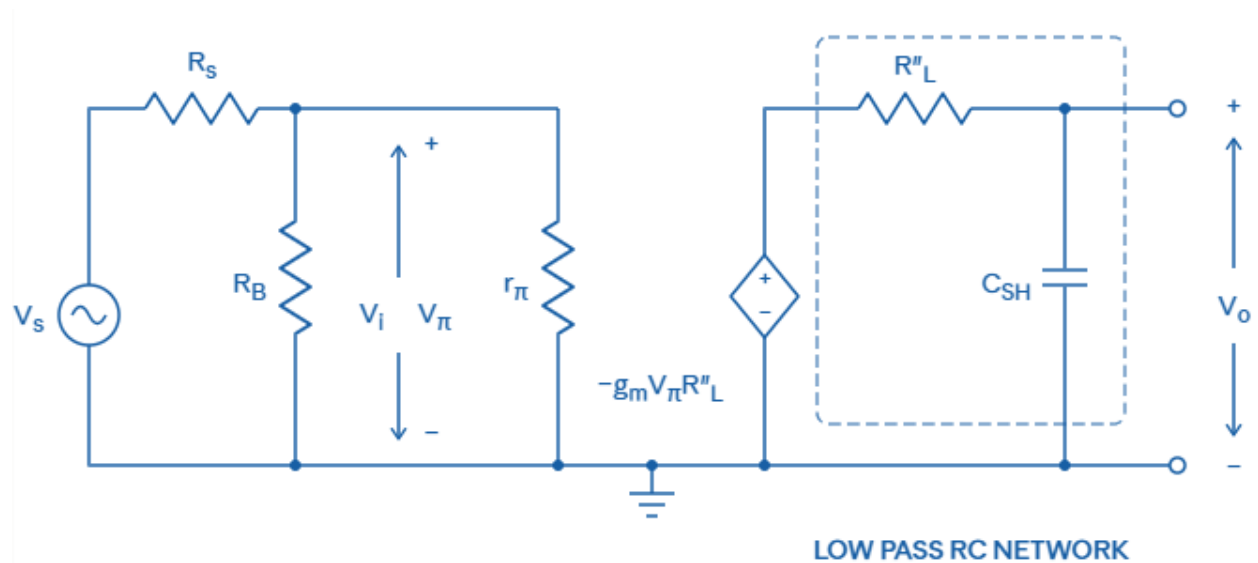


HIGH FREQUENCY AC EQUIVALENT

★ Replace the BJT with the π MODEL and ignore C_{in} :



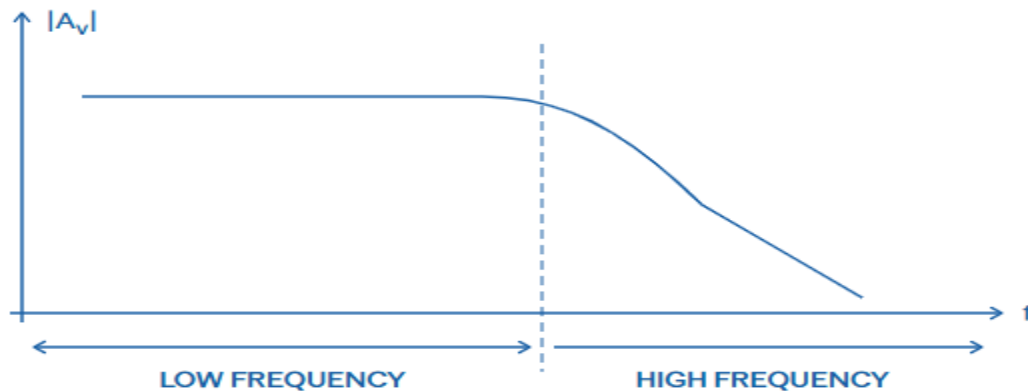
★ Replace the current source $g_m V_\pi$ with a voltage source:



★ C_{SH} and R_L'' form a **low pass RC circuit** at high frequencies.

★ If $f \uparrow \rightarrow$ reactance of the shunt capacitor $\downarrow \rightarrow$ voltage across C_{SH} (V_o) $\downarrow \rightarrow A_v = \frac{V_o}{V_i} \downarrow$.

★ Voltage gain of the RC coupled amplifier **decreases at high frequencies** due to small capacitors such as shunt capacitors and internal capacitances of the BJT.



Analysis

$$A_{VH} = \frac{V_o}{V_i} \quad (1)$$

★ By voltage division rule:

$$V_o = \frac{-g_m V_{\pi} R_L'' \times \frac{1}{j\omega C_{SH}}}{j\omega C_{SH}} \Rightarrow \frac{V_o}{V_{\pi}} = \frac{V_o}{V_i} = \frac{-g_m R_L'' \times \frac{1}{j\omega C_{SH}}}{j\omega C_{SH}} \frac{V_o}{V_i} = \frac{-g_m R_L''}{1 + j\omega R_L'' C_{SH}}$$

the numerator is A_{VM} , the mid-band gain

$$A_{VH} = \frac{A_{VM}}{1 + j\frac{\omega}{\omega_H}} = \frac{A_{VM}}{1 + j\frac{f}{f_H}}$$

$$|A_{VH}| = \frac{|A_{VM}|}{\sqrt{1 + \left(\frac{f}{f_H}\right)^2}}$$

voltage gain at high frequency

Also,

$$\angle A_{VH} = 180^\circ - \tan^{-1}\left(\frac{f}{f_H}\right)$$

NOTE

At $f = f_H$, the RC coupled CE amplifier provides a phase shift of 135° .

★ If $f = f_H$:

$$|A_{VH}| = \frac{|A_{VM}|}{\sqrt{2}}$$

★ f_H is called the **HIGHER CUTOFF** or **HIGHER 3 dB FREQUENCY** of the RC coupled CE amplifier. Mathematically:

$$f_{H1} = f_H = \frac{1}{2\pi R_L'' C_{SH}}$$

$$R_L'' = R_L' \parallel r_o ; \text{ mostly } r_o = \infty, \quad \text{so} \quad R_L'' = R_L'$$

NOTE

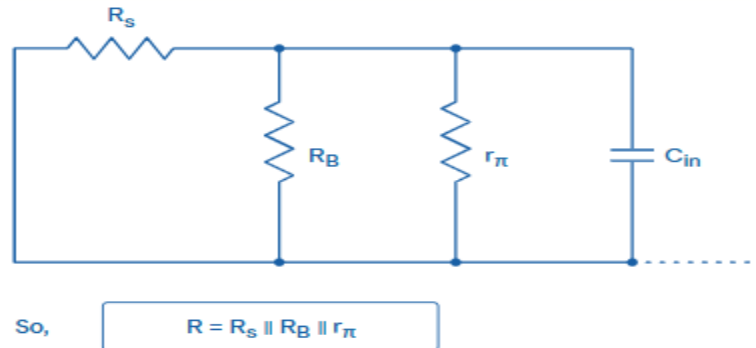
The higher cutoff frequency is $f_H = \frac{1}{2\pi RC}$ where R is the resistance seen by the capacitor when the source V_s is disabled.

ii) Effect of input capacitance

★ If the effect of C_{in} is considered, then:

$$f_H = \frac{1}{2\pi R C_{in}} \Rightarrow f_{H2}$$

R : resistance seen by C_{in}



NOTE

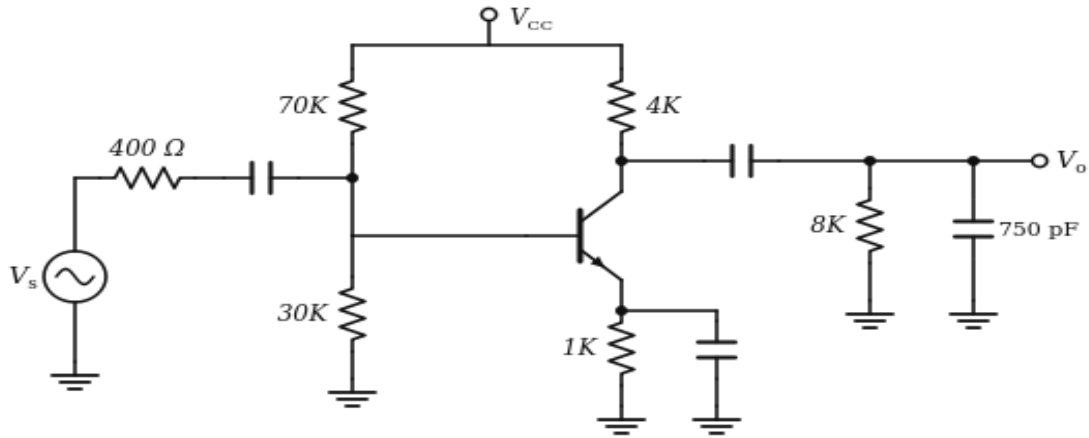
The net f_H is the smaller of the above two frequencies, if it is smaller by at least 4 times.

If no frequency is 4 times smaller, then the net f_H is calculated from:

$$\frac{1}{f_H} = 1.1 \times \sqrt{\frac{1}{f_{H1}^2} + \frac{1}{f_{H2}^2}}$$

PROBLEM 1

In the amplifier shown, the BJT has $g_m = 50 \text{ m}\Omega$, $r_\pi = 2 \text{ k}\Omega$, $C_\pi = 10 \text{ pF}$, $C_\mu = 1 \text{ pF}$ and $r_o = \infty$. Calculate: i) input and output capacitances of the BJT, ii) high cutoff frequency.



SOLUTION

i)

$$R_L' = 4k\Omega \parallel 8k\Omega = 2.67k\Omega \quad ; \quad A_V = -g_m R_L' = -50 \times 2.67k\Omega$$

$$A_V = -133.33$$

Now,

$$C_{in} = C_{\pi} + C_{\mu}(1 - A_V) = 10\text{pF} + 1 \times (1 + 133.33)$$

$$C_{in} = 144.33\text{pF}$$

$$C_{out} \approx C_{\mu} = 1\text{pF}$$

ii) If C_{SH} is considered, then:

$$f_{H1} = \frac{1}{2\pi R_L' C_{SH}} = \frac{1}{2\pi \times 2.67 \times 10^3 \times 750 \times 10^{-12}}$$

$$f_{H1} = 79.478\text{kHz}$$

If C_{in} is considered, then:

$$f_{H2} = \frac{1}{2\pi R C_{in}} \quad ; \quad R = R_s \parallel R_B \parallel r_{\pi} = 400\Omega \parallel 21000\Omega \parallel 2000\Omega$$

$$R = 0.328k\Omega$$

$$\text{So, } f_{H2} = \frac{1}{2\pi \times 328 \times 144.33 \times 10^{-12}}$$

$$f_{H2} = 3.36\text{MHz}$$

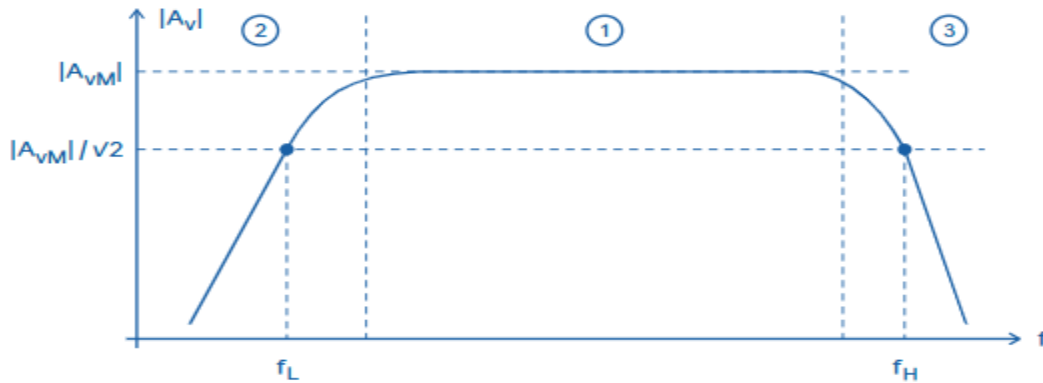
NOTE

Since $f_{H1} \leq \frac{f_{H2}}{4}$, hence:

$$\text{Net } f_H = f_{H1} = 79.5\text{kHz}$$

FREQUENCY RESPONSE OF RC COUPLED AMPLIFIER

★ Frequency response is a graph plotted between the magnitude of voltage gain and frequency.



NOTE

- **Region 1** is the medium or MID FREQUENCY REGION, in which the voltage gain is maximum and constant. A_{vM} : medium frequency or mid band voltage gain.
- **Region 2** is the low frequency region, where the voltage gain decreases due to large capacitors such as the coupling capacitor.
- **Region 3** is the high frequency region, where the voltage gain decreases due to small capacitors such as the shunt capacitor and the internal capacitance of the BJT.

★ The voltage gain becomes $\frac{A_{vM}}{\sqrt{2}}$ at two frequencies known as the lower cutoff f_L and the higher cutoff f_H .

★ $(f_H - f_L)$ is called the **3 dB BANDWIDTH**.

$$BW = (f_H - f_L)$$

★ For an RC coupled amplifier $f_H \gg f_L$, then:

$$BW \approx f_H = \frac{1}{2\pi R_L' C_{SH}} \text{ if } C_{SH} \text{ is present}$$

$$BW \propto \frac{1}{C_{SH}}$$

NOTE

- Bandwidth can be controlled or adjusted by changing the shunt capacitance.
- A shunt capacitor is used if a smaller BW, i.e. tens of kHz, is needed. For example, to amplify audio frequency signals a BW of 20 kHz is needed.

- If a large BW (MHz) is needed, then the shunt capacitor is avoided. In such circuits the bandwidth depends on the input capacitance:

$$f_H = \frac{1}{2\pi RC_{in}} \text{ (in MHz)} ; R = R_s \parallel R_B \parallel r_\pi$$

MILLER EFFECT

★ It occurs in **CE configuration only**.

★ It refers to the increase in input capacitance of the CE configuration due to the appearance of C_m in parallel to C_π .

$$C_{in} = C_m + C_\pi$$

C_m : Miller capacitance

$$C_m = C_\mu(1 - A_V) ; A_V = -g_m R_L'$$

$(1 - A_V)$: Miller multiplier

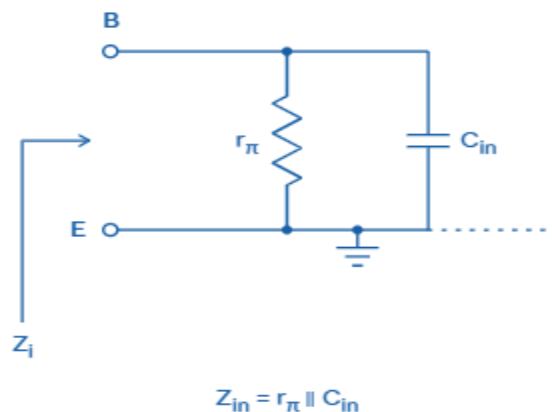
NOTE

The Miller effect is undesired. The increase in input capacitance has two undesired effects.

a) Bandwidth or higher 3 dB frequency of the CE amplifier decreases:

$$BW \propto \frac{1}{C_{in} \uparrow} \Rightarrow f_H \propto \frac{1}{C_{in} \uparrow}$$

b) Input impedance of the CE amplifier decreases at high frequencies:



★ If $C_{in} \uparrow \rightarrow X_{Cin} \downarrow \rightarrow Z_{in} \downarrow \rightarrow R_i' \downarrow \rightarrow \frac{V_o}{V_s} \downarrow$, since $\frac{V_o}{V_s} = \frac{A_V \times R_i'}{R_s + R_i'}$.

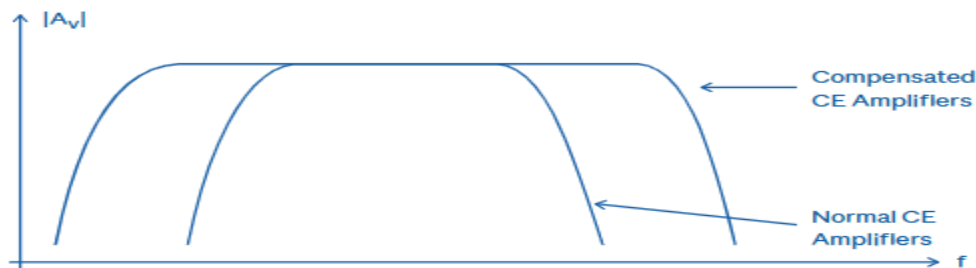
★ Hence the performance of the CE amplifier will degrade at high frequencies due to the **MILLER EFFECT**.

NOTE

Due to the Miller effect, the CE amplifier cannot have a very high BW; hence it cannot be used in VIDEO FREQUENCY AMPLIFICATION (video frequency signals have a BW of about 5 MHz).

★ Hence, to amplify video frequency signals the following amplifiers can be used:

- **CASCODE AMPLIFIER** (best)
- **COMMON BASE AMPLIFIER** (but it has smaller R_i and unity current gain)
- **COMPENSATED CE AMPLIFIER**, in which BW is increased by using frequency compensation techniques

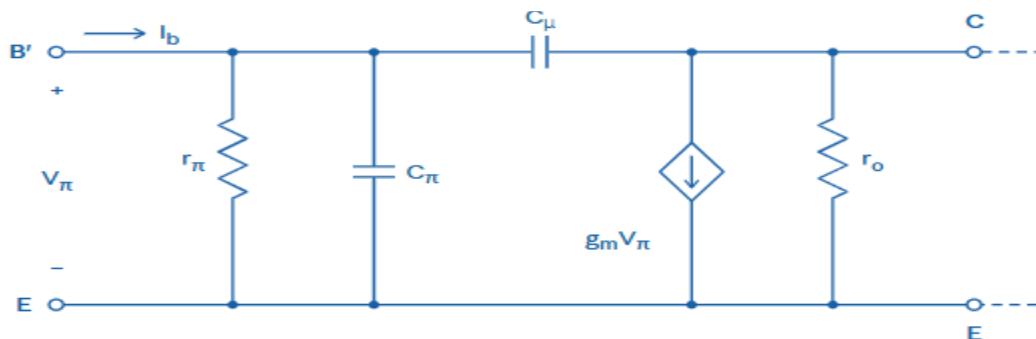


CE SHORT CIRCUIT CURRENT GAIN AT HIGH FREQUENCIES

★ If the current gain is calculated by taking $R_L = 0$, then it is called **SHORT CIRCUIT CURRENT GAIN**. It represents the maximum current gain, because for $R_L = 0$ maximum current flows.

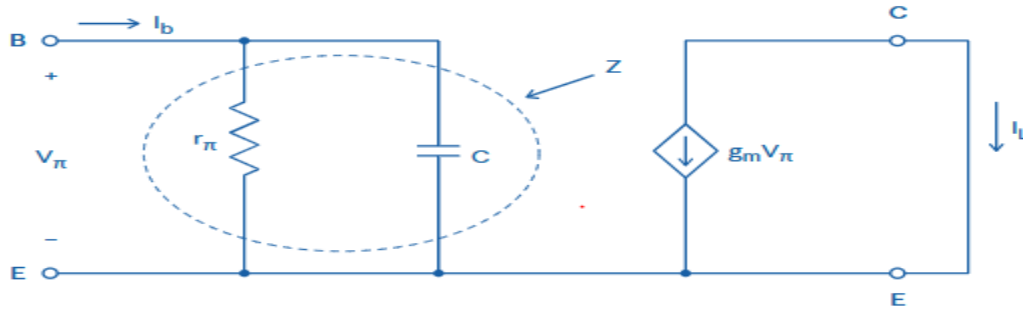
$$A_{ie} = \frac{I_L}{I_b} |_{R_L=0} \dots (I)$$

★ A_{ie} is calculated from the high frequency π model by shorting collector and emitter:



★ When the collector is shorted to the emitter:

- r_o becomes dead;
- C_μ appears in parallel to C_π and the net capacitance becomes $C = C_\pi + C_\mu$.



$$V_{\pi} = I_b \times Z \Rightarrow V_{\pi} = I_b \frac{r_{\pi} \times \frac{1}{j\omega C}}{(r_{\pi} + \frac{1}{j\omega C})} \Rightarrow V_{\pi} = I_b \frac{r_{\pi}}{1 + j\omega r_{\pi} C} \dots (2)$$

★ From the output circuit:

$$I_L = -g_m V_{\pi} = -g_m I_b \times \frac{r_{\pi}}{1 + j\omega r_{\pi} C} \Rightarrow \frac{I_L}{I_b} = \frac{-g_m r_{\pi}}{1 + j\frac{\omega}{\omega_{\beta}}} \left\{ r_{\pi} = \frac{\beta}{g_m} \right\} \Rightarrow \frac{I_L}{I_b} = \frac{-\beta}{1 + j\frac{\omega}{\omega_{\beta}}} = \frac{-\beta}{1 + j\frac{f}{f_{\beta}}} = A_{ie}$$

$$\Rightarrow |A_{ie}| = \frac{\beta}{\sqrt{1 + \left(\frac{f}{f_{\beta}}\right)^2}}$$

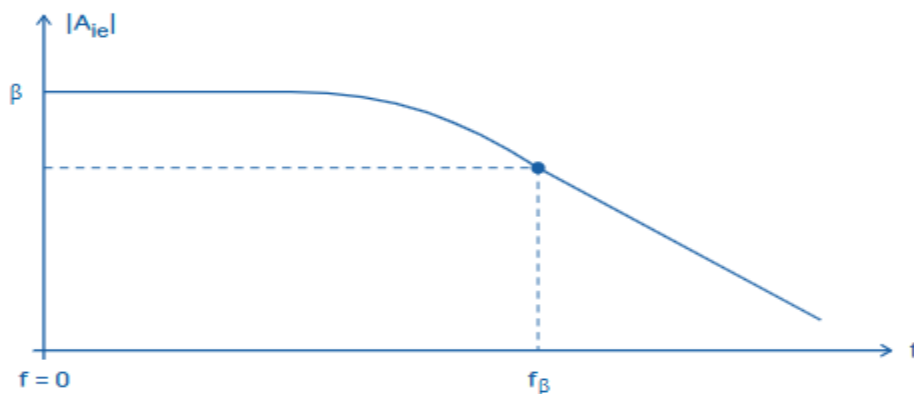
CE short circuit current gain at high frequency

NOTE

If $f = 0$, then $|A_{ie}| = \beta$.

If $f = \infty$, then $|A_{ie}| = 0$.

CE has current gain β at low and medium frequencies, but the current gain decreases at high frequencies due to the internal capacitances C_{π} and C_{μ} .



★ At $f = f_\beta \Rightarrow |A_{ie}| = \frac{\beta}{\sqrt{2}}$. ; f_β is the **β CUTOFF FREQUENCY** — the high frequency at which the CE short circuit current gain becomes $\frac{\beta}{\sqrt{2}}$.

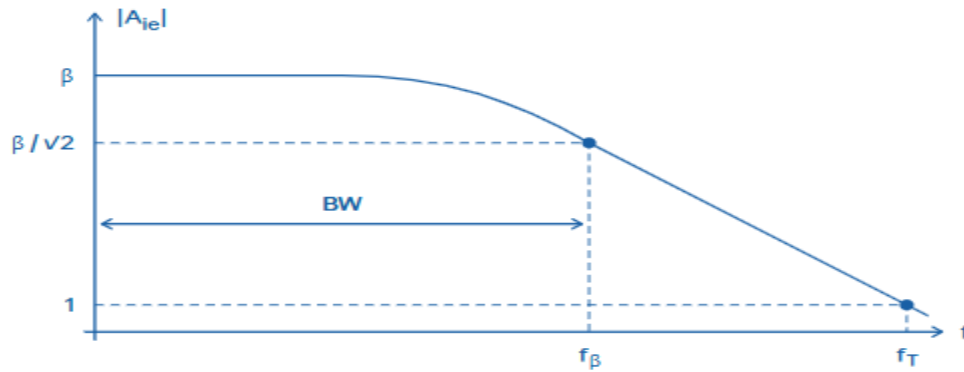
Also,

$$\omega_\beta = \frac{1}{r_\pi C} \quad ; \quad f_\beta = \frac{1}{2\pi r_\pi C}$$

$$f_\beta = \frac{1}{2\pi r_\pi (C_\pi + C_\mu)}$$

UNITY GAIN FREQUENCY (f_T)

★ It is the high frequency at which the CE short circuit current gain becomes unity, i.e. at $f = f_T$, $|A_{ie}| = 1$.



As

$$|A_{ie}| = \frac{\beta}{\sqrt{1 + \left(\frac{f}{f_\beta}\right)^2}} \quad \Rightarrow \quad 1 = \frac{\beta}{\sqrt{1 + \left(\frac{f_T}{f_\beta}\right)^2}} \quad \{ \text{at } f = f_T, |A_{ie}| = 1 \}$$

$$f_T \gg f_\beta \quad \text{or} \quad \frac{f_T}{f_\beta} \gg 1 \quad \text{So, } 1 = \frac{\beta}{\frac{f_T}{f_\beta}} \Rightarrow \frac{f_T}{f_\beta} = \beta$$

$$f_T = \beta f_\beta$$

★ As $f_T = \beta \times f_\beta = (\text{maximum current gain}) \times (\text{bandwidth})$, f_T is also called the **GAIN BANDWIDTH PRODUCT** of the BJT. Mathematically:

$$f_T = \beta \times \frac{1}{2\pi r_\pi (C_\pi + C_\mu)}$$

$$f_T = \frac{g_m}{2\pi (C_\pi + C_\mu)}$$

$$\left\{ r_\pi = \frac{\beta}{g_m} \Rightarrow g_m = \frac{\beta}{r_\pi} \right\}$$

NOTE

- If the BJT is used at a frequency greater than f_T ($f > f_T$), then the gain becomes less than 1 ($|A_{ie}| < 1$), hence amplification does not occur. So f_T represents the highest frequency up to which the BJT can provide amplification.
- If the BJT is to provide better amplification even at high frequencies, then f_T should be large. Hence the internal capacitances C_π and C_μ should be smaller.

★ Also for MOSFET and JFET:

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})}$$

NOTE

- As the BJT has greater transconductance (g_m), the gain bandwidth product f_T is greater for the BJT than for the MOSFET and JFET. Hence the BJT is preferred over the MOSFET and JFET in high frequency amplification.

PROBLEM 2

A BJT is operating at a collector current I_C of 5 mA. It has $\beta = 100$, $|A_{ie}| = 10$ at $f = 10$ MHz and $C_\mu = 5$ pF. Calculate f_β , f_T and C_π .

SOLUTION

Given: $\beta = 100$, $I_C = 5$ mA, $|A_{ie}| = 10$ at $f = 10$ MHz.

$$\text{Now, } g_m = \frac{I_C}{V_T} = \frac{5}{26}$$

$$g_m = 0.1922127$$

$$|A_{ie}| = \frac{\beta}{\sqrt{1 + \left(\frac{f}{f_\beta}\right)^2}} \Rightarrow 10 = \frac{100}{\sqrt{1 + \left(\frac{10}{f_\beta}\right)^2}} \Rightarrow 1 + \left(\frac{10}{f_\beta}\right)^2 = 100 \Rightarrow \left(\frac{10}{f_\beta}\right)^2 = 99 \Rightarrow f_\beta = \frac{10}{\sqrt{99}}$$

$$f_\beta \approx 1.005 \text{ MHz}$$

$$\text{So, } f_T = \beta f_\beta \approx 100 \text{ MHz}$$

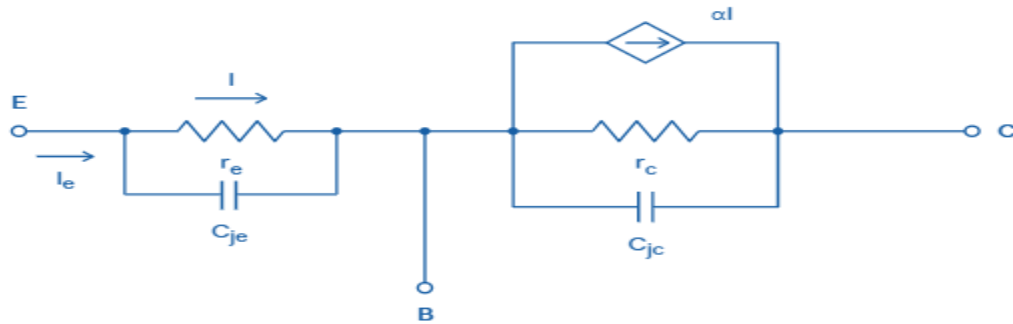
$$f_T \approx 100 \text{ MHz}$$

Also,

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)} \Rightarrow (C_\pi + C_\mu) = \frac{g_m}{2\pi f_T} = \frac{0.192}{2\pi \times 100 \times 10^6} = 305.578 \text{ pF} \Rightarrow C_\pi = 305.578 \text{ pF} - 5 \text{ pF}$$

$$\Rightarrow C_{\pi} = 300.578 \text{pF}$$

HIGH FREQUENCY T MODEL



★ C_{je} : diffusion capacitance of J_E

$$C_{je} = \frac{\tau_t}{r_e}$$

τ_t : base transit time — the average time taken by injected carriers to travel through the base

Mathematically:

$$\tau_t = \frac{W_B^2}{2D_B}$$

- W_B : base width
- D_B : diffusion constant — D_n for NPN, D_p for PNP
- r_e : dynamic resistance of J_E (common base input resistance)

$$r_e = \frac{V_T}{I_E}$$

As $\frac{\tau_t}{r_e} = C_{je}$, so:

$$C_{je} = \frac{\tau_t}{\frac{V_T}{I_E}}$$

$$C_{je} = \frac{\tau_t \cdot I_E}{V_T}$$

NOTE

- The high frequency response of the BJT is limited by the internal capacitances of the BJT. So, if C_{je} increases, the high frequency response of the BJT is also affected. So the BJT is operated at smaller currents at high frequencies.

- When the BJT is used in high frequency circuits, it is operated at smaller I_C or I_E so that the diffusion capacitance C_{je} remains smaller.

$$C_{je} \propto I_E \text{ or } I_C$$

★ C_{jc} : transition capacitance of the collector junction. Mathematically:

$$C_{jc} = \frac{\epsilon A}{W}$$

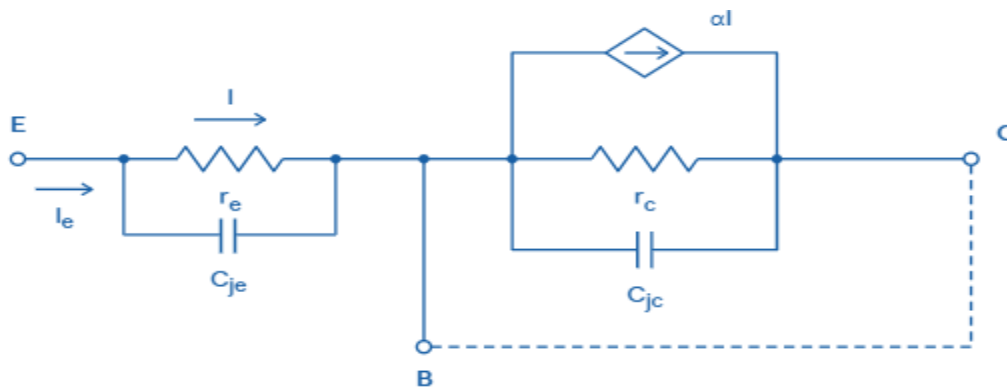
- ϵ : permittivity of the semiconductor
- A : area
- W : depletion width across J_C

CB short circuit current gain

★ For CB configuration, the short circuit current gain is defined as:

$$A_{ib} = \frac{I_L}{I_e} \bigg|_{R_L=0}$$

★ A_{ib} is calculated from the high frequency T MODEL by shorting collector and base:



Mathematically:

$$|A_{ib}| = \frac{\alpha}{\sqrt{1 + \left(\frac{f}{f_\alpha}\right)^2}}$$

★ CB configuration has current gain α at low and medium frequencies, but the current gain decreases at high frequencies due to internal capacitances.

★ If $f = f_\alpha \Rightarrow |A_{ib}| = \frac{\alpha}{\sqrt{2}}$, where f_α is the **α cutoff frequency** of the BJT — the high frequency at which the CB current gain becomes $\frac{\alpha}{\sqrt{2}}$.

$$f_{\alpha} = \frac{1}{2\pi r_e C_{je}}$$

NOTE

For better amplification at high frequencies, the BJT should have large values of f_{α} , f_{β} and f_T .

Also,

$$f_{\alpha} = \frac{1}{2\pi r_e C_{je}} = \frac{1}{2\pi r_e \times \frac{\tau_t}{r_e}}$$

$$f_{\alpha} = \frac{1}{2\pi \tau_t} \Rightarrow f_{\alpha} \propto \frac{1}{\tau_t}$$

the α cutoff has an inverse relation with the transit time

★ To achieve greater f_{α} , a BJT should have a smaller transit time.

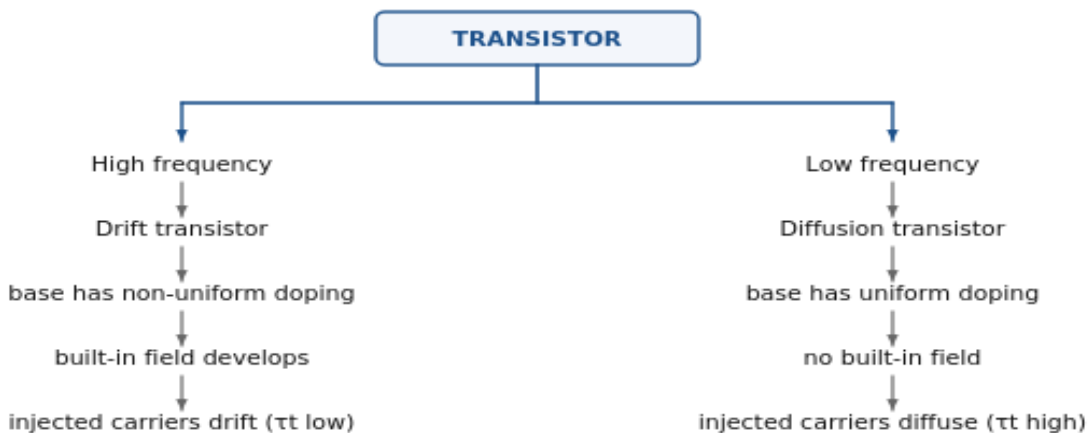
NOTE

- In high frequency transistors the transit time is reduced:

i) by decreasing the base width ($\tau_t \propto W_B^2$);

ii) by doping the base region **non-uniformly**, i.e. the base has greater doping towards the emitter and less doping towards the collector. Due to this a **built-in electric field** develops inside, which helps in rapid drift movement of the carriers injected by the emitter. Due to this the transit time decreases and such a BJT is called a **DRIFT TRANSISTOR**.

- In a normal BJT or low frequency BJT, the base is uniformly doped and hence a built-in field cannot develop. Hence injected carriers diffuse through the base and such a BJT is called a **DIFFUSION TRANSISTOR**.



Analysis

$$f_{\alpha} = \frac{1}{2\pi\tau_t} = \frac{1}{2\pi \times \frac{W_B^2}{2D_B}}$$

$$f_{\alpha} \uparrow = \frac{D_B}{\pi W_B^2} \downarrow$$

- $f_{\alpha} \uparrow$ when $W_B \downarrow$
- $f_{\alpha} \propto$ diffusion constant (D_B), where $D_B = D_n$ for NPN and D_p for PNP

NOTE

- As $D_n > D_p$, an NPN transistor has greater f_{α} than a PNP transistor of the same size and doping. Hence NPN is preferred over PNP in high frequency amplification.
- Ge has a greater diffusion constant value, hence a Ge transistor has greater f_{α} and f_T than a Si transistor. Hence Ge is preferred over Si in discrete high frequency amplifiers.
- In high frequency IC amplifiers, a GaAs NPN transistor is preferred because the electron mobility is the highest in GaAs ($\mu_n = 8500 \text{ cm}^2/\text{V}\cdot\text{s}$).

★ The gain bandwidth product of the BJT is given as:

$$f_T = \beta f_{\beta} = \alpha f_{\alpha}$$

$$\beta f_{\beta} = \alpha f_{\alpha} \Rightarrow \beta f_{\beta} = \frac{\beta}{(1 + \beta)} f_{\alpha}$$

$$f_{\beta} = \frac{f_{\alpha}}{(1 + \beta)} \Rightarrow f_{\alpha} = (1 + \beta) f_{\beta}$$

PROBLEM 3

A Ge PNP transistor has base width $W_B = 5 \mu\text{m}$ and is operating at an emitter current of 2 mA. If $D_p = 47 \text{ cm}^2/\text{sec}$, calculate: i) resistance of the emitter junction, ii) base transit time, iii) capacitance of J_E , iv) α cutoff frequency.

SOLUTION

Given: $W_B = 5 \mu\text{m} = 5 \times 10^{-4} \text{ cm}$, $I_E = 2 \text{ mA}$, $D_p = 47 \text{ cm}^2/\text{sec}$.

$$i) r_e = \frac{V_T}{I_E} = \frac{26\text{mV}}{2\text{mA}}$$

$$r_e = 13\Omega$$

$$\text{ii) } \tau_t = \frac{W_B^2}{2D_B} = \frac{(5 \times 10^{-4})^2}{2 \times 47}$$

$$\tau_t = 2.65 \text{ nsec}$$

$$\text{iii) } C_{je} = \frac{\tau_t}{r_e} = \frac{2.65 \times 10^{-9}}{13}$$

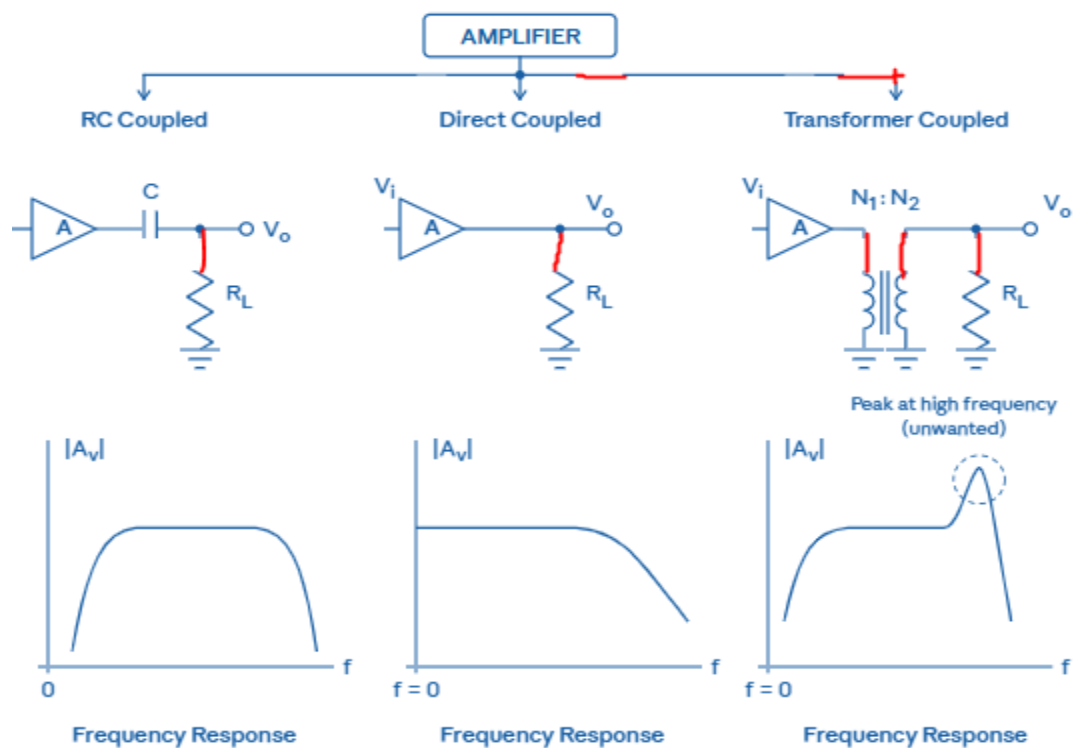
$$C_{je} = 203.8 \text{ pF}$$

$$\text{iv) } f_\alpha = \frac{1}{2\pi\tau_t} = \frac{1}{2\pi \times 2.65 \times 10^{-9}}$$

$$f_\alpha \approx 60 \text{ MHz}$$

TYPES OF AMPLIFIER COUPLING

★ According to the interconnection of amplifier and load, amplifiers are divided into three types: RC coupled amplifier, direct coupled amplifier and transformer coupled amplifier.



NOTE

For a direct coupled amplifier the gain is constant at low frequencies because it does not have any large capacitor, but the gain decreases at high frequencies due to the internal capacitances of the BJT.

Direct coupled amplifier

Advantages

- It is ideal for fabrication in integrated circuits, because it does not have any large capacitor (all IC amplifiers are direct coupled amplifiers). Large capacitors take a large area, hence direct coupling is used in ICs.
- It can amplify DC signals also, as it has no capacitor.

Disadvantages

- Maximum power cannot be transferred to the load due to impedance mismatch, i.e. $R_L \neq R_o$ (for amplifiers, the condition for maximum power transfer is $R_L = R_o$).
- No DC isolation between amplifier and load, i.e. the load can disturb the operating point.

Application

★ The direct coupled amplifier is used as a DC amplifier and also as a very low frequency amplifier, i.e. to amplify signals of 1 Hz, 2 Hz etc. Direct coupling is used in ICs also.

Transformer coupled amplifier

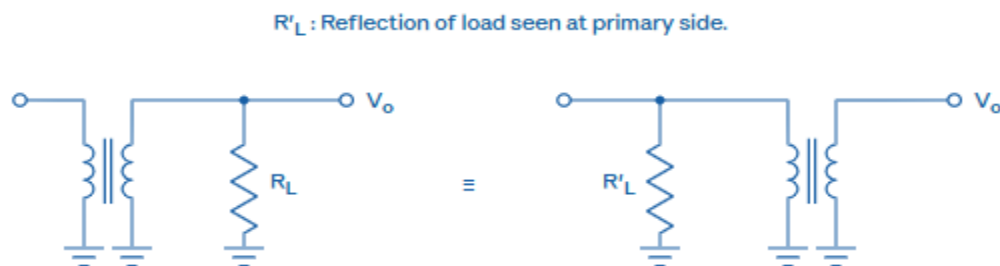
NOTE

Gain decreases at low frequencies due to the inductance of the transformer windings.

A peak appears in the frequency response at high frequencies due to leakage inductance and capacitances of the transformer, which result in PARALLEL RESONANCE.

Advantages

★ Maximum power can be transferred to the load due to impedance matching property of the transformer. For maximum power transfer, the reflected load should be the same as the output resistance of the amplifier, i.e. $R_L' = R_o$, where R_L' is the reflection of the load seen at the primary side.



$$R_L' = n^2 R_L ; n = \frac{N_1}{N_2}$$

$$\frac{V_1}{V_2} \times \frac{I_2}{I_1} = n^2$$

$$\Rightarrow \frac{\frac{V_1}{I_1}}{\frac{V_2}{I_2}} = n^2$$

$$\Rightarrow \frac{R_L'}{R_L} = n^2$$

For maximum power transfer:

$$R_L' = R_o \Rightarrow n^2 R_L = R_o$$

$$n = \frac{N_1}{N_2} = \sqrt{\frac{R_o}{R_L}}$$

turns ratio to achieve maximum power transfer

- Perfect DC isolation between amplifier and load (the transformer couples only the AC component, not DC).
- Greater efficiency (to be learned in power amplifiers).
- Greater voltage gain than the RC coupled amplifier if the turns ratio ($n > 2$).

$$A_V \text{ of transformer coupled amplifier} = \frac{n}{2} \times A_V \text{ of RC coupled amplifier} \Rightarrow \frac{n}{2} \times \frac{h_{fe} R_L}{h_{ie}}$$

Disadvantages

- Improper frequency response, due to the appearance of a peak at high frequencies.
- The transformer is bigger in size and costlier.

Application

★ Transformer coupling is used in **POWER AMPLIFIERS**.

RC coupled amplifier

Advantages

- Better frequency response, i.e. it can amplify the desired range of frequencies.
- DC isolation between amplifier and load.
- Less distortion in the output.

Disadvantages

- It cannot be fabricated in ICs because large capacitors (μF) require a large chip area.
- Maximum power cannot be transferred to the load due to impedance mismatch, i.e. $R_L \neq R_o$.

Application

★ The RC coupled amplifier is used as an audio and video frequency amplifier: RC CE amplifier → audio frequency amplifier; RC cascode amplifier → video frequency amplifier.

TYPES OF β OF A BJT

★ A BJT has three types of β values: DC β , large signal β and AC β .

i) DC β

★ It is defined as the ratio of DC collector current and DC base current.

$$\beta_{DC} = \frac{I_C}{I_B}$$

ii) Large signal β

★ It is defined as the ratio of change in collector current to change in base current when the BJT operation changes from **cutoff region to active region**.

$$\beta = \frac{\Delta I_C}{\Delta I_B} \Big|_{\text{cutoff to active}}$$

$$I_C = \beta I_B + (1 + \beta)I_{CO} \Rightarrow I_C - I_{CO} = \beta(I_B + I_{CO})$$

$$\Rightarrow \beta = \frac{I_C - I_{CO}}{I_B - (-I_{CO})}$$

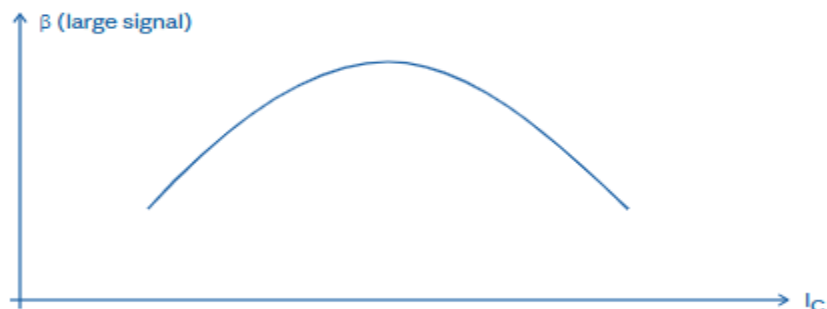
★ Since I_{CO} is a negligible value, hence:

$$\beta \approx \frac{I_C}{I_B} = \beta_{DC}$$

★ Large signal β is equal to DC β when I_{CO} is neglected as it is small.

NOTE

Large signal β varies with collector current. It initially increases with I_C , becomes maximum and then decreases with I_C .



iii) AC β

★ Also called **SMALL SIGNAL β** . Mathematically:

$$\beta_{AC} = \frac{\partial I_C}{\partial I_B} = \frac{\Delta I_C}{\Delta I_B} \big|_{\Delta I_B \rightarrow 0}$$

As $I_C = \beta I_B + (1 + \beta)I_{CO}$, differentiating with respect to I_C we get:

$$1 = \beta \frac{\partial I_B}{\partial I_C} + I_B \frac{\partial \beta}{\partial I_C} + I_{CO} \frac{\partial \beta}{\partial I_C} \quad \Rightarrow \quad 1 = \beta \frac{\partial I_B}{\partial I_C} + \frac{\partial \beta}{\partial I_C} (I_B + I_{CO}) \Rightarrow 1 - (I_B + I_{CO}) \frac{\partial \beta}{\partial I_C} = \beta \frac{\partial I_B}{\partial I_C}$$

$$\frac{\partial I_C}{\partial I_B} = \frac{\beta}{1 - (I_B + I_{CO}) \frac{\partial \beta}{\partial I_C}}$$

So,

$$\beta_{AC} = \frac{\beta}{1 - (I_B + I_{CO}) \frac{\partial \beta}{\partial I_C}}$$

$\frac{\partial \beta}{\partial I_C}$: slope of the β vs I_C graph

Cases

- If $\frac{\partial \beta}{\partial I_C} = 0 \Rightarrow \beta_{AC} = \beta$
- If $\frac{\partial \beta}{\partial I_C} = +ve \Rightarrow \beta_{AC} > \beta$
- If $\frac{\partial \beta}{\partial I_C} = -ve \Rightarrow \beta_{AC} < \beta$

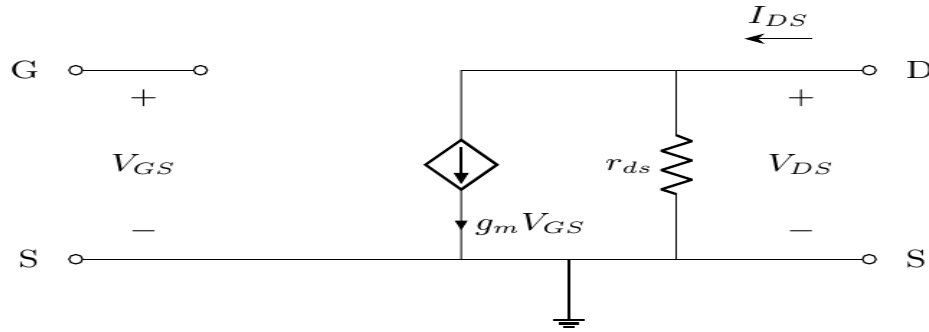
NOTE

- In numerical, all β values are considered to be the same if they are not provided separately.
- If DC β and AC β are given in the question, then take: (i) β_{DC} for the analysis of the operating point (DC analysis); (ii) β_{AC} for the analysis of voltage gain, R_i , R_o and A_i (AC analysis).
- But most of the time $\beta_{DC} = \beta_{AC} = \beta$, and they are not given separately.

1. & 2. Small Signal Model of MOSFET and JFET

The small signal model is the **equivalent circuit of a MOSFET or JFET**, used in the analysis of small signal FET amplifiers. It is obtained from the small signal model equation:

$$I_{ds} = g_m V_{GS} + \frac{V_{ds}}{r_{ds}}$$



- The **open circuit between gate and source** indicates that zero current flows from gate to source, i.e. $I_G = 0$.
- g_m and r_{ds} are the small signal parameters of the FET.

3. Transconductance g_m

Transconductance is the rate of change of drain current with respect to V_{GS} :

$$g_m = \frac{\partial I_{DS}}{\partial V_{GS}} \quad (1)$$

It is also defined as the **slope of the transfer characteristics**.

For a JFET

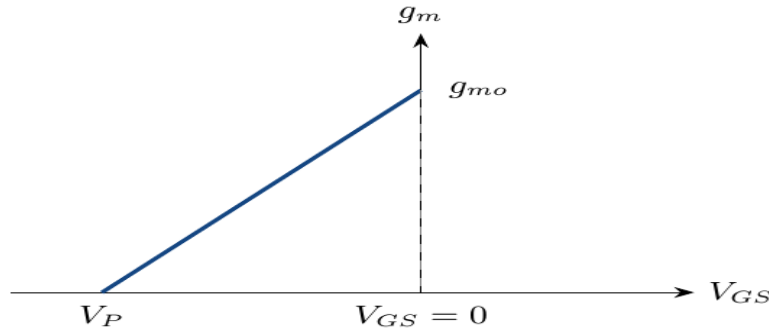
$$I_{DS} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$\frac{\partial I_{DS}}{\partial V_{GS}} = I_{DSS} \times 2 \left(1 - \frac{V_{GS}}{V_P}\right) \times \left(-\frac{1}{V_P}\right) = \frac{-2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right) \quad (2)$$

Taking the magnitude, and writing $g_{mo} = 2I_{DSS}/|V_P|$:

$$g_m = g_{mo} \left(1 - \frac{V_{GS}}{V_P}\right) \quad g_{mo} = \frac{2I_{DSS}}{|V_P|}$$

- As V_{GS} lies between 0 and V_P , the transconductance lies between 0 and g_{mo} .
- g_{mo} is the **maximum transconductance** of the JFET; it occurs at $V_{GS} = 0$.



Expressing g_m in terms of the drain current

Since $\left(1 - \frac{V_{GS}}{V_P}\right) = \sqrt{\frac{I_{DS}}{I_{DSS}}}$, substituting into equation (2):

$$g_m = \frac{2I_{DSS}}{|V_P|} \times \sqrt{\frac{I_{DS}}{I_{DSS}}} = \frac{2}{|V_P|} \sqrt{I_{DS} \times I_{DSS}}$$

For a MOSFET in saturation

$$I_{DS} = \frac{K_n}{2} (V_{GS} - V_T)^2$$

$$\Rightarrow g_m = \frac{K_n}{2} \times 2(V_{GS} - V_T)$$

$$g_m = K_n(V_{GS} - V_T)$$

$$\text{where } K_n = \mu_n C_{ox} \frac{W}{L}$$

Also, since $(V_{GS} - V_T) = \sqrt{2I_{DS}/K_n}$:

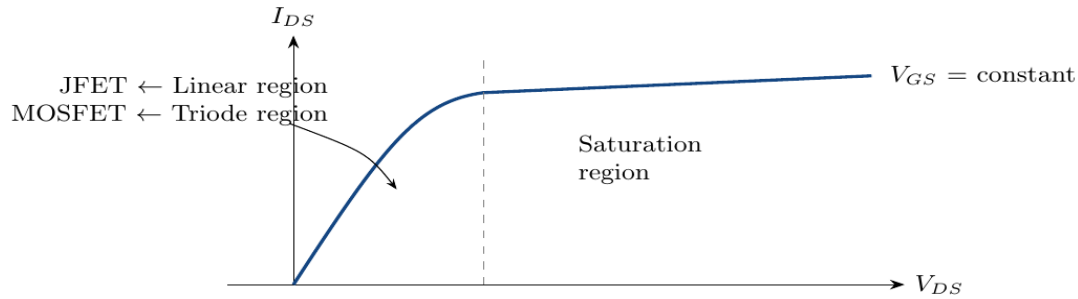
$$g_m = \sqrt{2K_n I_{DS}}$$

4. Dynamic Drain Resistance r_{ds} (or r_o)

r_{ds} is the **reciprocal of the slope of the drain characteristics**.

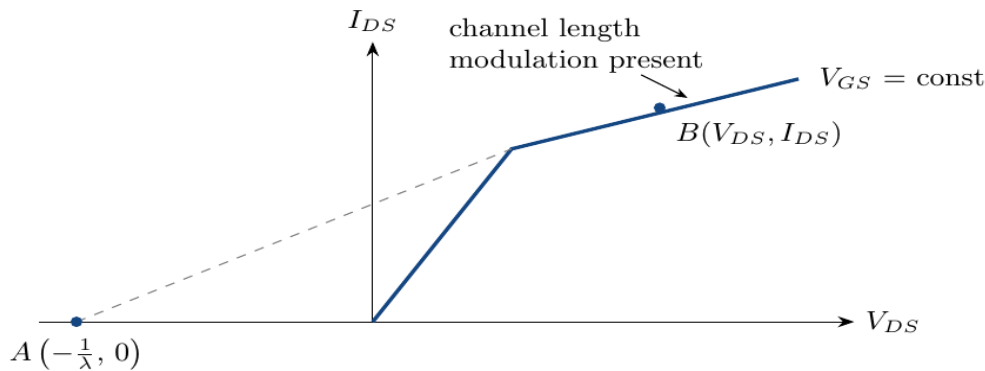
$$\text{Slope} = \frac{\partial I_{DS}}{\partial V_{DS}} \Rightarrow r_{ds} = \frac{1}{\text{slope}} = \frac{\partial V_{DS}}{\partial I_{DS}}$$

For a JFET amplifier, if r_{ds} is not provided then it is considered to be ∞ .



5. Channel Length Modulation

If a MOSFET has the **channel length modulation effect**, the drain current increases even in the saturation region. If the drain characteristics are projected backwards, they cut the voltage axis at $(-1/\lambda, 0)$, where λ is the **channel length modulation parameter**.



Considering the straight line AB :

$$\text{Slope} = \frac{\partial I_{DS}}{\partial V_{DS}} = \frac{I_{DS} - 0}{V_{DS} - \left(-\frac{1}{\lambda}\right)} = \frac{I_{DS}}{V_{DS} + \frac{1}{\lambda}}$$

$$r_{ds} = \frac{1}{\text{slope}} = \frac{V_{DS} + \frac{1}{\lambda}}{I_{DS}}$$

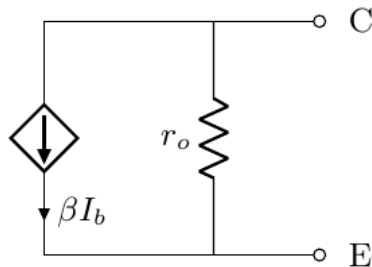
Since $\frac{1}{\lambda} \gg V_{DS}$:

$$r_{ds} = \frac{1}{\lambda I_{DS}}$$

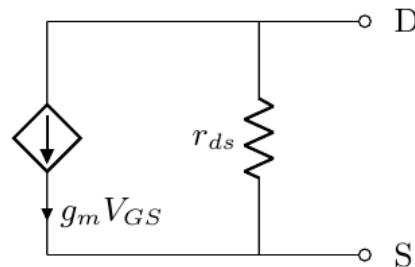
If the MOSFET does not have the channel length modulation effect, then $\lambda = 0$ and $r_{ds} = \infty$. Also, $V_A = 1/\lambda$ is the **Early voltage** of the MOSFET.

6. Non-Ideal Current Sources

The BJT and the MOSFET are **ideal current sources** if there is no Early effect and no channel length modulation effect. But due to the Early effect in the BJT and channel length modulation in the MOSFET, they become **non-ideal current sources**.



BJT



MOSFET

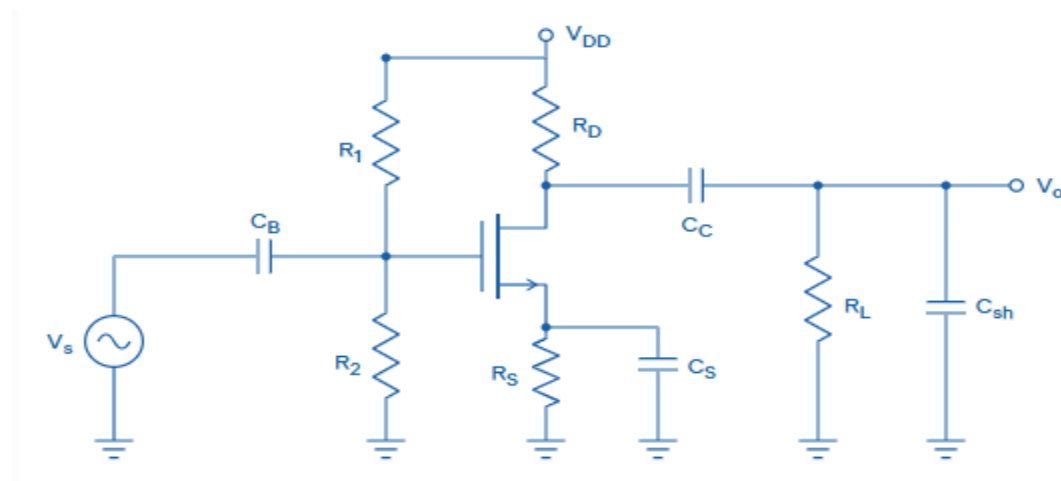
7. Analysis of FET Amplifiers

The analysis is the same for MOSFET and JFET. A FET can be used in three configurations:

Configuration	Input node	Output node
Common Source	G	D
Common Drain	G	S
Common Gate	S	D

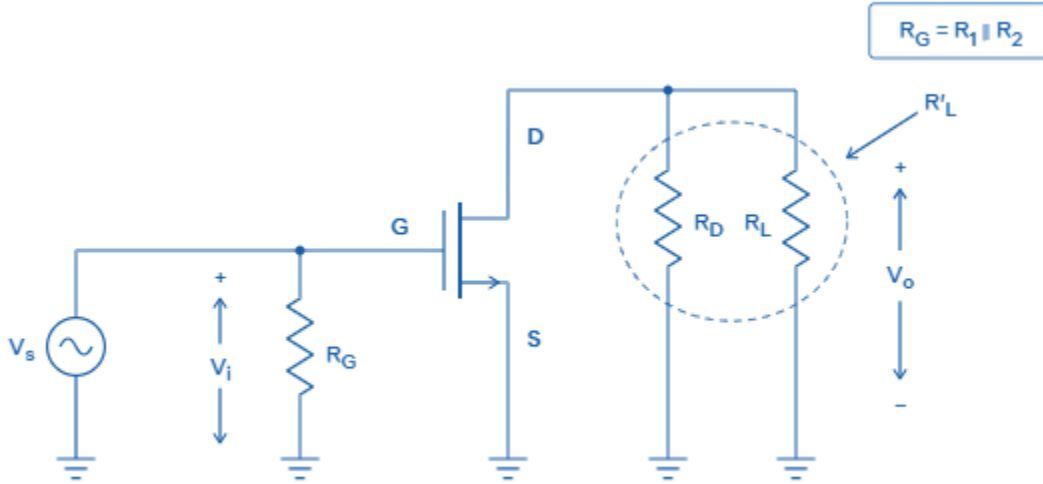
Source $\equiv$ Emitter • Drain $\equiv$ Collector • Gate $\equiv$ Base

8. Common Source Amplifier



Draw the medium frequency AC equivalent circuit by replacing C_B , C_S and C_C with a short circuit and C_{SH} with an open circuit, and by connecting the V_{DD} node to ground.

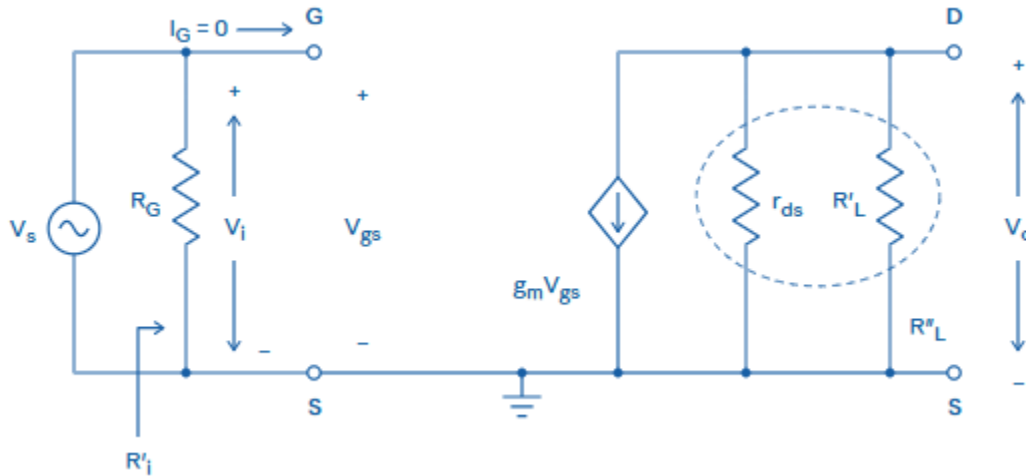
The source resistance r_s is neglected, because $r_s \approx \text{k}\Omega$ while $R_G \approx \text{M}\Omega$. Hence $R_G \gg r_s$.



$$R_G = R_1 \parallel R_2$$

$$R'_L = R_D \parallel R_L$$

Replacing the MOSFET with its small signal model



(a) Input resistance R_i

$$R_i = \frac{V_i}{I_G} = \frac{V_i}{0} = \infty$$

$$R'_i = R_i \parallel R_G = \infty \parallel R_G = R_G$$

(b) Voltage gain A_V

$$A_V = \frac{V_o}{V_i} \quad \text{and} \quad V_o = -g_m V_{GS} \times R'_L$$

Since $V_i = V_{GS}$:

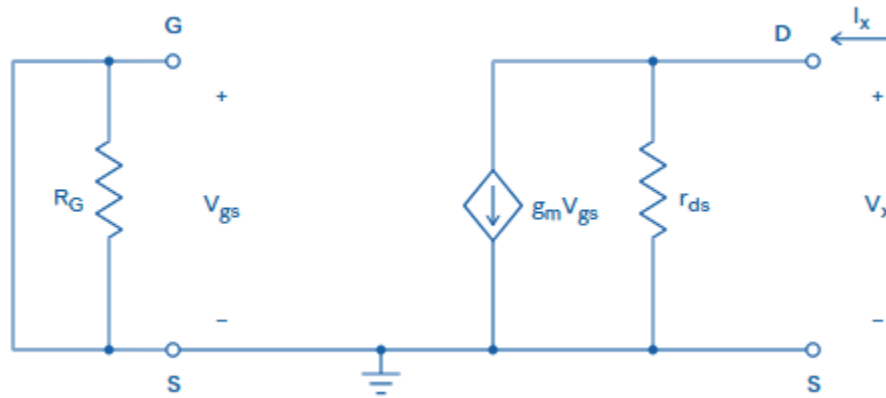
$$A_V = \frac{-g_m V_{GS} R_L''}{V_{GS}} = -g_m R_L'' \quad \text{where} \quad R_L'' = R_L' \parallel r_{ds}$$

If $r_{ds} = \infty$, then $A_V = -g_m R_L'$.

Because $R_i = \infty$, no current flows through r_s , so $V_i = V_s$ and therefore $V_o/V_s = V_o/V_i = A_V$. The negative sign indicates a phase shift of **180°** between V_o and V_i .

(c) Output resistance R_o

To calculate R_o : replace V_s with a short circuit, disconnect the load R_L' , and apply a voltage V_x across the output port.



As the gate is shorted to the source terminal, $V_{GS} = 0$. Applying KCL at the drain:

$$I_x = \frac{V_x}{r_{ds}} + g_m V_{GS} = \frac{V_x}{r_{ds}}$$

$$R_o = \frac{V_x}{I_x} = r_{ds} \quad R_o' = R_o \parallel R_D = r_{ds} \parallel R_D$$

Summary — common source amplifier

- iv. $R_i' = R_G$
- v. $A_V = -g_m R_L''$
- vi. $R_o' = r_{ds} \parallel R_D$

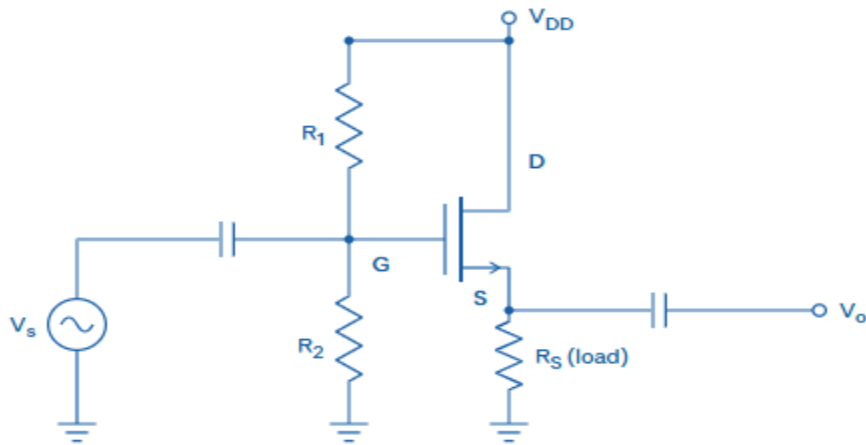
If the bypass capacitor C_S is disconnected, then $R_i = \infty$ and $R_i' = R_G$;

$$A_V = \frac{-g_m R_L''}{1 + g_m R_S}; \text{ and}$$

$$R_o = r_{ds} + (1 + \mu) R_S,$$

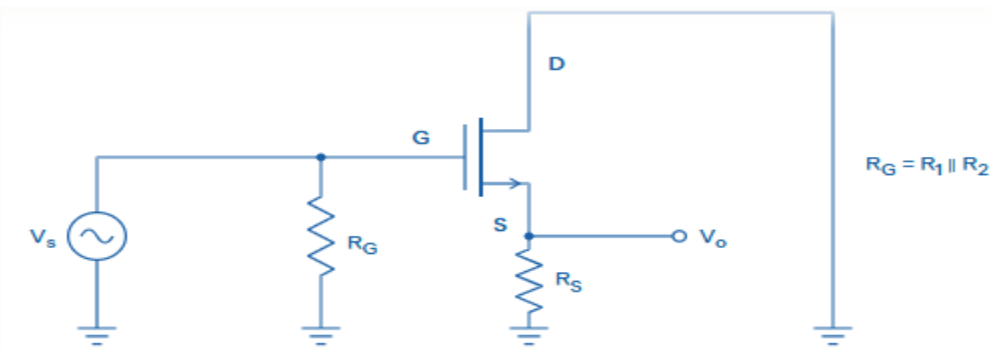
where $\mu = g_m r_{ds}$ is the **amplification factor**.

9. Common Drain Amplifier

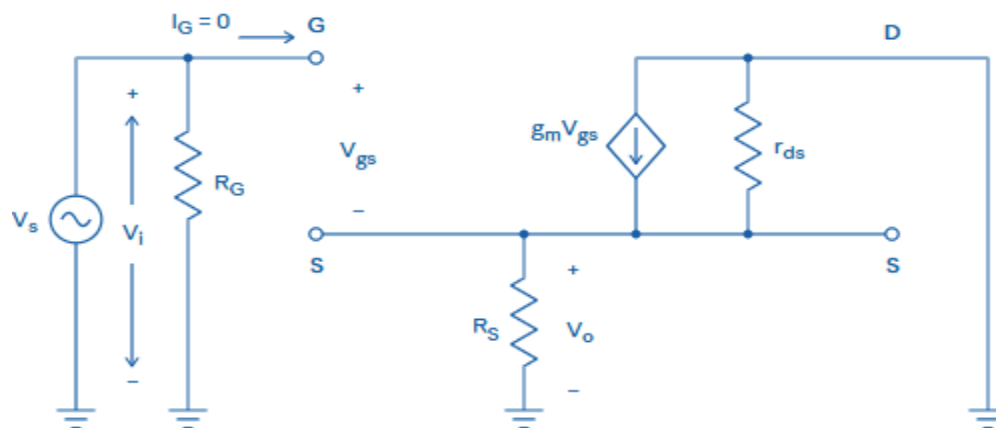


R_S is the load, and $R_G = R_1 \parallel R_2$.

AC equivalent circuit



Replacing the MOSFET with its small signal model

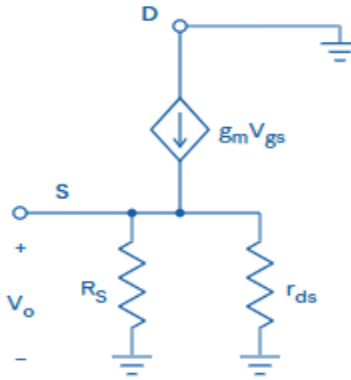


(a) Input resistance R_i

$$R_i = \frac{V_i}{I_G} = \frac{V_i}{0} = \infty \quad R_i' = R_i \parallel R_G = R_G$$

(b) Voltage gain A_V

As the drain is grounded, r_{ds} appears in parallel with R_S :



$$V_o = (R_S \parallel r_{ds}) g_m V_{GS} \quad (1)$$

From the input loop, $V_i = V_{GS} + V_o$:

$$V_i = V_{GS} + g_m V_{GS} (R_S \parallel r_{ds}) = V_{GS} \{1 + g_m (R_S \parallel r_{ds})\} \quad (2)$$

Dividing (1) by (2):

$$A_V = \frac{V_o}{V_i} = \frac{(R_S \parallel r_{ds}) g_m}{1 + g_m (R_S \parallel r_{ds})}$$

Since $g_m (R_S \parallel r_{ds}) \gg 1$, $A_V \approx 1$ — the common drain amplifier has **unity voltage gain**.

The AC output voltage is the same as the AC input voltage. The common drain amplifier is therefore also called a **source follower**, because the voltage at the source terminal follows the input voltage.

(c) Output resistance R_o

$$R_o = \frac{V_x}{I_x} = \frac{1}{g_m} \parallel r_{ds}$$

If $r_{ds} = \infty$, then:

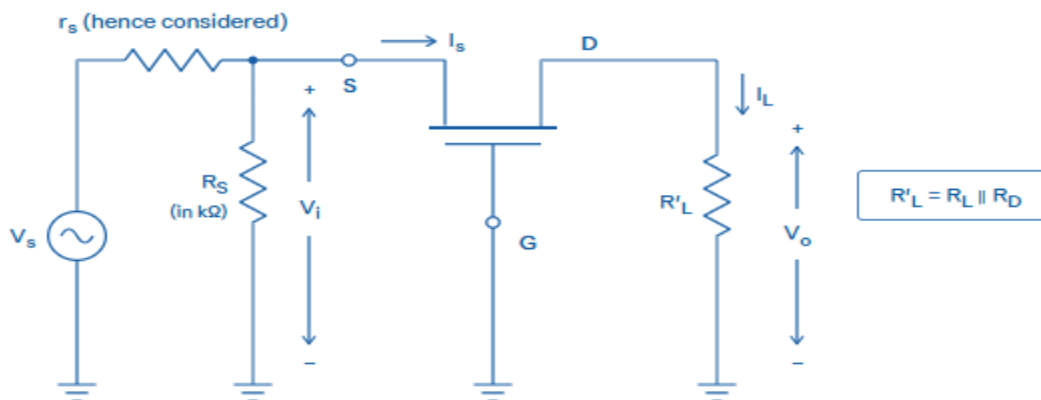
$$A_V = \frac{g_m R_S}{1 + g_m R_S} \quad R_o = \frac{1}{g_m}$$

Summary — common drain amplifier

- vii. Unity voltage gain
- viii. Large R_i
- ix. Small R_o

It is used as a **voltage buffer** for impedance matching.

10. Common Gate Amplifier



with $R'_L = R_L \parallel R_D$. Note that here r_s is taken into account.

- $A_I = \frac{I_L}{I_S} = 1$
- $R_i = \frac{V_i}{I_S} = \frac{1}{g_m}$
- $A_V = \frac{V_o}{V_i} = g_m R'_L$
- $R_o = \infty$

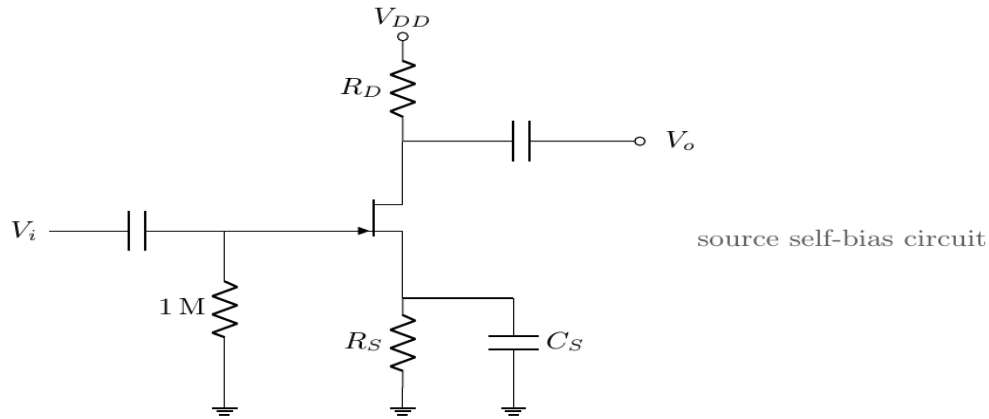
These are the same as for the common base amplifier (with $r_{ds} = \infty$).

The common gate amplifier has unity current gain, large voltage gain, small R_i and large R_o . It is used as a **current buffer** and as a **high frequency amplifier**.

11. Worked Problems

Q6 — JFET self-bias amplifier

Q. In the amplifier shown, the JFET has $I_{DSS} = 1.65$ mA, $V_P = -2$ V, and it is operating at a drain current of 0.8 mA. Calculate (i) V_{GS} , (ii) R_S , (iii) g_m , (iv) R_D if the voltage gain is 20 dB.



(i) From $I_{DS} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$:

$$V_{GS} = V_P \left[1 - \sqrt{\frac{I_{DS}}{I_{DSS}}}\right] = -2 \left[1 - \sqrt{\frac{0.8}{1.65}}\right] = -0.607 \text{ V}$$

(ii) For the self-bias circuit, $V_{GS} = -I_{DS}R_S$:

$$R_S = \frac{-V_{GS}}{I_{DS}} = \frac{0.607}{0.8} = 0.758 \text{ k}\Omega$$

(iii)

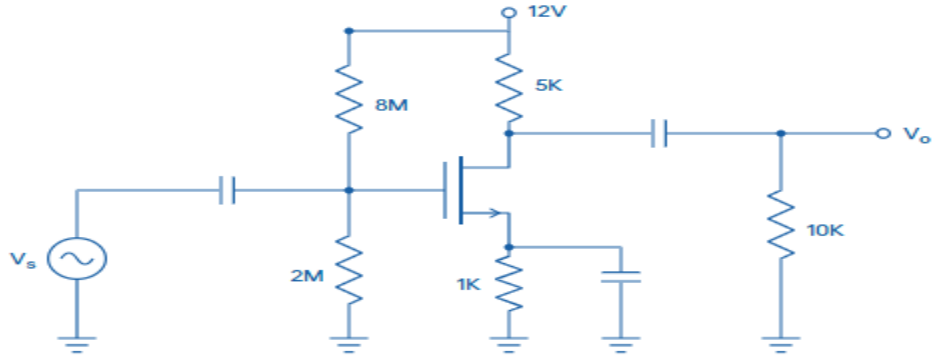
$$g_m = \frac{2}{|V_P|} \sqrt{I_{DS} \times I_{DSS}} = \frac{2}{2} \sqrt{0.8 \times 1.65} = 1.148 \text{ m}\Omega$$

(iv) $20\log|A_V| = 20 \Rightarrow A_V = 10$. The circuit is a common source stage, so $|A_V| = g_m R_D$:

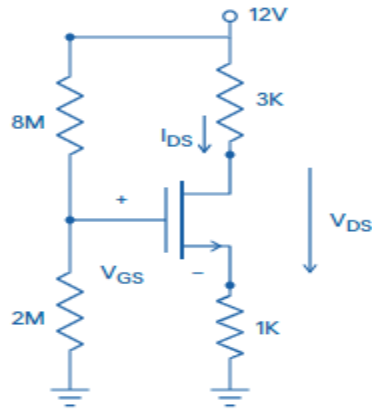
$$R_D = \frac{|A_V|}{g_m} = \frac{10}{1.148} = 8.7 \text{ k}\Omega$$

Q7 — NMOS common source amplifier

Q. In the amplifier shown, the MOSFET has $\mu_n C_{ox}(W/L) = 2 \text{ mA/V}^2$, $V_T = 1 \text{ V}$ and $\lambda = 0$. Calculate (i) the operating point, (ii) the small signal voltage gain.



(i) **Operating point.** For the DC calculation all capacitors are open circuits.



$$V_{GS} = V_G - V_S = \frac{12 \times 2}{10} - I_{DS} \times 1 = 2.4 - I_{DS} \quad \Rightarrow \quad I_{DS} = 2.4 - V_{GS} \quad (1)$$

Also, in saturation:

$$2.4 - V_{GS} = \frac{2}{2} (V_{GS} - 1)^2 \quad \Rightarrow \quad V_{GS}^2 - V_{GS} - 1.4 = 0$$

$$V_{GS} = 1.78 \text{ V} \quad \text{or} \quad -0.78 \text{ V}$$

Since $V_{GS} > V_T$ is required, $V_{GS} = 1.78 \text{ V}$.

From (1), $I_{DS} = 2.4 - 1.78 = 0.62 \text{ mA}$.

$$V_{DS} = V_{DD} - I_{DS}(R_D + R_S) = 12 - 0.62(5 + 1) = 8.28 \text{ V}$$

Operating point Q (8.28 V, 0.62 mA).

(ii) **Small signal voltage gain.**

$$g_m = K_n(V_{GS} - V_T) = 2(1.78 - 1) = 1.56 \text{ mS} \quad r_{ds} = \infty \quad (\lambda = 0)$$

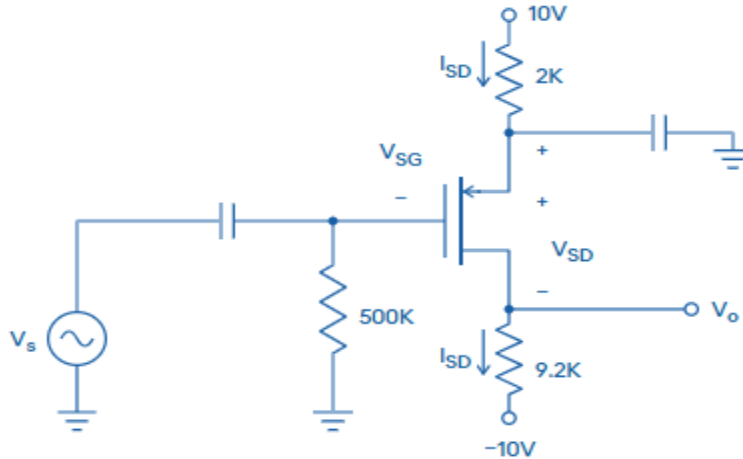
$$A_V = -g_m R_L' = -1.56 \times (10 \text{ k} \parallel 5 \text{ k}) = -1.56 \times 3.33 = -5.2$$

the same amplifier with the bypass capacitor disconnected

$$A_V = \frac{-g_m R_L'}{1 + g_m R_S} = \frac{-1.56 \times 3.33}{1 + 1.56 \times 1} = -2.02$$

Q9 — PMOS amplifier

Q. In the amplifier shown, the MOSFET has $\mu_p C_{ox}(W/L) = 4 \text{ mA/V}^2$, $V_T = -1.5 \text{ V}$ and $\lambda = 0$. Calculate (i) I_{SD} and V_{SD} , (ii) the small signal voltage gain.



(i)

$$V_{SG} = V_S - V_G = (10 - 2I_{SD}) - 0 \Rightarrow I_{SD} = \frac{10 - V_{SG}}{2} \quad (1)$$

For a p-channel MOSFET in saturation:

$$I_{SD} = \frac{K_p}{2} \{V_{SG} - |V_T|\}^2$$

$$\frac{10 - V_{SG}}{2} = \frac{4}{2} (V_{SG} - 1.5)^2$$

$$10 - V_{SG} = 4V_{SG}^2 + 9 - 12V_{SG} \Rightarrow 4V_{SG}^2 - 11V_{SG} - 1 = 0$$

$$V_{SG} = 2.83 \text{ V} \quad \text{or} \quad -0.08 \text{ V}$$

Since $V_{SG} > |V_T|$ is required, $V_{SG} = 2.83 \text{ V}$.

From (1):

$$I_{SD} = \frac{10 - 2.83}{2} = 3.585 \text{ mA}$$

The supply spans +10 V to -10 V, a total of 20 V, so:

$$V_{SD} = 20 - (2 + 2.2)I_{SD} = 20 - 4.2 \times 3.585 = 4.943 \text{ V}$$

(ii) The circuit is a CS amplifier with a bypass:

$$g_m = K_p(V_{SG} - |V_T|) = 4(2.83 - 1.5) = 5.32 \text{ m}\mathcal{U}$$

$$r_{ds} = \infty$$

$$A_V = -g_m R_L' = -5.32 \times 2.2 = -11.704$$

12. Frequency Response of the RC-Coupled CS Amplifier

Low and high frequency analysis of the RC-coupled CS amplifier is the same as for the RC-coupled CE amplifier.

$$(i) C_C: f_L = \frac{1}{2\pi C_C(R_D + R_L)}$$

$$(ii) C_S: f_L = \frac{1}{2\pi C_S R}, \quad R = \frac{1}{g_m} \parallel R_S$$

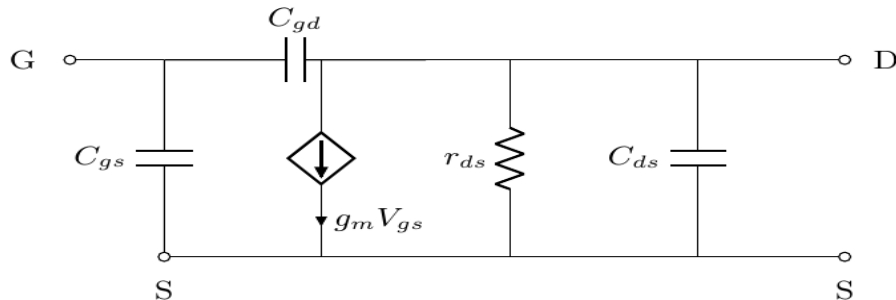
$$(iii) C_B: f_L = \frac{1}{2\pi C_B R}, \quad R = r_s + R_i' \approx R_G$$

$$(iv) C_{SH}: f_H = \frac{1}{2\pi C_{SH} R_L''}, \quad R_L'' = R_L' \parallel r_{ds}$$

$$(v) C_{in}: f_H = \frac{1}{2\pi C_{in} R}, \quad R = r_s \parallel R_G$$

13. High Frequency Small Signal Model of the FET

The high frequency model is obtained by including the internal capacitances in the low frequency model.



For a JFET

- C_{gs} and C_{gd} are the **transition capacitance of the gate–channel junction**. They lie between 1 pF and 10 pF.
- C_{ds} is the **internal capacitance of the channel** (0.1 pF to 1 pF).

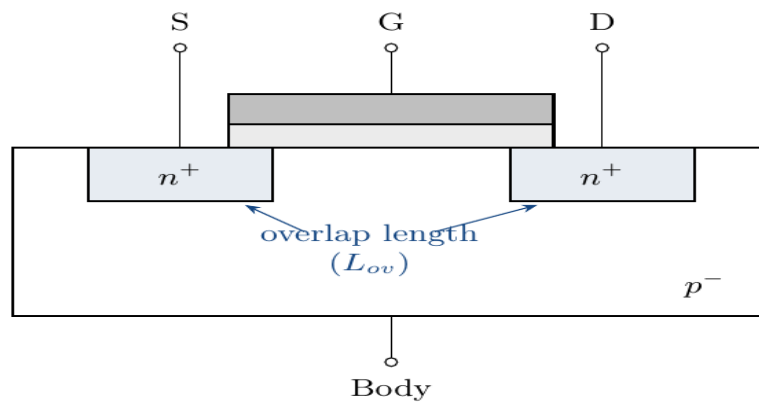
For a MOSFET in saturation

C_{gs} is the sum of the gate-to-source oxide capacitance and the overlap capacitance:

$$C_{gs} = C_{gs-ox} + C_{gs-ov} = \frac{2}{3}(C_{ox})(WL) + C_{ox} \times WL_{ov}$$

C_{gd} is the gate-to-drain overlap capacitance:

$$C_{gd} = C_{gd-ov} = (C_{ox}) \times (WL_{ov}) \quad L_{ov} = \text{overlap length}$$

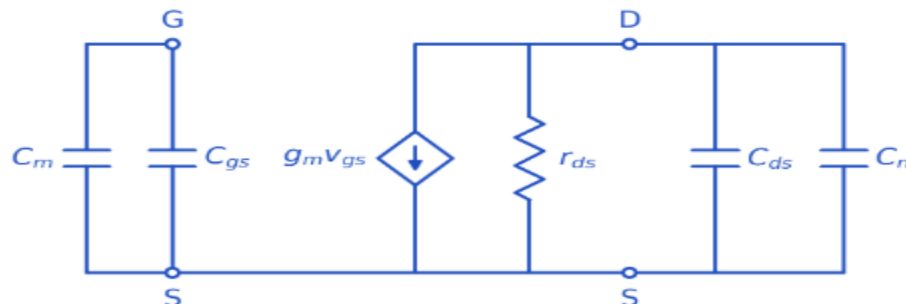


If a FET is used in the common source configuration, then C_{gd} appears between the input node (gate) and the output node (drain). Hence C_{gd} can be replaced with C_m and C_n by using **Miller's theorem**.

$$C_m = C_{gd}(1 - A_V) \quad C_n = C_{gd} \left(1 - \frac{1}{A_V}\right) \quad A_V = -g_m R_L''$$

The gain $A_V = -g_m R_L''$ is not very high, hence $1/A_V$ cannot be neglected.

Redrawing the circuit:



Net input capacitance of the CS configuration:

$$C_{in} = C_{gs} + C_m = C_{gs} + C_{gd}(1 - A_V)$$

Net output capacitance of the CS amplifier:

$$C_{out} = C_{ds} + C_n = C_{ds} + C_{gd}\left(1 - \frac{1}{A_V}\right)$$

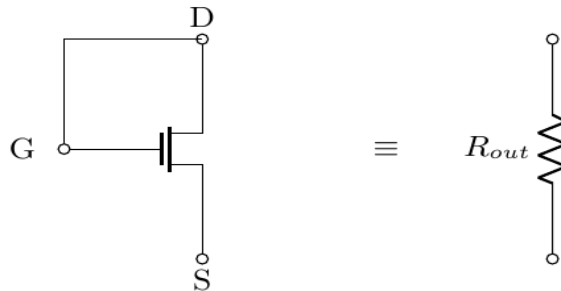
When C_m appears in parallel with C_{gs} , the input capacitance of the CS configuration increases. This is known as the **Miller effect**.

15. Active Load in MOSFET IC Amplifiers

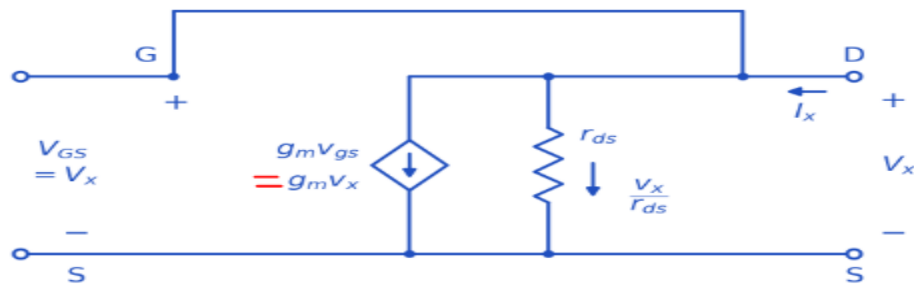
An active load can be:

- A MOSFET with drain shorted to gate (the same as base shorted to collector in a BJT)
- A depletion MOSFET with gate shorted to source
- A MOSFET current mirror

(i) MOSFET with drain shorted to gate



Replacing the MOSFET with its small signal model, and noting that $V_{GS} = V_x$:



By KCL at the drain:

$$I_x = g_m V_x + \frac{V_x}{r_{ds}} = V_x \left\{ g_m + \frac{1}{r_{ds}} \right\}$$

$$\frac{I_x}{V_x} = g_m + \frac{1}{r_{ds}} = \frac{1}{R_{out}} \quad \Rightarrow \quad R_{out} = \frac{1}{g_m} \parallel r_{ds}$$

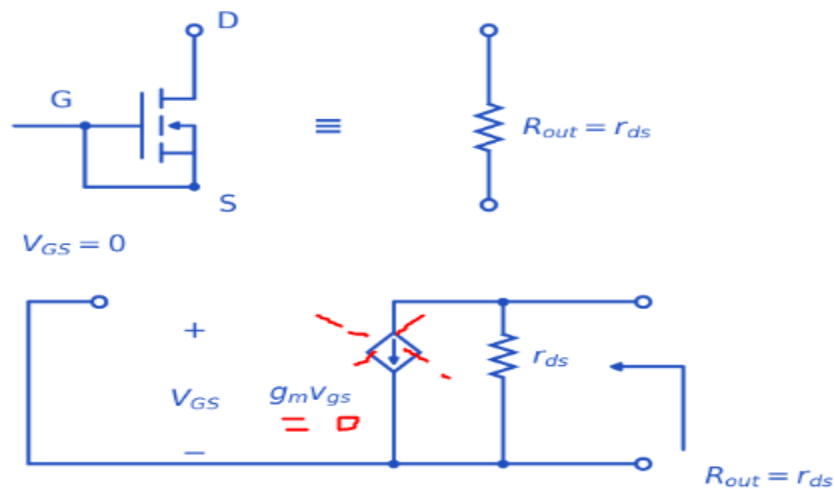
If $r_{ds} = \infty$ (i.e. $\lambda = 0$),

then $R_{out} = 1/g_m$.

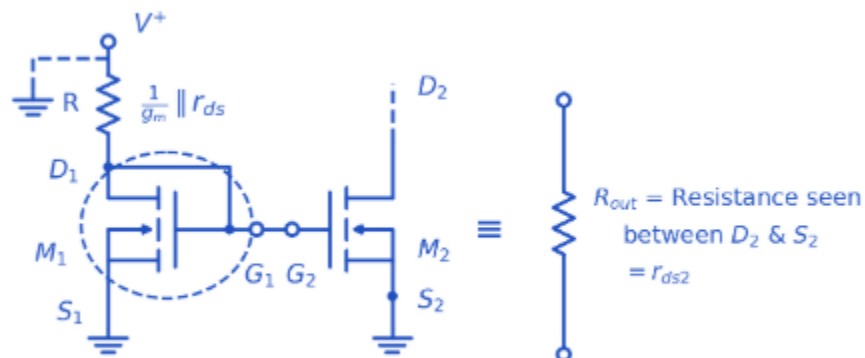
(ii) Depletion MOSFET with gate shorted to source

Here $V_{GS} = 0$, so the dependent source $g_m V_{GS}$ is zero and only r_{ds} remains:

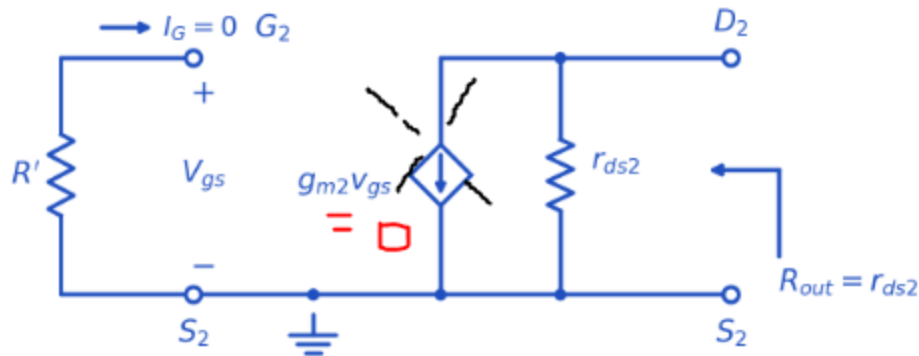
$$R_{out} = r_{ds}$$



(iii) MOSFET current mirror



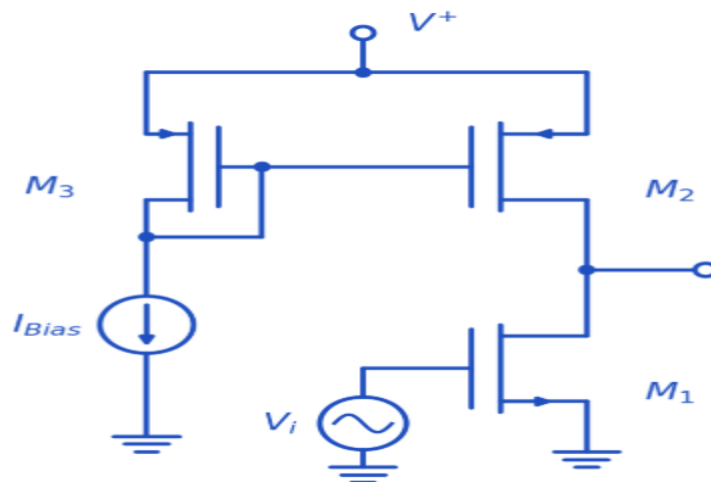
Replacing M_2 with its small signal model, the gate of M_2 sees $R' = \frac{1}{g_m} \parallel r_{ds} \parallel R$, and because $I_G = 0$ there is no signal at the gate, so $V_{gs} = 0$:



$$R_{out} = r_{ds2} \quad R' = \frac{1}{g_m} \parallel r_{ds} \parallel R$$

Q10 — CMOS common source amplifier with an active load

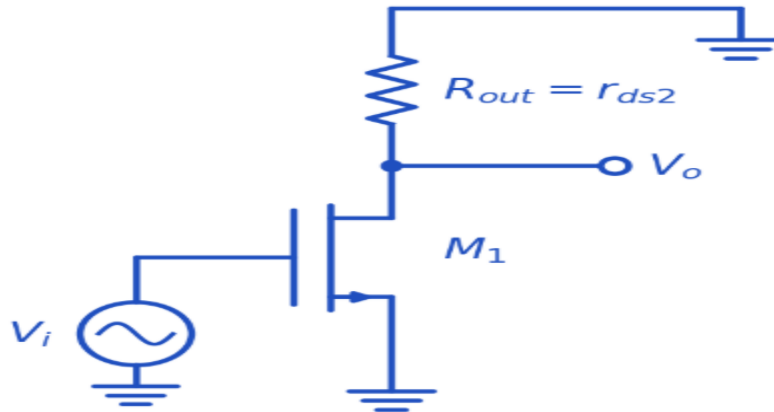
Q. Calculate the small signal voltage gain of the amplifier shown, if the MOSFETs have $\mu_n C_{ox} = 80 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 40 \mu\text{A}/\text{V}^2$, $(W/L)_N = 15$, $(W/L)_P = 30$, $\lambda_n = 0.01 \text{ V}^{-1}$, $\lambda_p = 0.02 \text{ V}^{-1}$ and $I_{Bias} = 0.2 \text{ mA}$.



Solution.

- The n-channel MOSFET M_1 is a **common source amplifier**.
- The p-channel MOSFETs M_2 and M_3 form a **current mirror**, used as an **active load** for M_1 .
- The output resistance of the current mirror acts as the load of M_1 .

Redrawing the circuit,



the load of M_1 is $R_{out} = r_{ds2}$, so:

$$A_V = -g_m R_L'' \quad \text{where} \quad R_L'' = R_L' \parallel r_{ds} = r_{ds2} \parallel r_{ds1}$$

$$A_V = -g_{m1}(r_{ds2} \parallel r_{ds1})$$

Since the same bias current flows in both branches,

$$I_{DSN} = I_{SDP} = I_{Bias} = 0.2 \text{ mA.}$$

$$g_{m1} = \sqrt{2K_n I_{DS}} = \sqrt{2 \times 80 \times 10^{-6} \times 15 \times 0.2 \times 10^{-3}} = 0.692 \text{ mS}$$

$$r_{ds1} = \frac{1}{\lambda_N I_{DSN}} = \frac{1}{0.01 \times 0.2 \times 10^{-3}} = 500 \text{ k}\Omega$$

$$r_{ds2} = \frac{1}{\lambda_P I_{SDP}} = \frac{1}{0.02 \times 0.2 \times 10^{-3}} = 250 \text{ k}\Omega$$

$$A_V = -0.692 \times (250 \parallel 500) = -115.33$$

- When a current mirror is used as an active load, a **high voltage gain** can be achieved, because the output resistance of the current mirror is high.
- The above circuit is called a **CMOS common source amplifier**, in which the NMOS is used as the amplifier and the PMOS current mirror is the load.
- NMOS is preferred as the amplifier over PMOS because it has greater transconductance, since g_m is proportional to mobility and $\mu_N > \mu_P$.

Revisiting Single-Stage MOSFET Amplifiers to get deep insight:

Common Source (four loads) • Common Drain • Common Gate

1. Overview and a Useful Shortcut

Every single-stage amplifier is characterized by the same three small-signal quantities: the voltage gain A_v , the input resistance R_{in} and the output resistance R_{out} . For each stage we first fix the operating point (so that the amplifying transistor sits in saturation) and then replace the transistor by its small-signal model, in which the drain current is

$$i_d = g_m v_{gs} + \frac{v_{ds}}{r_o}$$

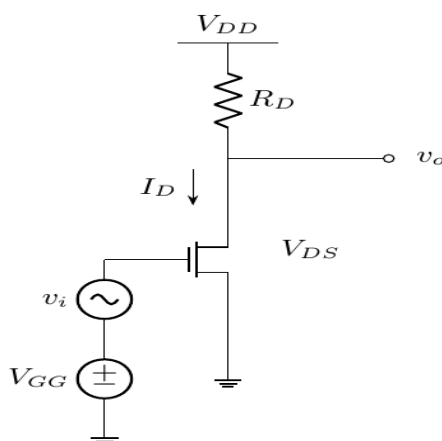
with r_o (also written r_d) the output resistance due to channel length modulation.

Gain as $-G_m R_{out}$. For a linear stage, the voltage gain can always be found as

$$A_v = -G_m R_{out}, \quad G_m = \left. \frac{i_{sc}}{v_{in}} \right|_{v_{out}=0}, \quad R_{out} = \left. \frac{v_x}{i_x} \right|_{v_{in}=0}$$

where i_{sc} is the current that flows *into* the output node from a short circuit placed there. Finding G_m and R_{out} separately is often easier than solving the whole circuit at once.

2. CS Stage with a Resistive Load

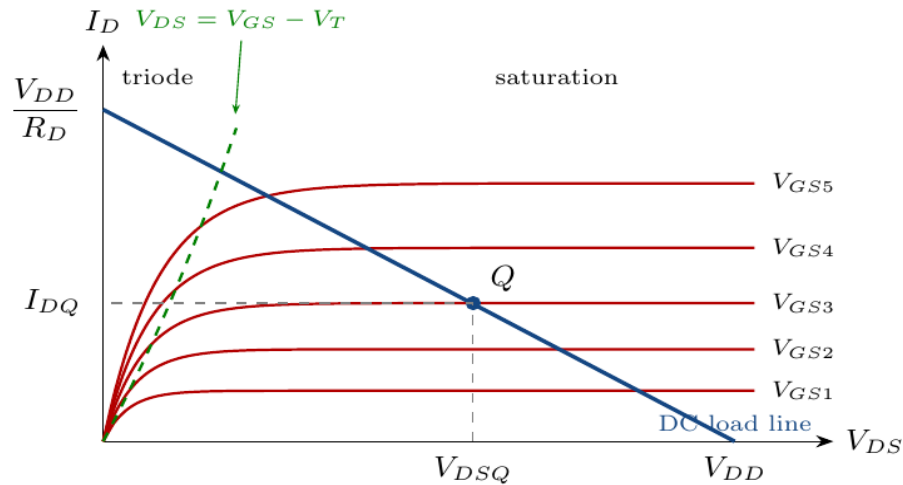


DC operating point

Setting the signal to zero ($v_i = 0$), the gate sits at $V_{GS} = V_{GG}$ and KVL around the output loop gives the **DC load line**:

$$I_D = \frac{V_{DD} - V_{DS}}{R_D}$$

Its end points are $I_D = V_{DD}/R_D$ at $V_{DS} = 0$, and $I_D = 0$ at $V_{DS} = V_{DD}$. The quiescent point Q is where this line meets the transistor curve for the chosen V_{GS} .



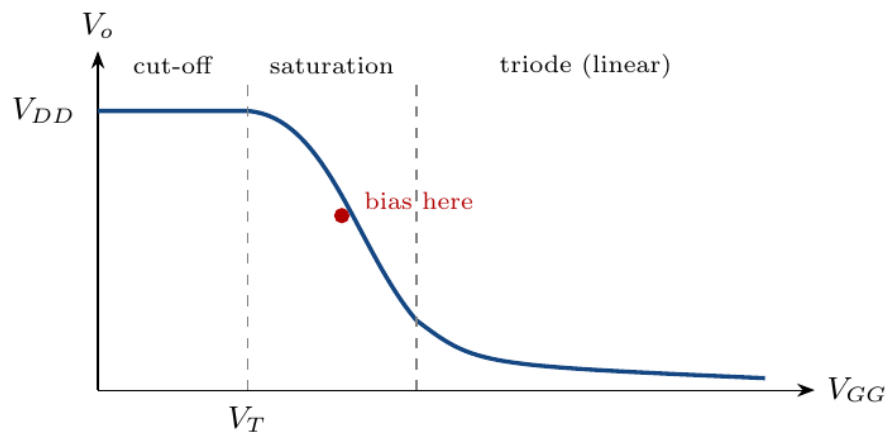
For linear amplification the transistor must stay in **saturation**:

$$V_{DS} \geq V_{GS} - V_T \quad \Leftrightarrow \quad V_D \geq V_G - V_T$$

i.e. the drain may sit at most one threshold below the gate.

Transfer characteristic

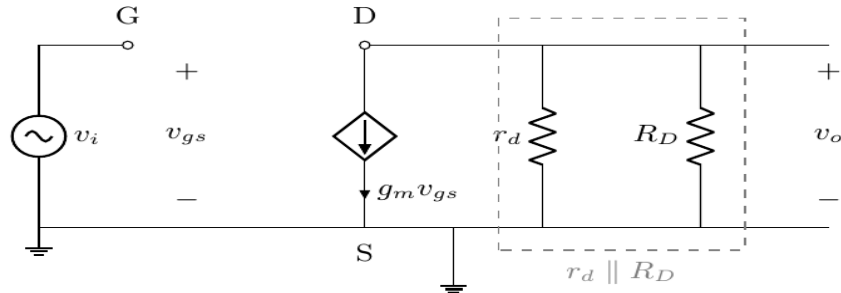
Sweeping V_{GG} shows three regions. Below V_T the device is off, no current flows and the output rests at V_{DD} . Just above V_T the device enters saturation and the output falls steeply — this is the useful high-gain region. Once V_o drops below $V_{GG} - V_T$ the device enters the triode region and the slope flattens out.



The bias point is placed in the middle of the steep (saturation) segment. A small sinusoid superimposed on V_{GG} then produces a large, inverted, undistorted swing at the output. If the bias drifts toward V_T the negative peaks cut off; if it drifts too high, the positive peaks push the device into triode.

Small-signal analysis

For AC analysis all DC sources are set to zero ($V_{GG} = 0$, $V_{DD} = 0$), so R_D now connects the drain to ground.



The gate is driven directly, so $v_{gs} = v_i$. The dependent source pushes its current through the parallel combination of r_d and R_D , and because that current flows downward (out of the output node) the output goes negative:

$$v_o = -g_m v_{gs} (r_d \parallel R_D) = -g_m (r_d \parallel R_D) v_i$$

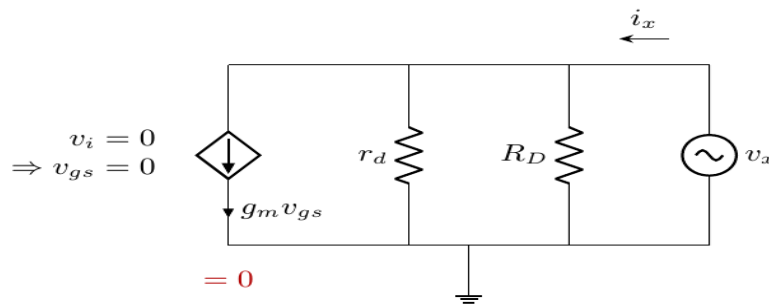
$$A_v = \frac{v_o}{v_i} = -g_m (r_d \parallel R_D)$$

The minus sign is the 180° phase inversion characteristic of the CS stage.

$$\text{Input Resistance } R_{in} = \infty$$

Output resistance

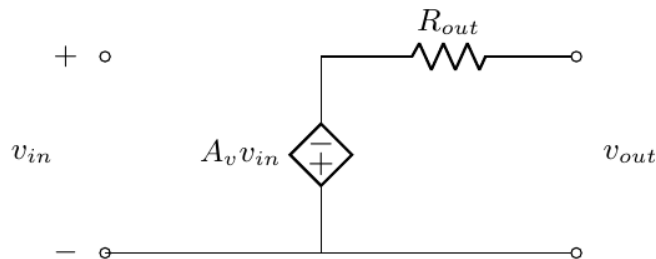
Kill the input ($v_i = 0$, hence $v_{gs} = 0$ and the dependent source is dead), then apply a test source v_x at the output:



With the dependent source off, the test source simply sees r_d and R_D in parallel: $v_x = (r_d \parallel R_D) i_x$.

$$R_{out} = \left. \frac{v_x}{i_x} \right|_{v_i=0} = r_d \parallel R_D$$

Taken together, the stage behaves at its output like a voltage source $A_v v_{in}$ in series with R_{out} — a Thevenin model that makes it easy to predict what happens when the next stage loads it.



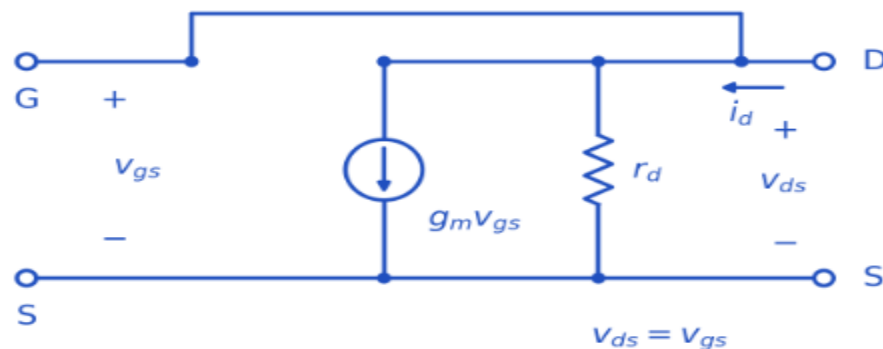
3. CS Stage with a Diode-Connected Load

The diode-connected MOSFET

Tying the gate of a MOSFET to its own drain forces $V_{DS} = V_{GS}$. Since the saturation condition is $V_{DS} \geq V_{GS} - V_T$, and V_{GS} is always greater than $V_{GS} - V_T$, a diode-connected device is **always in saturation** whenever it conducts. It behaves like a two-terminal element — hence the name.



To find its small-signal resistance, note that $v_{gs} = v_{ds}$ because gate and drain are the same node:

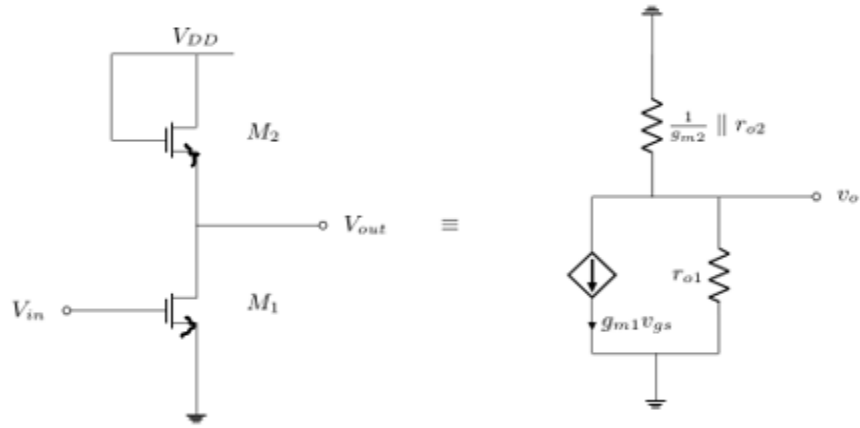


$$i_d = g_m v_{gs} + \frac{v_{ds}}{r_d} = g_m v_{ds} + \frac{v_{ds}}{r_d} = \left(g_m + \frac{1}{r_d} \right) v_{ds}$$

$$\frac{v_{ds}}{i_d} = \frac{1}{g_m + \frac{1}{r_d}} = \frac{\frac{1}{g_m} r_d}{\frac{1}{g_m} + r_d}$$

$$R_{\text{diode}} = \frac{1}{g_m} \parallel r_d \approx \frac{1}{g_m} \quad (\text{since normally } r_d \gg 1/g_m)$$

Using it as the load



M_1 is the amplifying device; M_2 is diode-connected and appears at the output as the resistance $\frac{1}{g_{m2}} \parallel r_{o2}$ to AC ground. (Body effect in M_2 is neglected here; including it would replace g_{m2} by $g_{m2} + g_{mb2}$.)

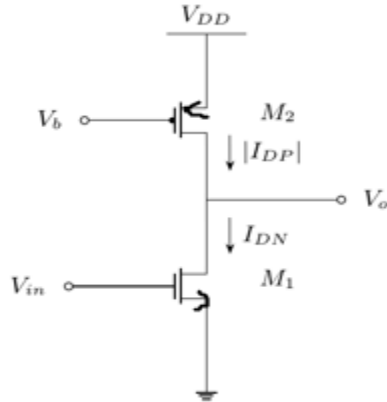
The load for M_1 is therefore r_{o1} in parallel with the diode resistance, and with $v_{gs} = v_{in}$:

$$v_o = -g_{m1} v_{gs} \left[r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2} \right) \right]$$

$$A_v = \frac{v_o}{v_{in}} = -g_{m1} \left[r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2} \right) \right]$$

$$R_{in} = \infty \quad R_{out} = r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2} \right)$$

4. CS Stage with a Triode Load



Operating conditions

Here the PMOS M_2 is biased by a fixed gate voltage V_b so that it operates in the **triode region**, where it behaves like a resistor R_{on2} . M_1 must stay in saturation. The same current flows through both, $|I_{DP}| = I_{DN}$.

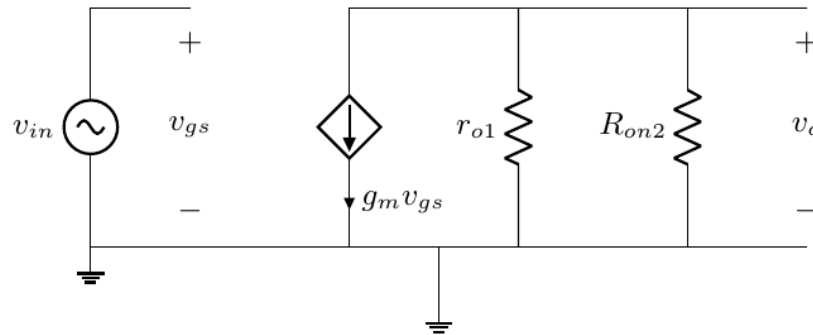
$$M_2 \text{ (triode): } |V_{DSP}| < |V_{GSP}| - |V_{TP}| \quad M_1 \text{ (saturation): } V_o > V_{IN} - V_{TN}$$

Together these fix the allowed output range:

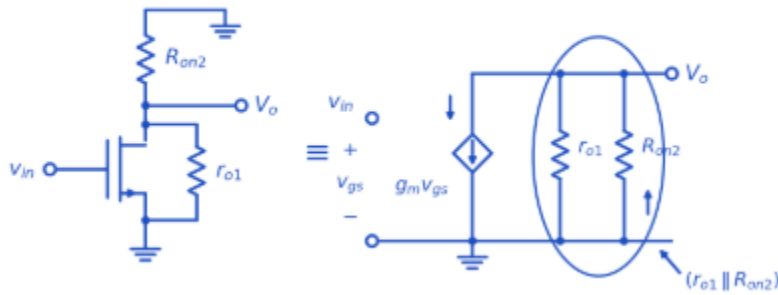
$$V_{IN} - V_{TN} < V_o < V_{DD} - |V_{DSP}|$$

Small-signal analysis

Replacing M_2 by its on-resistance R_{on2} and M_1 by its model:



Voltage Gain



$$V_o = -g_m v_{in} (r_{o1} \parallel R_{on2})$$

$$A_v = -g_m (r_{o1} \parallel R_{on2})$$

$$v_o = -g_m v_{in} (r_{o1} \parallel R_{on2}) \Rightarrow A_v = -g_m (r_{o1} \parallel R_{on2})$$

Input and output resistance

The input drives only a gate, so $R_{in} = \infty$. For R_{out} , setting $v_{in} = 0$ disables the dependent source, leaving the test source facing r_{o1} and R_{on2} in parallel:

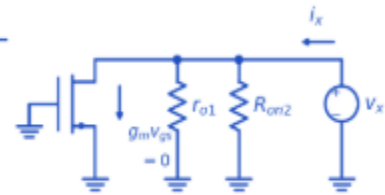
$$R_{in} = \infty \quad R_{out} = \left. \frac{v_x}{i_x} \right|_{v_{in}=0} = r_{o1} \parallel R_{on2}$$

I/p Resistance

$$R_{in} = \infty$$

Output Resistance

$$R_{out} = \left. \frac{v_x}{i_x} \right|_{v_{in}=0}$$

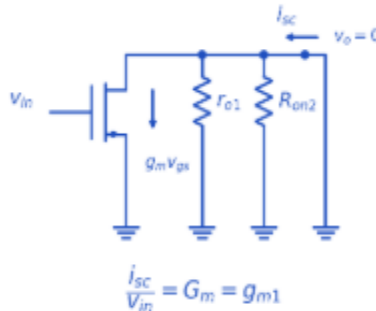


$$v_x = (r_{o1} \parallel R_{on2}) i_x$$

$$R_{out} = (r_{o1} \parallel R_{on2})$$

Transconductance

$$G_m = \left. \frac{i_{sc}}{v_{in}} \right|_{v_{out}=0}$$



$$i_{sc} = g_m v_{in}$$

$$\frac{i_{sc}}{v_{in}} = G_m = g_m$$

$$A_v = -G_m R_{out}$$

$$A_v = -g_m (r_{o1} \parallel R_{on2})$$

Transconductance — confirming the gain

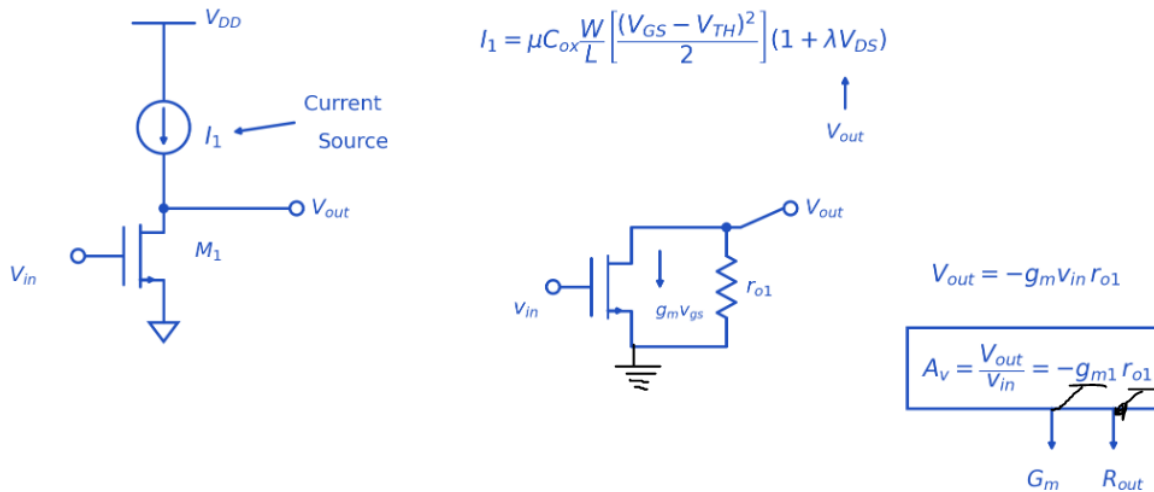
Short the output to ground. Now no current flows in r_{o1} or R_{on2} (both have zero volts across them), so the whole dependent-source current $g_{m1}v_{in}$ is drawn from the short: $i_{sc} = g_{m1}v_{in}$.

$$G_m = \left. \frac{i_{sc}}{v_{in}} \right|_{v_{out}=0} = g_{m1} \quad \Rightarrow \quad A_v = -G_m R_{out} = -g_{m1}(r_{o1} \parallel R_{on2})$$

which agrees with the direct calculation above.

5. CS Stage with a Current-Source Load

Ideal current source



An ideal current source has infinite small-signal resistance. The bias current must then be supplied entirely by M_1 in saturation, including the channel length modulation term:

$$I_1 = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS}), \quad V_{DS} = V_{out}$$

Because I_1 is fixed, it is the $(1 + \lambda V_{out})$ factor that absorbs any mismatch: the output voltage settles wherever M_1 's current exactly equals I_1 . With λ small, this makes V_{out} very sensitive to V_{in} — which is exactly why the gain is high.

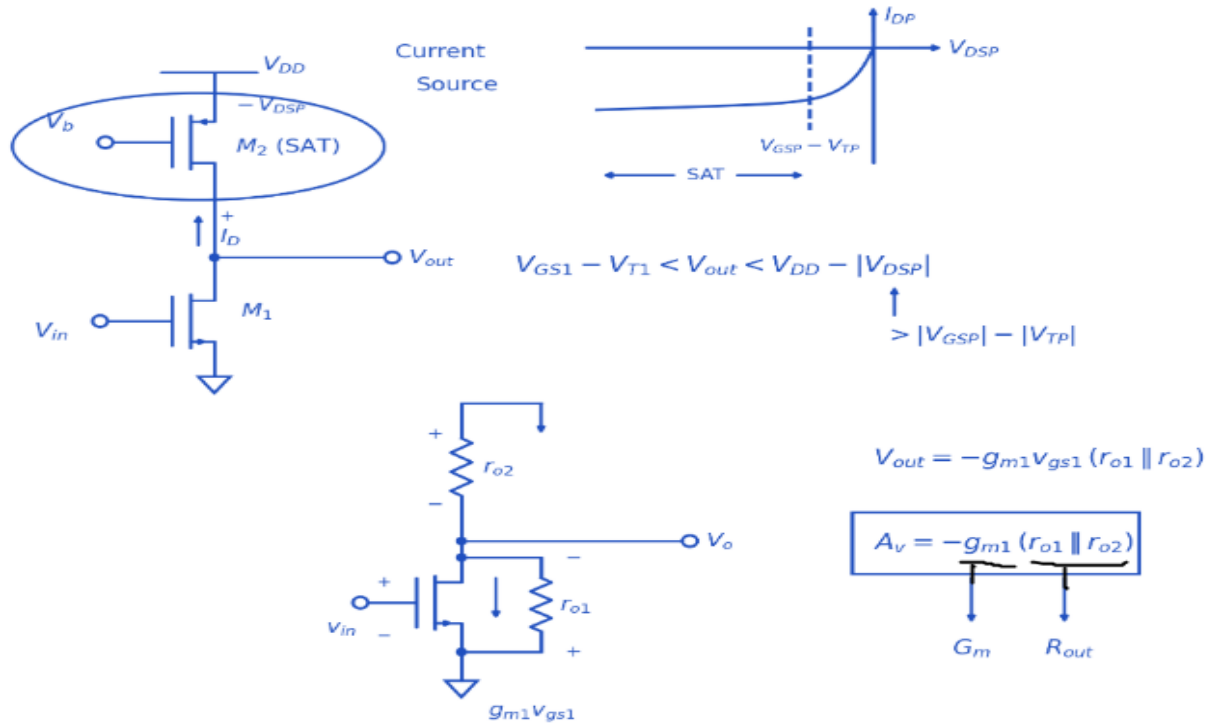
The only resistance at the output is r_{o1} , so:

$$v_{out} = -g_{m1} v_{in} r_{o1} \quad \Rightarrow \quad A_v = -g_{m1} r_{o1} \quad (G_m = g_{m1}, R_{out} = r_{o1})$$

This is the largest gain a single transistor can deliver — its **intrinsic gain**.

Practical current source — a PMOS in saturation

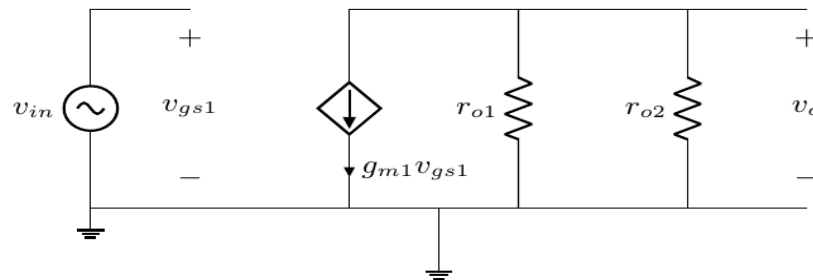
In practice the current source is a PMOS M_2 whose gate is held at a bias voltage V_b . Unlike the triode-load case, M_2 is now kept in **saturation**, where it behaves as a current source of output resistance r_{o2} .



Both transistors must be in saturation, which bounds the output swing:

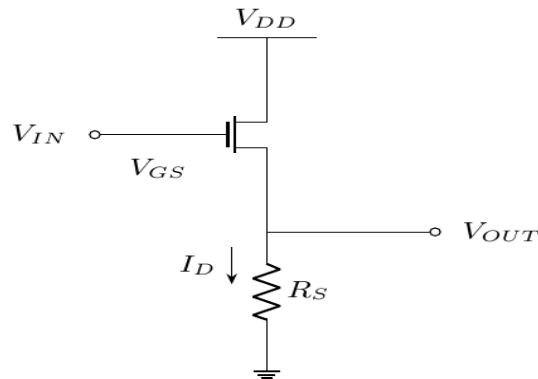
$$V_{GS1} - V_{T1} < V_{out} < V_{DD} - |V_{DSP}|, \quad \text{with} \quad |V_{DSP}| > |V_{GS2}| - |V_{TP}|$$

In the small-signal model M_2 contributes only r_{o2} to AC ground, which appears in parallel with r_{o1} :



$$v_{out} = -g_{m1}v_{gs1}(r_{o1} \parallel r_{o2}) \quad \Rightarrow \quad A_v = -\underbrace{g_{m1}}_{G_m}(\underbrace{r_{o1} \parallel r_{o2}}_{R_{out}})$$

6. Common Drain Stage (Source Follower)



The input drives the gate and the output is taken from the source; the drain is tied to the supply (AC ground). Because the output follows the input, the stage is also called a **source follower**.

Large-signal behaviour

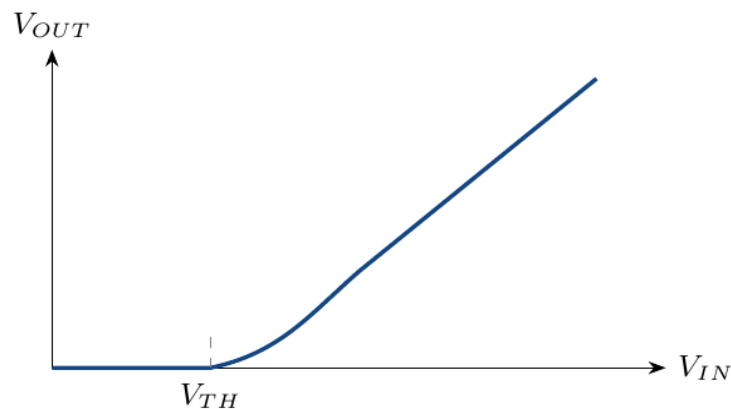
The gate-to-source voltage is whatever is left of the input after the output takes its share, and the output voltage is set by the drain current through R_S :

$$V_{IN} = V_{GS} + V_{OUT} \quad V_{OUT} = I_D R_S = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{IN} - V_{OUT} - V_{TH})^2 R_S$$

Writing $K = \frac{\mu_n C_{ox}}{2} \frac{W}{L}$ and rearranging:

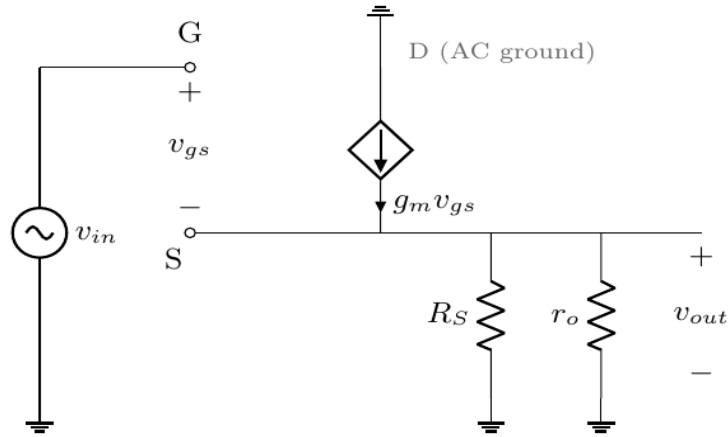
$$\sqrt{\frac{V_{OUT}}{KR_S}} + V_{OUT} = V_{IN} - V_{TH}$$

For $V_{IN} < V_{TH}$ no current flows and $V_{OUT} = 0$; above threshold the output rises, staying roughly one V_{GS} below the input.



Voltage gain

With the drain at AC ground, r_o appears from the source to ground, directly in parallel with R_S .



Now $v_{gs} = v_{in} - v_{out}$, and the dependent source current flows down into $R_S \parallel r_o$:

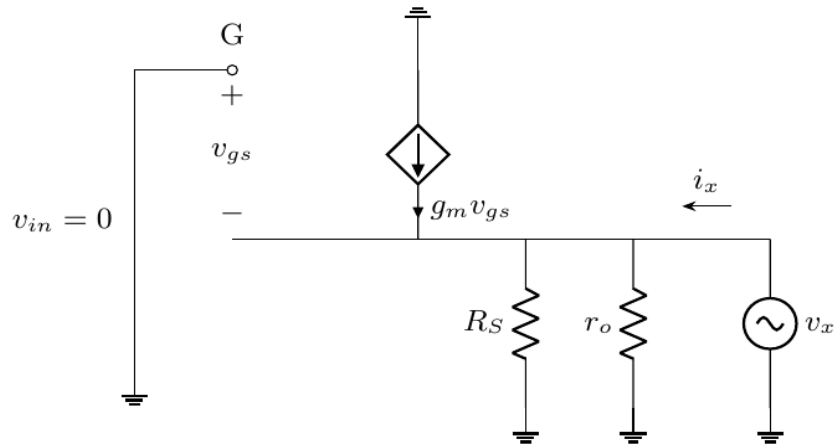
$$v_{out} = (R_S \parallel r_o) g_m v_{gs} = (R_S \parallel r_o) g_m (v_{in} - v_{out})$$

$$v_{out} [1 + g_m (r_o \parallel R_S)] = g_m (r_o \parallel R_S) v_{in}$$

$$A_v = \frac{v_{out}}{v_{in}} = \frac{g_m (r_o \parallel R_S)}{1 + g_m (r_o \parallel R_S)} < 1$$

The gain is positive and approaches unity when $g_m (r_o \parallel R_S) \gg 1$.

Output resistance



With the gate grounded, $v_{gs} = 0 - v_x = -v_x$. The test current plus the dependent source current together flow through $r_o \parallel R_S$:

$$v_x = (i_x + g_m v_{gs}) (r_o \parallel R_S) = (i_x - g_m v_x) (r_o \parallel R_S)$$

$$v_x[1 + g_m(r_o \parallel R_S)] = i_x(r_o \parallel R_S)$$

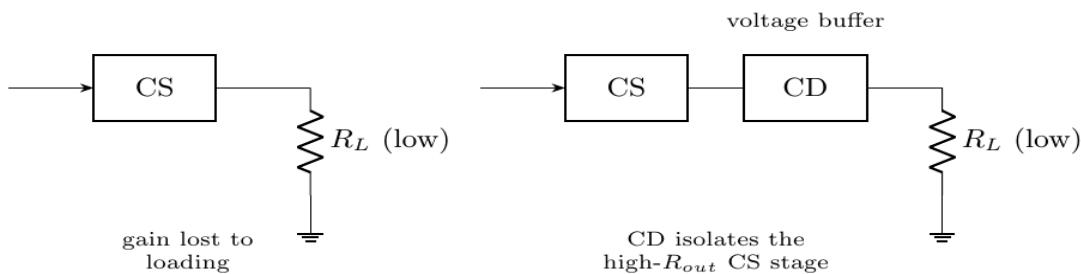
$$R_{out} = \frac{v_x}{i_x} = \frac{r_o \parallel R_S}{1 + g_m(r_o \parallel R_S)} = \frac{1}{g_m} \parallel r_o \parallel R_S \approx \frac{1}{g_m}$$

Checking with $A_v = -G_m R_{out}$. Short the output: then $v_{gs} = v_{in}$ and the transistor pushes $g_m v_{in}$ out of the source into the short, so the current flowing into the output node is $-g_m v_{in}$ and $G_m = -g_m$. Then $A_v = -G_m R_{out} = g_m \cdot \frac{r_o \parallel R_S}{1 + g_m(r_o \parallel R_S)}$, matching the direct result.

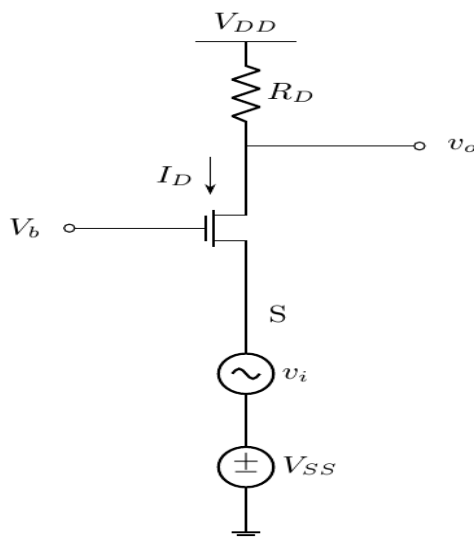
Why use a stage with gain below one?

$$A_v < 1 \quad R_{out} \text{ small} \quad R_{in} = \infty$$

A CS stage has a large output resistance, so connecting a small load resistor directly to it collapses the gain. Inserting a source follower between them solves this: its infinite input resistance does not load the CS stage, and its small output resistance can drive the low-resistance load with little loss. Used this way, the CD stage is a **voltage buffer**.



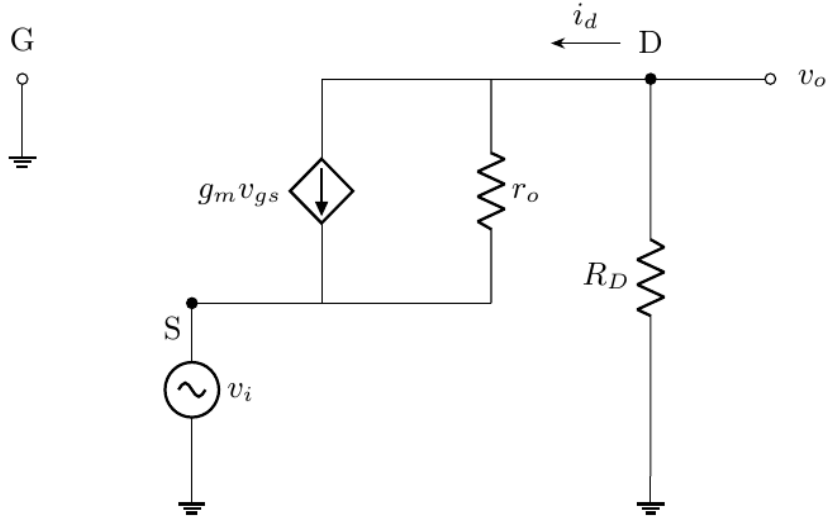
7. Common Gate Stage



Here the roles are swapped: the gate is held at a fixed bias V_b (AC ground), the signal enters at the source, and the output is taken from the drain. The DC relations are:

$$V_{DD} = I_D R_D + V_o, \quad V_o = V_{DS} + V_{SS}, \quad V_b = V_{GS} + V_{SS}$$

Voltage gain



Since the gate is at AC ground and the source carries v_i , the gate-source voltage is $v_{gs} = 0 - v_i = -v_i$.

Let i_d be the current flowing from the output node down into the drain. The current in r_o is what remains after the dependent source takes its share, $i_d - g_m v_{gs}$, and the voltage across r_o is $v_o - v_i$:

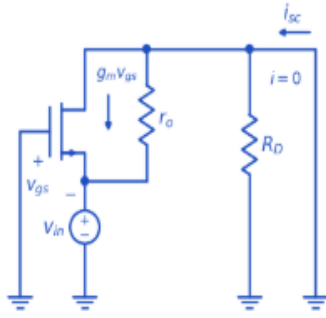
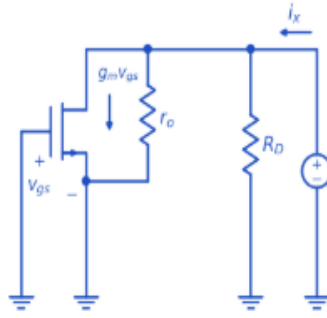
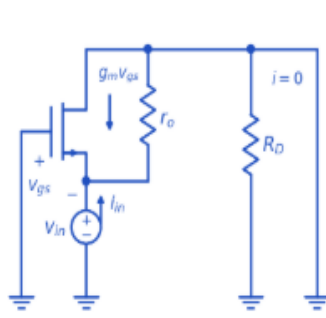
$$v_o = (i_d - g_m v_{gs})r_o + v_i = r_o i_d + g_m r_o v_i + v_i$$

The same i_d is drawn from AC ground through R_D , so $v_o = -i_d R_D$, i.e. $i_d = -v_o/R_D$. Substituting:

$$v_o = -\frac{r_o}{R_D} v_o + (1 + g_m r_o) v_i \quad \Rightarrow \quad v_o \left(1 + \frac{r_o}{R_D} \right) = (1 + g_m r_o) v_i$$

$$A_v = \frac{v_o}{v_i} = \frac{1 + g_m r_o}{1 + \frac{r_o}{R_D}} = \frac{(1 + g_m r_o) R_D}{r_o + R_D}$$

The gain is **positive** (no inversion). As $r_o \rightarrow \infty$ it reduces to $A_v \approx g_m R_D$.

Transconductance	O/P Resistance	I/P Resistance
$G_m = \frac{i_{sc}}{V_{in}} \Big _{v_{out}=0}$  $(i_{sc} - g_m v_{gs}) r_o + v_{in} = 0$ $v_{gs} = -v_{in}$ $i_{sc} r_o + v_{in} (1 + g_m r_o) = 0$ $\frac{i_{sc}}{V_{in}} = -\frac{(1 + g_m r_o)}{r_o}$	$R_{out} = \frac{v_x}{i_x} \Big _{v_{in}=0}$  $R_{out} = (r_o \parallel R_D)$ $A_v = -G_m R_{out}$ $= \frac{(1 + g_m r_o) (r_o \parallel R_D)}{r_o}$	$R_{in} = \frac{v_{in}}{i_{in}} \Big _{v_{out}=0}$  $v_{gs} = -v_{in}$ $v_{in} = r_o (g_m v_{gs} + i_{in})$ $v_{in} = r_o g_m (-v_{in}) + r_o i_{in}$ $\frac{v_{in}}{i_{in}} = \frac{r_o}{1 + g_m r_o} = \left(\frac{1}{g_m} \parallel r_o \right)$

Transconductance

Short the output ($v_{out} = 0$). The voltage across r_o is now $0 - v_i$, and with i_{sc} the current into the output node:

$$(i_{sc} - g_m v_{gs}) r_o + v_i = 0, \quad v_{gs} = -v_i$$

$$i_{sc} r_o + v_i (1 + g_m r_o) = 0 \quad \Rightarrow \quad G_m = \frac{i_{sc}}{v_i} = -\frac{1 + g_m r_o}{r_o}$$

Output resistance

With the input source killed, the source terminal is grounded, so $v_{gs} = 0$ and the dependent source is off. The test source sees r_o (to ground through the source) in parallel with R_D :

$$R_{out} = \frac{v_x}{i_x} \Big|_{v_{in}=0} = r_o \parallel R_D$$

The gain via the shortcut is then

$$A_v = -G_m R_{out} = \frac{1 + g_m r_o}{r_o} (r_o \parallel R_D) = \frac{(1 + g_m r_o) R_D}{r_o + R_D}$$

agreeing with the direct derivation.

Input resistance

Looking into the source with the drain at AC ground ($v_{out} = 0$), apply v_{in} and find the current i_{in} it drives into the source. With $v_{gs} = -v_{in}$, the voltage across r_o is v_{in} and its current is $g_m v_{gs} + i_{in}$:

$$v_{in} = r_o(g_m v_{gs} + i_{in}) = r_o(-g_m v_{in}) + r_o i_{in} \quad \Rightarrow \quad v_{in}(1 + g_m r_o) = r_o i_{in}$$

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{r_o}{1 + g_m r_o} = \frac{1}{g_m} \parallel r_o \approx \frac{1}{g_m}$$

Unlike the CS and CD stages, the common gate stage has a **low** input resistance.

8. Summary

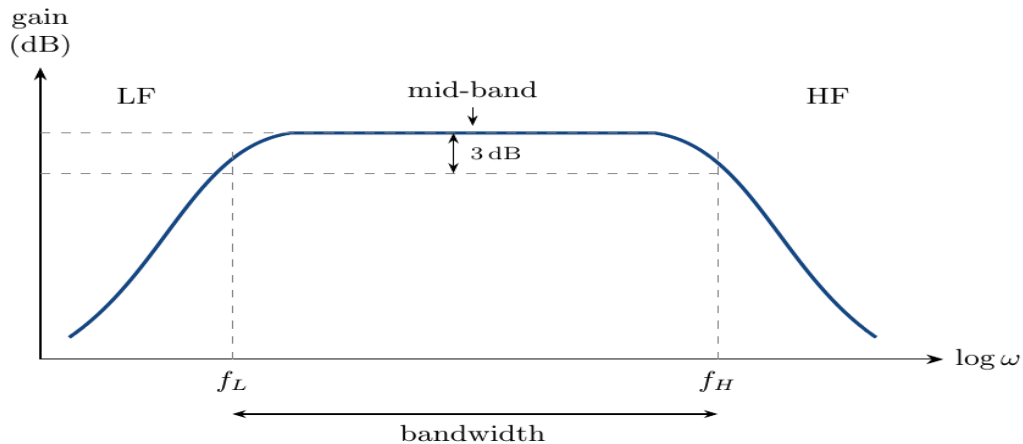
Stage	A_v	R_{in}	R_{out}
CS, resistive load	$-g_m(r_d \parallel R_D)$	∞	$r_d \parallel R_D$
CS, diode load	$-g_{m1} \left[r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2} \right) \right]$	∞	$r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2} \right)$
CS, triode load	$-g_{m1}(r_{o1} \parallel R_{on2})$	∞	$r_{o1} \parallel R_{on2}$
CS, ideal current source	$-g_{m1} r_{o1}$	∞	r_{o1}
CS, PMOS current source	$-g_{m1}(r_{o1} \parallel r_{o2})$	∞	$r_{o1} \parallel r_{o2}$
Common drain	$\frac{g_m(r_o \parallel R_S)}{1 + g_m(r_o \parallel R_S)}$	∞	$\frac{1}{g_m} \parallel r_o \parallel R_S$
Common gate	$\frac{(1 + g_m r_o) R_D}{r_o + R_D}$	$\frac{1}{g_m} \parallel r_o$	$r_o \parallel R_D$

Building Intuition: A Second Look

Frequency Response of MOSFET Amplifiers

1. The Shape of the Response

The gain of a MOSFET amplifier is constant only over a middle band of frequencies. It falls off at both ends, but for two quite different reasons.



At low frequencies the gain is reduced by the **external** capacitors deliberately placed in the circuit:

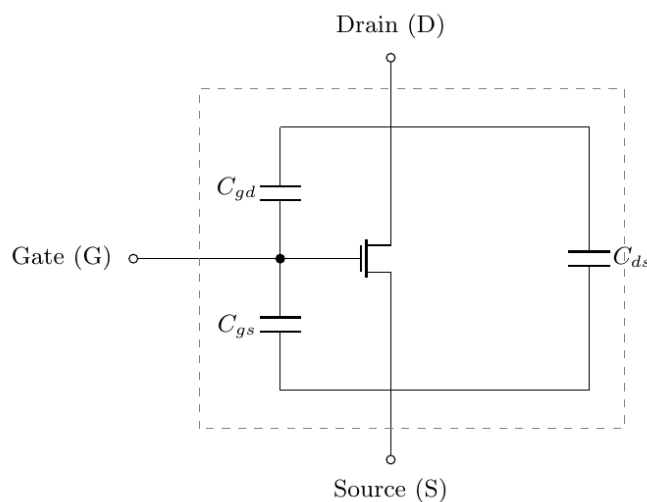
- coupling capacitors
- bypass capacitors

At high frequencies the gain is reduced by capacitances that cannot be avoided:

- internal (parasitic) capacitances of the device: C_{gs} , C_{gd} , C_{ds}
- wiring and lead capacitances

f_L and f_H are the frequencies at which the gain has fallen 3 dB below its mid-band value. The **bandwidth** is $f_H - f_L$.

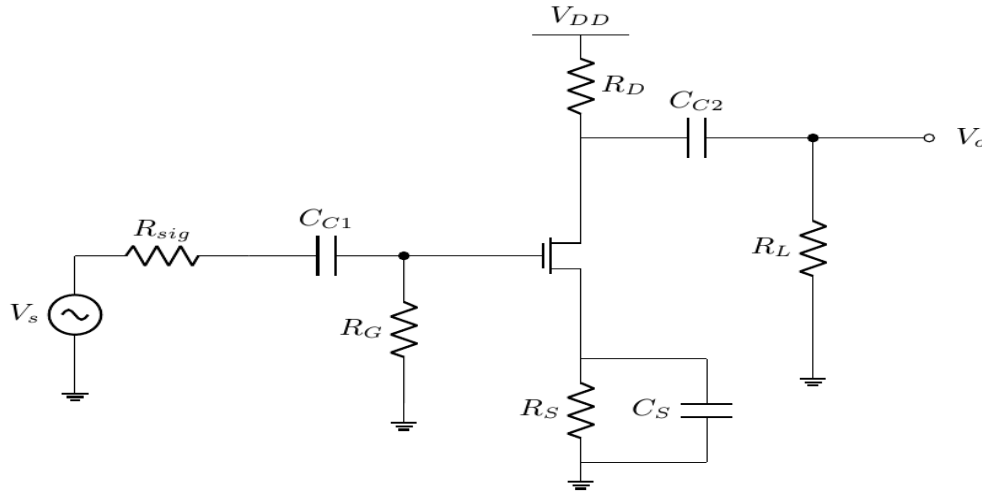
Where the internal capacitances sit



2. Low Frequency Response

The amplifier below uses two coupling capacitors and one bypass capacitor. C_{C1} couples the source to the gate, C_{C2} couples the drain to the load, and C_S bypasses the source resistor.

Notation. R_{sig} is the resistance of the signal source and R_S is the source (degeneration) resistor — both are often written simply as R_S , which can be confusing, so they are kept distinct here.



Each capacitor produces one low-frequency pole:

$$f_L = \frac{1}{2\pi RC}$$

where R is the **equivalent resistance seen across that capacitor**, with all the *other* capacitors shorted.

Because the highest pole is the one that determines where the gain has already fallen,

$$f_L = \max(f_{L1}, f_{L2}, f_{L3})$$

f_{L1} — due to the input coupling capacitor C_{C1}

Looking left and right from C_{C1} , the signal source resistance is in series with the gate bias resistor (the gate itself draws no current):

$$R = R_{sig} + R_G \quad \Rightarrow \quad f_{L1} = \frac{1}{2\pi(R_{sig} + R_G)C_{C1}}$$

f_{L2} — due to the output coupling capacitor C_{C2}

On the drain side the capacitor sees r_d in parallel with R_D , and on the load side it sees R_L :

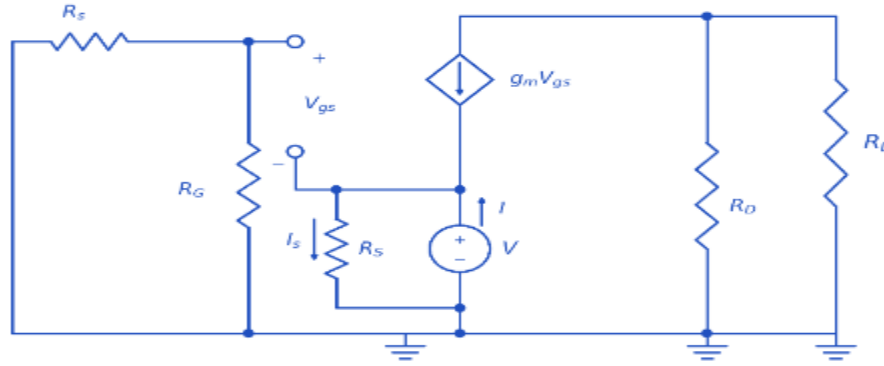
$$R = (r_d \parallel R_D) + R_L$$

$$f_{L2} = \frac{1}{2\pi[(r_d \parallel R_D) + R_L]C_{C2}} \approx \frac{1}{2\pi(R_D + R_L)C_{C2}}$$

the approximation holding because r_d is normally much larger than R_D .

f_{L3} — due to the bypass capacitor C_S

This one needs a little work, because the transistor itself is part of the resistance seen by C_S . Short the other capacitors (so the gate is at AC ground) and apply a test voltage V at the source node.



With the gate grounded and the source raised to V :

$$V_{gs} = 0 - V = -V$$

The test source must supply the current through R_S plus the current the transistor pulls out of the node. The dependent source contributes $g_m V_{gs} = -g_m V$, which flows *into* the source node, so:

$$I = \frac{V}{R_S} + g_m V = V \left(g_m + \frac{1}{R_S} \right)$$

$$\frac{V}{I} = R = \frac{1}{g_m + \frac{1}{R_S}} = R_S \parallel \frac{1}{g_m}$$

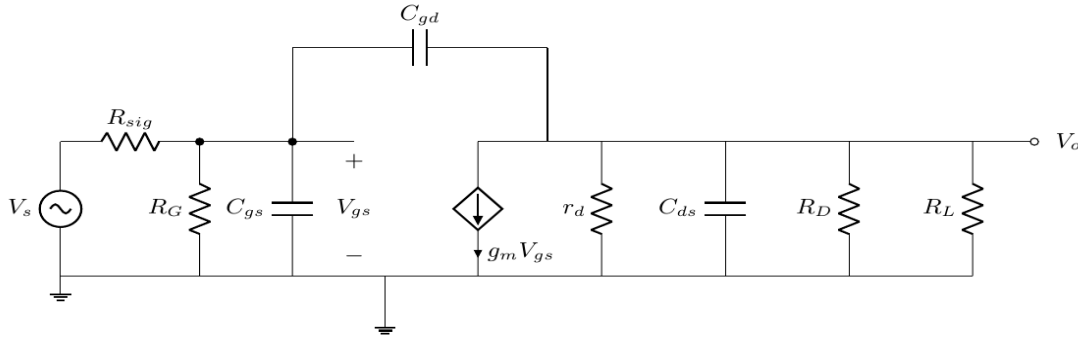
$$f_{L3} = \frac{1}{2\pi \left(R_S \parallel \frac{1}{g_m} \right) C_S}$$

Since $R_S \parallel 1/g_m$ is a *small* resistance, f_{L3} tends to be the highest of the three poles — which is why the bypass capacitor usually sets f_L , and why it has to be made large.

3. High Frequency Response

At high frequencies the roles reverse: the coupling and bypass capacitors are so large that they behave as short circuits, while the small internal capacitances C_{gs} , C_{gd} and C_{ds} can no longer be ignored.

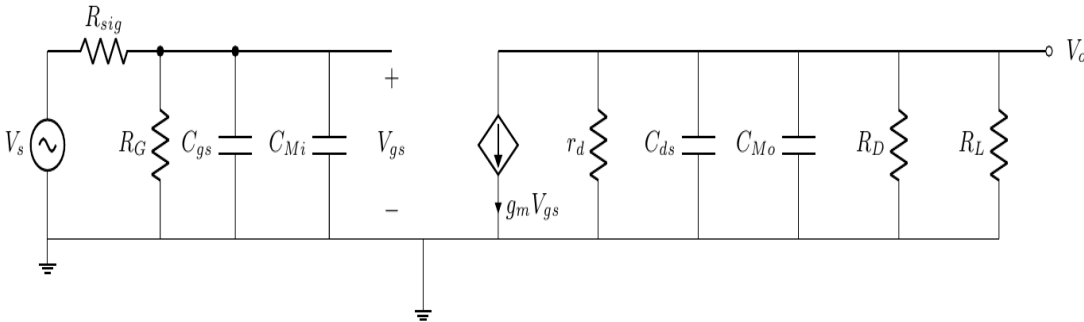
Short C_{C1} , C_{C2} and C_S ; retain C_{gs} , C_{gd} and C_{ds} .



Splitting C_{gd} with Miller's theorem

C_{gd} is awkward because it bridges the input and the output. Miller's theorem replaces it with two grounded capacitors, one at each end:

$$C_{Mi} = C_{gd}(1 - A_v) \quad C_{Mo} = C_{gd}\left(1 - \frac{1}{A_v}\right)$$



Now the input and output nodes each carry a single lumped capacitance:

$$C_i = C_{gs} + C_{Mi} \quad C_o = C_{ds} + C_{Mo}$$

Input node.

$$R_i = R_{sig} \parallel R_G \quad f_{H1} = \frac{1}{2\pi R_i C_i}, \quad C_i = C_{gs} + C_{gd}(1 - A_v)$$

Output node.

$$R_o = r_d \parallel R_D \parallel R_L \quad f_{H2} = \frac{1}{2\pi R_o C_o}, \quad C_o = C_{ds} + C_{gd} \left(1 - \frac{1}{A_v}\right)$$

The lower of the two poles is the one that starts rolling the gain off first:

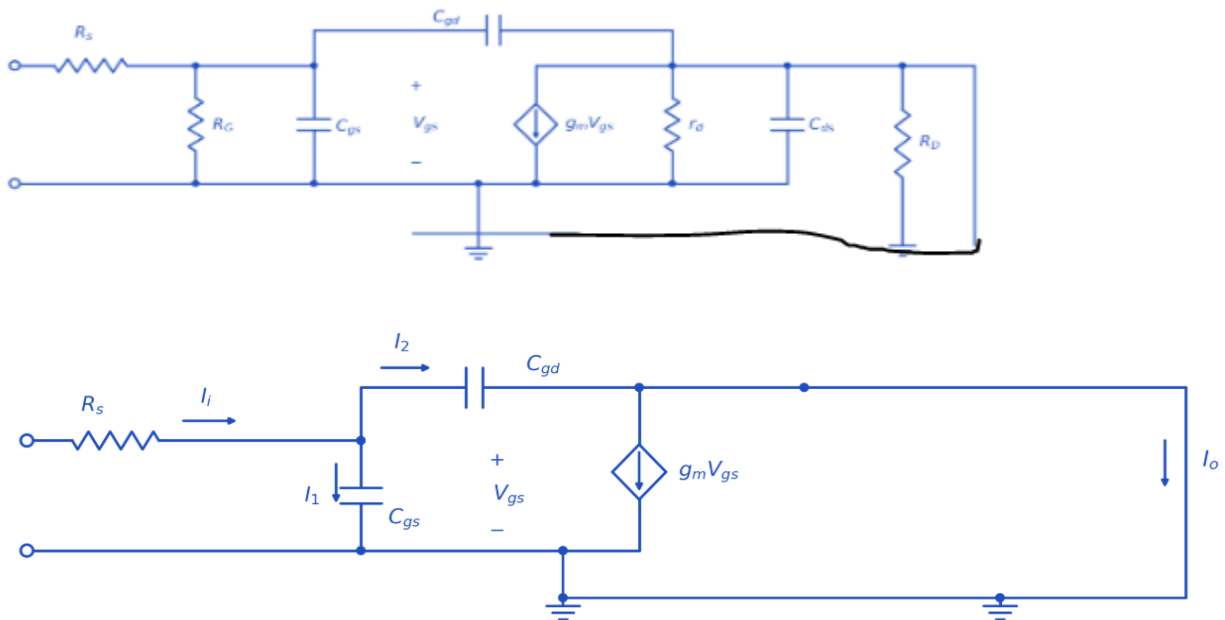
$$f_H = \min(f_{H1}, f_{H2})$$

Since A_v is large and negative for a CS stage, $(1 - A_v)$ is large and C_{Mi} is much bigger than C_{gd} itself. This **Miller multiplication** of C_{gd} is usually what limits the bandwidth of a common source amplifier. Note also that $C_{Mo} \approx C_{gd}$ when $|A_v| \gg 1$.

4. Unity Gain Bandwidth f_T

f_T is the frequency at which the **short-circuit current gain** of the common source stage falls to unity.

To find it, drive the gate with a current and short the drain to AC ground, so that the output current can be measured directly.



Input current

The input current splits between C_{gs} and C_{gd} . With the output shorted, both capacitors have V_{gs} across them:

$$I_1 = \frac{V_{gs}}{\frac{1}{j\omega C_{gs}}} = j\omega C_{gs} V_{gs} \quad I_2 = \frac{V_{gs} - 0}{\frac{1}{j\omega C_{gd}}} = j\omega C_{gd} V_{gs}$$

$$I_i = I_1 + I_2 = j\omega (C_{gs} + C_{gd}) V_{gs}$$

Output current

At the shorted output, the current I_2 arriving through C_{gd} works against the transistor's own current $g_m V_{gs}$:

$$I_o = I_2 - g_m V_{gs} = j\omega C_{gd} V_{gs} - g_m V_{gs} = -(g_m - j\omega C_{gd}) V_{gs}$$

Current gain

$$A_I = \frac{I_o}{I_i} = \frac{-(g_m - j\omega C_{gd})}{j\omega(C_{gs} + C_{gd})} \approx \frac{-g_m}{j\omega(C_{gs} + C_{gd})}$$

the approximation holding because $g_m \gg \omega C_{gd}$ at any frequency of interest.

Setting $|A_I| = 1$ at $\omega = \omega_T$:

$$1 = \frac{g_m}{\omega_T(C_{gs} + C_{gd})} \quad \Rightarrow \quad \omega_T = \frac{g_m}{C_{gs} + C_{gd}} \quad f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})}$$

f_T depends only on the device — its transconductance and its input capacitances — and not on the surrounding circuit. It is therefore the standard figure of merit for comparing the speed of transistors and of a fabrication technology.

5. Summary

Pole	Caused by	Expression
f_{L1}	input coupling C_{C1}	$\frac{1}{2\pi(R_{sig} + R_G)C_{C1}}$
f_{L2}	output coupling C_{C2}	$\frac{1}{2\pi[(r_d \parallel R_D) + R_L]C_{C2}}$
f_{L3}	source bypass C_S	$\frac{1}{2\pi(R_S \parallel 1/g_m)C_S}$
f_{H1}	input node, $C_i = C_{gs} + C_{Mi}$	$\frac{1}{2\pi(R_{sig} \parallel R_G)C_i}$
f_{H2}	output node, $C_o = C_{ds} + C_{Mo}$	$\frac{1}{2\pi(r_d \parallel R_D \parallel R_L)C_o}$
f_T	device only	$\frac{g_m}{2\pi(C_{gs} + C_{gd})}$

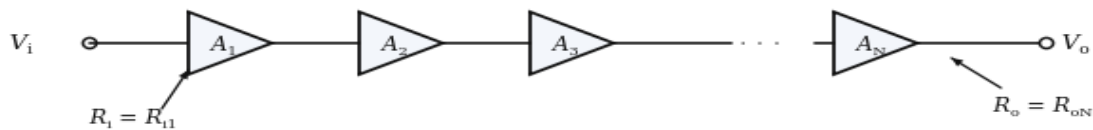
$$f_L = \max(f_{L1}, f_{L2}, f_{L3})$$

$$f_H = \min(f_{H1}, f_{H2})$$

MULTISTAGE AMPLIFIERS — CASCADING, DARLINGTON & CASCODE, CURRENT MIRRORS

CASCADING

★ Cascading refers to connecting the **output of one amplifier as the input to another amplifier**.



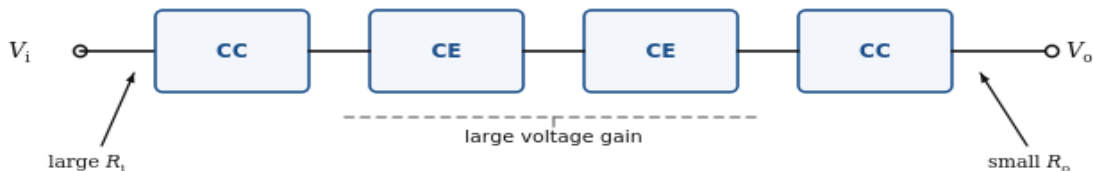
CASCADED OR MULTISTAGE AMPLIFIER

Need of cascading

Amplifiers are cascaded:

- to obtain **greater gain** or better amplification;
- to obtain **desired values of input and output impedances**.

★ For example, the following cascade of CC and CE stages can be used to obtain a large voltage gain, a large R_i and a smaller R_o :



★ High voltage gain is achieved by cascading **CE amplifiers**.

When amplifiers are cascaded

- The overall input resistance is the same as the input resistance of the **first** amplifier.
- The overall output resistance is the same as the output resistance of the **last** amplifier.
- The overall voltage gain is the **multiplication** of the voltage gains of the individual amplifiers.

$$A_V = A_{V1} \times A_{V2} \times A_{V3} \times \dots \times A_{VN}$$

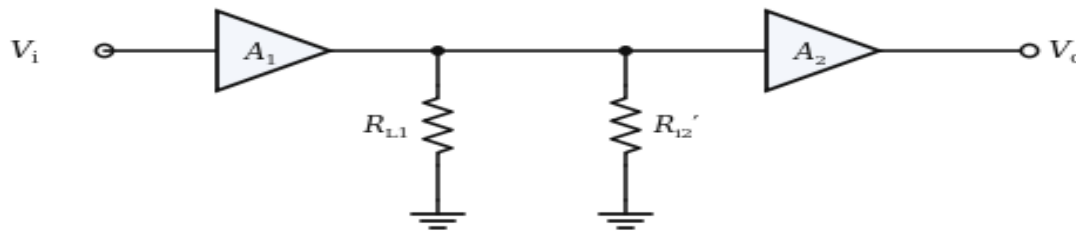
NOTE

$A_{V1}, A_{V2}, \dots$ are calculated by considering the **loading effect**.

LOADING EFFECT

★ It occurs in multistage amplifiers.

★ The loading effect refers to the **decrease in voltage gain of one amplifier due to the input impedance of the next amplifier.**



★ If two amplifiers A_1 and A_2 are cascaded, then the net load for A_1 is calculated by including the effect of the input impedance of A_2 :

$$R_{L1}' = R_{L1} \parallel R_{i2}'$$

net load for A_1

Case 1 — R_{i2}' is small or medium (Ω or $k\Omega$)

$$R_{L1}' = R_{L1} \parallel R_{i2}' \Rightarrow R_{L1}' \text{ decreases}$$

Hence, since $A_V \propto R_L'$, the voltage gain of A_1 reduces.

★ This decrease in the voltage gain of A_1 due to the input impedance of A_2 is known as the **LOADING EFFECT.**

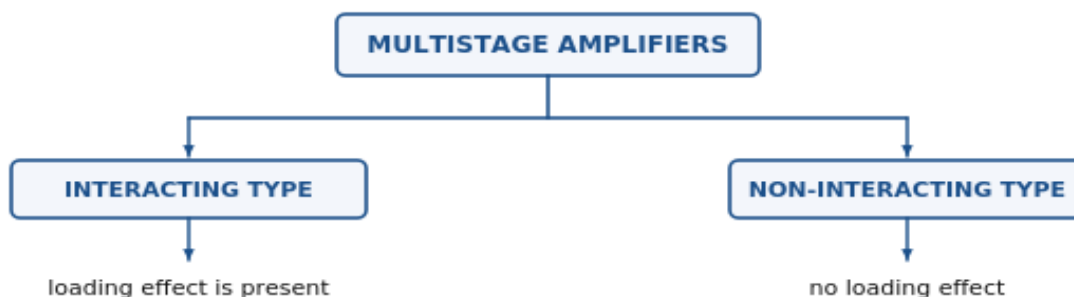
★ If amplifiers having low and medium input resistances are cascaded, then the voltage gain of the individual amplifiers will decrease, i.e. the loading effect will occur. Such amplifiers are called **INTERACTING AMPLIFIERS.**

Case 2 — R_{i2}' is large ($M\Omega$)

$$R_{L1}' \approx R_{L1}$$

★ The voltage gain of A_1 does not decrease, i.e. the loading effect does not occur.

★ Hence, if amplifiers having **high input impedance** are cascaded, the loading effect does not occur, and such amplifiers are called **NON-INTERACTING AMPLIFIERS.**



EFFECT OF CASCADING ON CUTOFF FREQUENCIES

- ★ Consider a cascade of N identical **non-interacting** amplifiers.
- ★ Assume that each amplifier has a lower cutoff f_{L1} and a higher cutoff f_{H1} .

Case 1 — effect on the lower cutoff frequency

- ★ For a single stage amplifier, the low frequency voltage gain is given as:

$$A_{VL1} = \frac{A_{VM1}}{\sqrt{1 + \left(\frac{f_{L1}}{f}\right)^2}} \quad \{\text{voltage gain of one amplifier}\}$$

- ★ If N such amplifiers are cascaded, then the overall voltage gain is the product of the individual gains:

$$A_{VL} = A_{VL1} \times A_{VL1} \times \dots (N \text{ times}) = (A_{VL1})^N$$
$$A_{VL} = \frac{A_{VM}}{\left[1 + \left(\frac{f_{L1}}{f}\right)^2\right]^{N/2}} ; A_{VM} = (A_{VM1})^N \dots (i)$$

- ★ Assume that the overall lower cutoff frequency of the cascade is f_L . Hence at $f = f_L$:

$$A_{VL} = \frac{A_{VM}}{\sqrt{2}}$$

Applying the above condition to equation (i):

$$\begin{aligned} \frac{A_{VM}}{\sqrt{2}} &= \frac{A_{VM}}{\left[1 + \left(\frac{f_{L1}}{f_L}\right)^2\right]^{N/2}} \\ \Rightarrow \left[1 + \left(\frac{f_{L1}}{f_L}\right)^2\right]^{N/2} &= \sqrt{2} \\ \Rightarrow \left[1 + \left(\frac{f_{L1}}{f_L}\right)^2\right]^N &= 2 \\ \Rightarrow 1 + \left(\frac{f_{L1}}{f_L}\right)^2 &= 2^{1/N} \\ \Rightarrow f_{L1} &= f_L \sqrt{2^{1/N} - 1} \end{aligned}$$

So,

$$f_L = \frac{f_{L1}}{\sqrt{2^{1/N} - 1}}$$

overall lower cutoff

★ For $N \geq 2 \Rightarrow \sqrt{2^{1/N} - 1} < 1$, hence:

$$f_L > f_{L1}$$

NOTE

The overall lower cutoff is greater than the individual lower cutoff frequencies, i.e. **cascading increases the lower cutoff frequency**.

Case 2 — effect on the higher cutoff frequency

$$A_{VH1} = \frac{A_{VM1}}{\sqrt{1 + \left(\frac{f}{f_{H1}}\right)^2}}$$

★ If N such amplifiers are cascaded, then:

$$A_{VH} = (A_{VH1})^N = \frac{(A_{VM1})^N}{\left[1 + \left(\frac{f}{f_{H1}}\right)^2\right]^{N/2}}$$

Let $(A_{VM1})^N = A_{VM}$, so:

$$A_{VH} = \frac{A_{VM}}{\left[1 + \left(\frac{f}{f_{H1}}\right)^2\right]^{N/2}} \dots (ii)$$

At $f = f_H$, $A_{VH} = \frac{A_{VM}}{\sqrt{2}}$. Applying this condition to equation (ii):

$$\begin{aligned} \left[1 + \left(\frac{f}{f_{H1}}\right)^2\right]^N &= 2 \\ \Rightarrow 1 + \left(\frac{f}{f_{H1}}\right)^2 &= 2^{1/N} \\ \Rightarrow \left(\frac{f}{f_{H1}}\right)^2 &= 2^{1/N} - 1 \\ \Rightarrow f_H &= f_{H1} \times \sqrt{2^{1/N} - 1} \end{aligned}$$

NOTE

$\sqrt{2^{1/N} - 1} < 1$ for $N \geq 2$, hence $f_H < f_{H1}$.

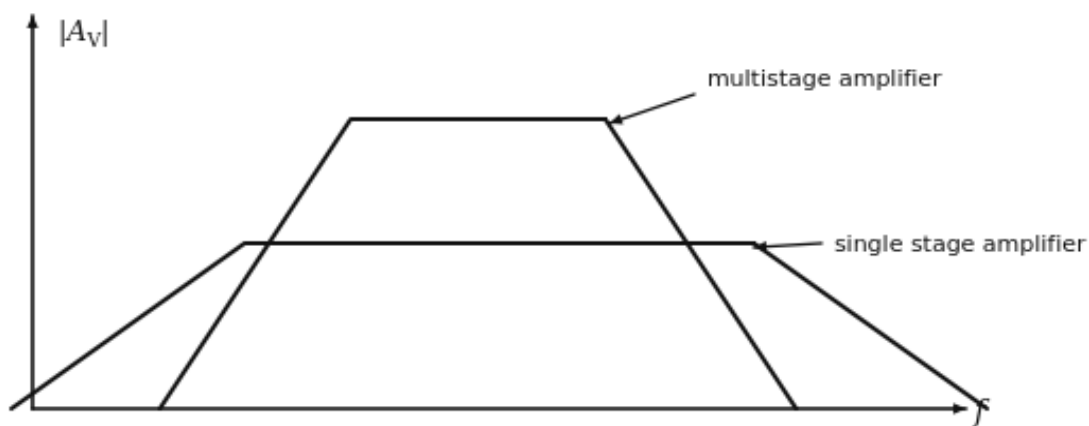
The overall higher cutoff is less than the individual higher cutoff frequency, i.e. **cascading decreases the higher cutoff frequency**.

NOTE

As f_L increases and f_H decreases, **cascading reduces the bandwidth**.

★ If N identical amplifiers having a bandwidth $(BW)_1$ are cascaded, then:

$$(BW)_{\text{overall}} = (BW)_1 \times \sqrt{2^{1/N} - 1}$$



Frequency response of a single stage and of a multistage amplifier

NOTE

$\sqrt{2^{1/N} - 1}$: for $N = 2$ it is $\sqrt{2^{1/2} - 1} \approx 0.645$, and for $N = 3$ it is $\sqrt{2^{1/3} - 1} \approx 0.5$; also $\frac{1}{\sqrt{2}} = 0.707$.

Cascade of interacting amplifiers

★ For a cascade of **interacting** amplifiers, the overall higher cutoff frequency is calculated from:

$$\frac{1}{f_H} = 1.1 \times \sqrt{\frac{1}{f_1^2} + \frac{1}{f_2^2} + \dots}$$

exact formula

$$\frac{1}{f_H} \approx \frac{1}{f_1} + \frac{1}{f_2} + \dots$$

approximate formula

★ $f_1, f_2, \dots$ are called **POLE FREQUENCIES**, which are obtained from the transfer function of the cascaded amplifier.

★ The overall lower cutoff is calculated from:

$$f_L = 1.1 \times \sqrt{f_{L1}^2 + f_{L2}^2 + \dots}$$

$f_{L1}, f_{L2}, \dots$ are the lower cutoff frequencies of the individual amplifiers

PROBLEM 1

Three identical non-interacting amplifiers having cutoff frequencies of 25 Hz and 40 kHz are cascaded. Calculate the overall cutoff frequencies of the cascade.

SOLUTION

$$f_L = \frac{f_{L1}}{\sqrt{2^{1/N} - 1}} = \frac{25}{\sqrt{2^{1/3} - 1}} = \frac{25}{0.5}$$

$$f_L \approx 50\text{Hz}$$

$$f_H = f_{H1} \times \sqrt{2^{1/3} - 1} = 40k \times 0.5$$

$$f_H \approx 20\text{kHz}$$

SPECIAL MULTISTAGE AMPLIFIERS

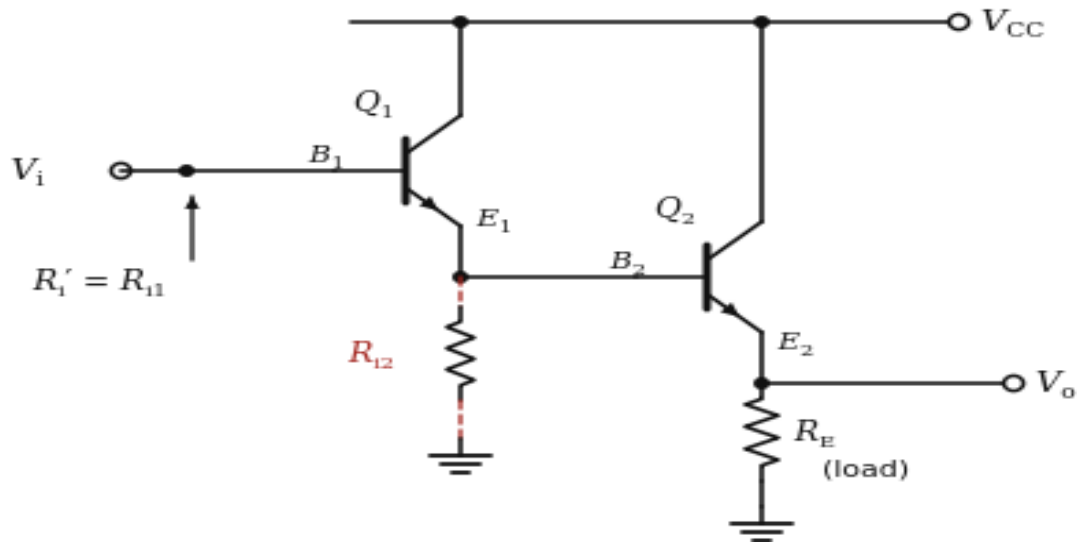
- Darlington amplifier
- Cascode amplifier

DARLINGTON AMPLIFIER

★ It is a cascade of two common collector amplifiers (CC – CC).

- very high input resistance (its most special property)
- unity voltage gain
- low output resistance
- high current gain

★ It is used as a **VOLTAGE BUFFER**.



DARLINGTON AMPLIFIER

★ Both Q_1 and Q_2 are in common collector configuration:

- R_E : load for Q_2
- R_{i2} : load for Q_1

$$R_{E1} = R_{i2}$$

$$R_{E1} = R_{i2} = h_{ie} + (1 + h_{fe})R_E$$

$$R_{E1} = R_{i2} \approx (1 + h_{fe})R_E \dots (i)$$

Also,

$$R_i = R_{i1} = h_{ie} + (1 + h_{fe})R_{E1}$$

$$\Rightarrow R_i = R_{i1} \approx (1 + h_{fe})R_{E1}$$

$$\Rightarrow R_i = R_{i1} \approx (1 + h_{fe}) \times (1 + h_{fe})R_E$$

$$\Rightarrow R_i = R_{i1} \approx (1 + h_{fe})^2 R_E$$

R_i is very large (tens of $M\Omega$)

NOTE

- $R_o = R_{o2} \approx \frac{h_{ie}}{1 + h_{fe}}$ (tens of Ω)
- $A_V = A_{V1} \times A_{V2} \approx 1 \times 1 \approx 1$
- $A_I = A_{I1} \times A_{I2} = (1 + h_{fe})(1 + h_{fe}) = (1 + h_{fe})^2$ if $h_{oe} \approx 0$

BOOTSTRAP BIASING

★ It is a special type of biasing used in the Darlington amplifier to **prevent a decrease in input resistance**.

★ If a self bias circuit is used in a Darlington amplifier, then the input resistance R_i' will decrease because of the biasing resistors R_1 and R_2 :

$$R_i' = R_i \parallel R_x$$

$$R_i \approx (1 + h_{fe})^2 R_E \{ \approx M\Omega \}$$

$$R_x = R_B = R_1 \parallel R_2 \{ \approx \text{hundreds of } k\Omega \text{ at most} \}$$

$$R_i' \approx R_x$$

at most a few hundred $k\Omega$

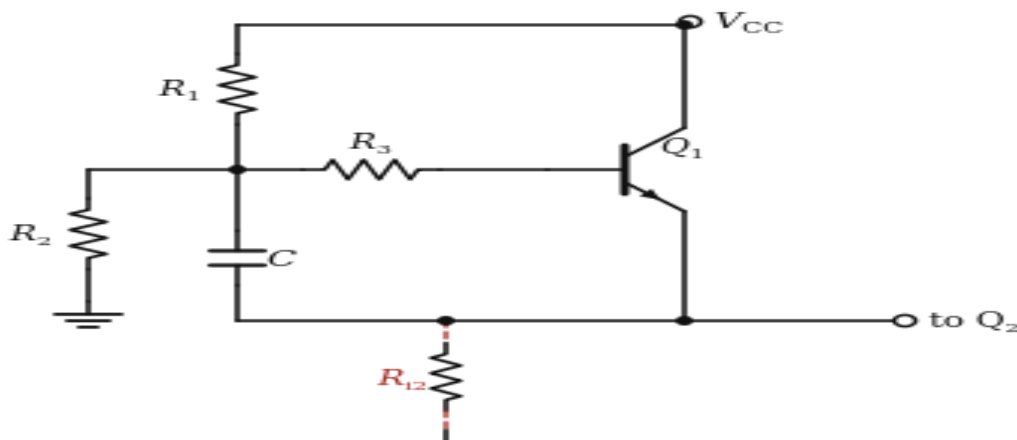
NOTE

R_1 and R_2 cannot be in $M\Omega$, because then I_B would become very small and the BJT could not conduct.

Also, if R_1 and R_2 are in $M\Omega$ they occupy a large area on the chip.

★ Hence the self bias circuit is **not** used in the Darlington amplifier.

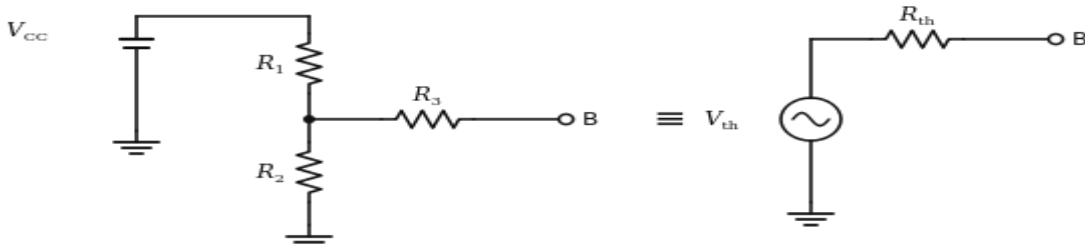
★ To prevent the decrease in input impedance, a **BOOTSTRAP BIASING CIRCUIT** is used.



Bootstrap biasing circuit

DC analysis

- The capacitor is replaced with an open circuit.
- The R_1 , R_2 , R_3 network is replaced by its Thevenin equivalent.



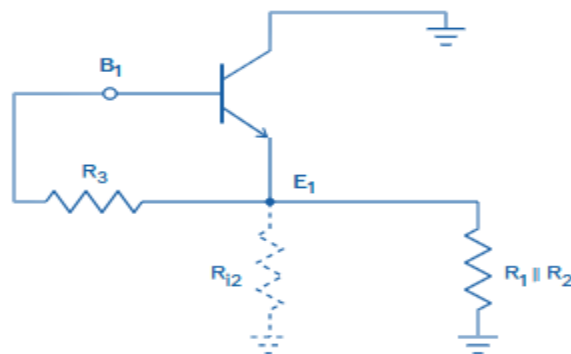
$$V_{th} = \frac{V_{CC} \times R_2}{R_1 + R_2}$$

$$R_{th} = R_3 + (R_1 \parallel R_2)$$

AC analysis

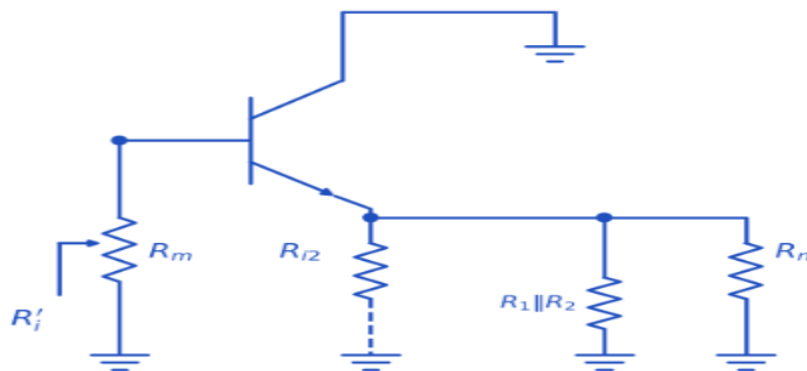
- The capacitor is replaced with a short circuit.
- V_{CC} is grounded.

★ Then R_1 and R_2 appear in parallel between the emitter and ground, whereas R_3 appears between the base and the emitter.



AC equivalent of the bootstrap biasing circuit

★ R_3 is between the input node B_1 and the output node E_1 . Hence R_3 can be replaced with R_m and R_n by using Miller's theorem.



where

$$R_m = \frac{R_3}{1 - A_V}$$

$$R_n = \frac{R_3}{1 - \frac{1}{A_V}}$$

★ For a CC amplifier $A_V \approx 1$. If $A_V \rightarrow 1$, then $R_m \approx \infty$ and $R_n \approx \infty$, i.e. both behave as **open circuits**.

$$R_i' = R_i \parallel R_x = R_i \parallel R_m \quad \{R_m \rightarrow \infty\}$$

$$R_i' \approx R_i$$

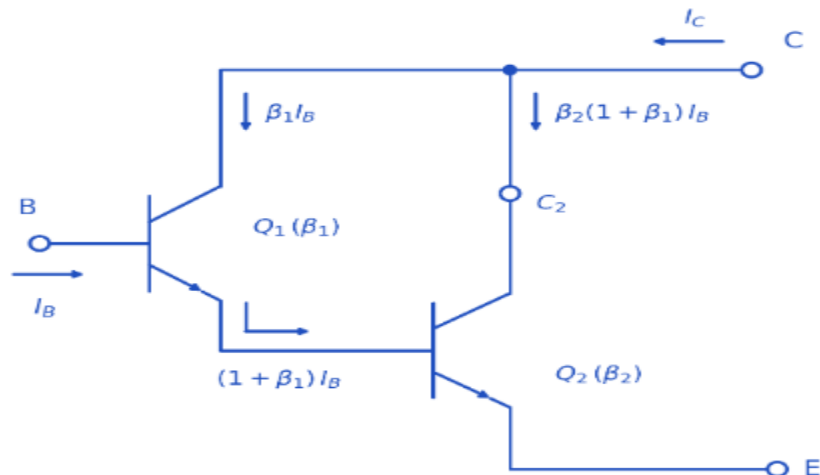
which is very high

NOTE

Hence bootstrap biasing is useful in preventing the decrease in R_i' .

DARLINGTON PAIR

★ It is a compounded BJT, i.e. an interconnection of two BJTs.



★ Applying KCL:

$$I_C = \beta_1 I_B + \beta_2 (1 + \beta_1) I_B$$

$$I_C = I_B [\beta_1 + \beta_2 (1 + \beta_1)]$$

$$I_C = I_B (\beta_1 + \beta_2 + \beta_1 \beta_2)$$

$$\frac{I_C}{I_B} = \beta_1 + \beta_2 + \beta_1\beta_2$$

overall β if node C is used as the collector

★ If C_2 is used as the collector, then:

$$I_{C2} = \beta_2(1 + \beta_1)I_B$$

$$\frac{I_{C2}}{I_B} = \beta_2(1 + \beta_1)$$

β if C_2 is the collector

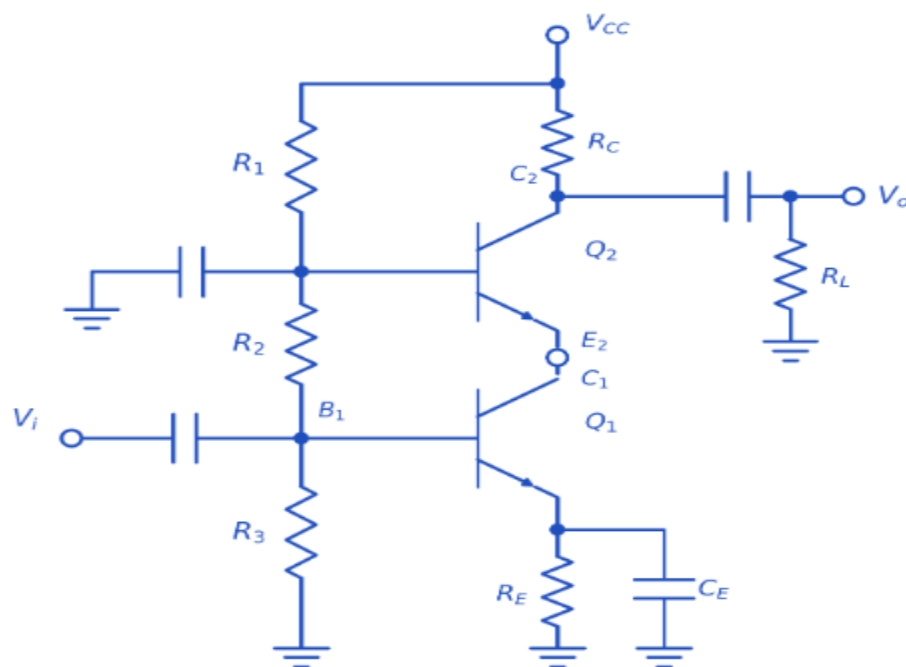
NOTE

- The Darlington pair has a very high β value. If a transistor of very high β is needed, then a Darlington pair is used.
- The Darlington pair is available as an IC and is sometimes also called a **SUPER- β BJT**.
- In a Darlington pair the overall transconductance is the same as the transconductance of the output BJT, i.e. $g_m = g_{m2}$.

CASCODE AMPLIFIER

★ It is a cascade of a CE – CB amplifier, or a cascade of a CS – CG amplifier.

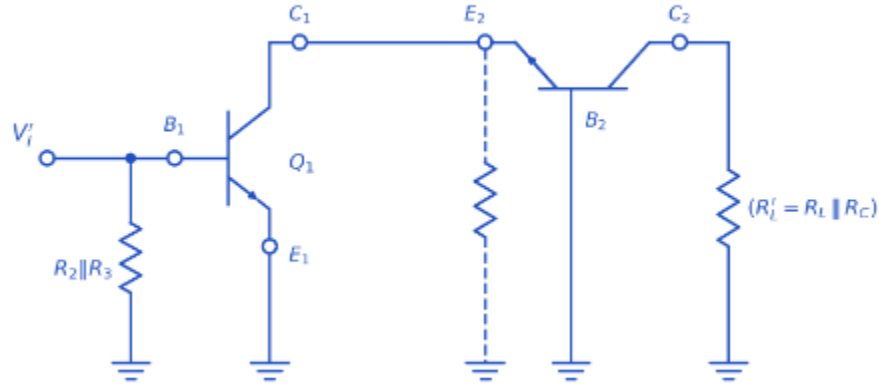
★ It has a **large bandwidth** (MHz), hence it is used as a **VIDEO FREQUENCY AMPLIFIER**.



CASCODE AMPLIFIER

NOTE

- Q_1 is in CE configuration and Q_2 is in CB configuration.
- The AC equivalent circuit is obtained by replacing the capacitors with short circuits and by grounding the V_{CC} node.



AC EQUIVALENT OF THE CASCODE

★ R_L' is the load for Q_2 and R_{i2} is the load for Q_1 . Hence:

$$R_{L1} = R_{i2} = \frac{h_{ie}}{1 + h_{fe}} \approx \frac{h_{ie}}{h_{fe}}$$

Analysis of the cascode

i) Input resistance

$$R_i = h_{ie} ;$$

$$R_i' = h_{ie} \parallel R_2 \parallel R_3 = R_i \parallel R_x$$

ii) Output resistance

$$R_o = R_{o2} = \infty ;$$

$$R_o' = R_o \parallel R_y = \infty \parallel R_C = R_C$$

iii) Current gain

$$A_I = A_{I1} \times A_{I2} = -h_{fe} \times 1$$

$$A_I = -h_{fe}$$

iv) Voltage gain

$$A_V = A_{V1} \times A_{V2} = \frac{-h_{fe}R_{L1}}{h_{ie}} \times \frac{h_{fe}R_L'}{h_{ie}}$$

$$\Rightarrow A_V = \frac{-h_{fe}}{h_{ie}} \times \frac{h_{ie}}{h_{fe}} \times \frac{h_{fe}}{h_{ie}} \times R_L'$$

$$\Rightarrow A_V = \frac{-h_{fe}}{h_{ie}} \times R_L'$$

NOTE

- A_I and A_V of the cascode amplifier are the same as those of a single stage CE amplifier. But the cascode amplifier has a much larger bandwidth in comparison to the CE amplifier.
- In a cascode amplifier the **MILLER EFFECT** becomes negligible, i.e. the increase in the input capacitance is negligible.

$$C_{in} = C_{\pi} + C_{\mu}(1 - A_{V1})$$

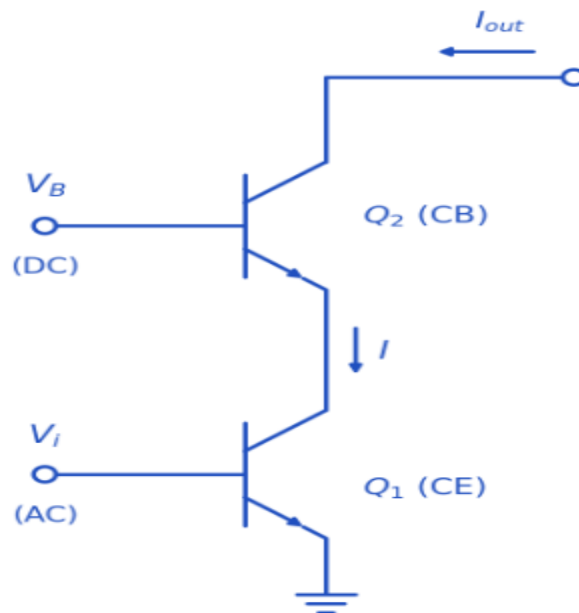
$$\text{As } \{A_{V1} = -1\}$$

$$C_{in} = C_{\pi} + C_{\mu}(1 + 1)$$

$$= C_{\pi} + 2C_{\mu} \approx C_{\pi}$$

★ Due to the negligible Miller effect, C_{in} remains smaller and hence a greater bandwidth can be achieved, since $BW \propto \frac{1}{C_{in}}$.

CASCODE PAIR



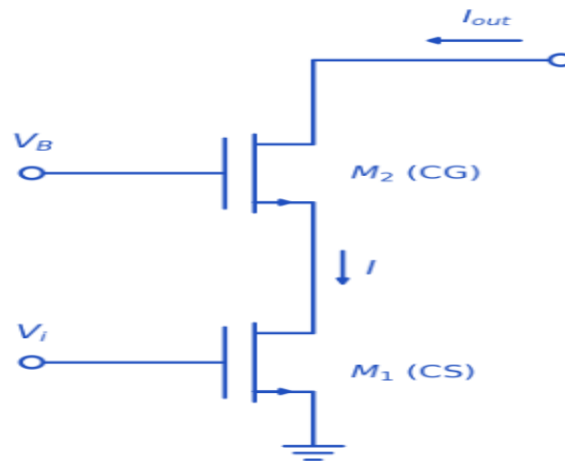
★ The transconductance of the cascode pair is defined as:

$$g_m = \frac{\partial I_{out}}{\partial V_i}$$
$$g_m = \frac{\partial I_{out}}{\partial I} \times \frac{\partial I}{\partial V_i}$$
$$g_m = \alpha_2 \times g_{m1}$$

★ If $\alpha_2 = 1$, then $g_m = g_{m1}$.

NOTE

The cascode pair can also be modelled using MOSFETs.



MOSFET CASCODE

$$g_m = \frac{\partial I_{out}}{\partial V_i} = \frac{\partial I_{out}}{\partial I} \times \frac{\partial I}{\partial V_i}$$
$$g_m = 1 \times g_{m1}$$

NOTE

In a cascode pair the overall transconductance is the same as the transconductance of the **input device**.

BJT BIASING IN INTEGRATED CIRCUITS

NOTE

For a PNP transistor both V_{BE} and V_{CE} are **negative**.

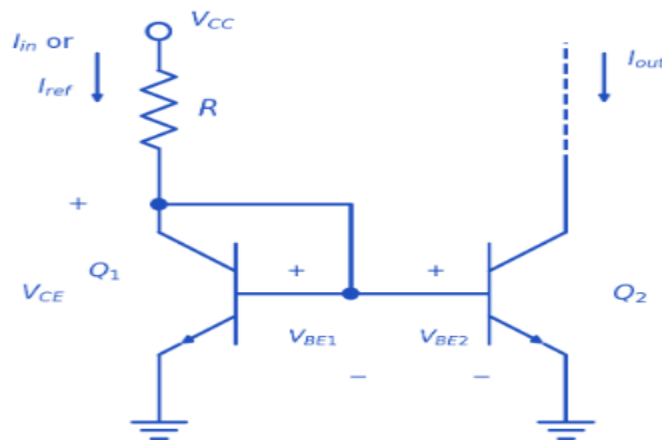
For an NPN transistor both V_{BE} and V_{CE} are **positive**.

★ In IC amplifiers, the BJT is biased by using **CURRENT MIRRORS**.

★ The self bias circuit is preferred in **discrete** design.

★ The self bias circuit is not preferred in ICs, because it has large resistances R_1 and R_2 which require a greater area on the silicon chip.

BASIC CURRENT MIRROR



Basic current mirror

NOTE

Q_1 and Q_2 are identical transistors.

ICs are made of silicon only.

★ For Q_1 , the collector is shorted to the base, i.e. $V_{CE} = V_{BE}$. Hence Q_1 always remains in the **ACTIVE REGION**.

$$V_{CB} = V_{CE} - V_{BE}$$

- if $V_{CE} \geq V_{BE} \Rightarrow V_{CB} \geq 0$, the BJT is in the active region;
- if $V_{CE} < V_{BE} \Rightarrow V_{CB} < 0$, the BJT is in the saturation region.

$$I_{in} = \frac{V_{CC} - V_{BE1}}{R}$$

★ V_{CC} , R and V_{BE1} are constant values, hence I_{in} remains constant.

★ For Q_1 and Q_2 the base terminals are shorted and the emitter terminals are also shorted. Hence their V_{BE} values become perfectly equal:

$$V_{BE1} = V_{BE2}$$

★ If two transistors are identical and their V_{BE} values are equal, then their currents will be equal:

$$I_{B1} = I_{B2} ; I_{C1} = I_{C2} \dots (i)$$

Now, by KCL:

$$I_{in} = I_{C1} + 2I_{B1}$$

$$\Rightarrow I_{C1} + 2 \frac{I_{C1}}{\beta}$$

$$\Rightarrow I_{in} = I_{C1} \left(\frac{2}{\beta} + 1 \right)$$

$$\Rightarrow I_{C1} = \frac{I_{in}}{1 + \frac{2}{\beta}} \dots (ii)$$

$$I_{C2} = I_{out} \dots (iii)$$

★ Combining equations (i), (ii) and (iii):

$$I_{out} = \frac{I_{in}}{1 + \frac{2}{\beta}}$$

★ If β is large for both transistors, then $\frac{2}{\beta}$ can be neglected:

$$I_{out} \approx I_{in}$$

NOTE

- Since the output current is a mirror image of the input current, the circuit is known as a **current mirror circuit**.
- As the input current is constant, the output current also remains constant, i.e. a current mirror supplies a constant output current. Therefore, it is also called a **CONSTANT CURRENT SOURCE CIRCUIT**.

★ The **output resistance** of a current mirror is the resistance seen between the collector of Q_2 and ground. It is the same as the dynamic output resistance of Q_2 :

$$R_{out} = r_o \text{ of } Q_2 = \frac{V_A + V_{CE}}{I_{out}}$$

V_A = Early voltage of the BJT (≈ 100 V);

as $V_A \gg V_{CE}$:

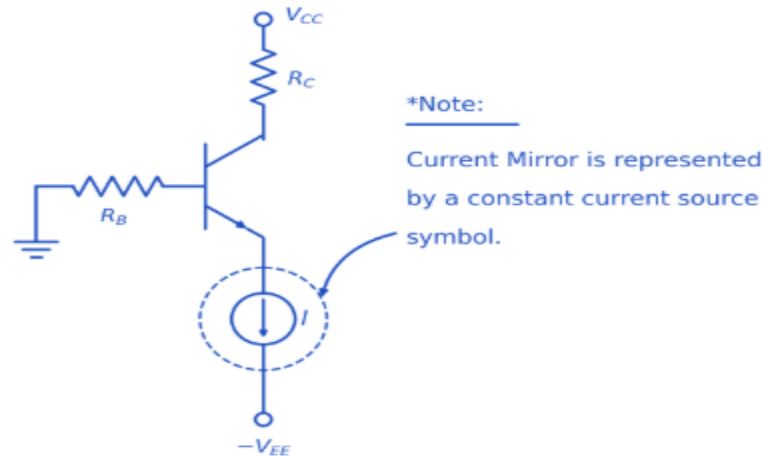
$$R_{out} = \frac{V_A}{I_{out}}$$

★ A current mirror has a large output resistance (hundreds of k Ω). For example, with $V_A = 100$ V and $I_{out} = 1$ mA:

$$R_{out} = \frac{100V}{1mA} = 100k\Omega$$

Applications of the current mirror

★ (i) The current mirror is used as a constant current source to bias a BJT in IC amplifiers.



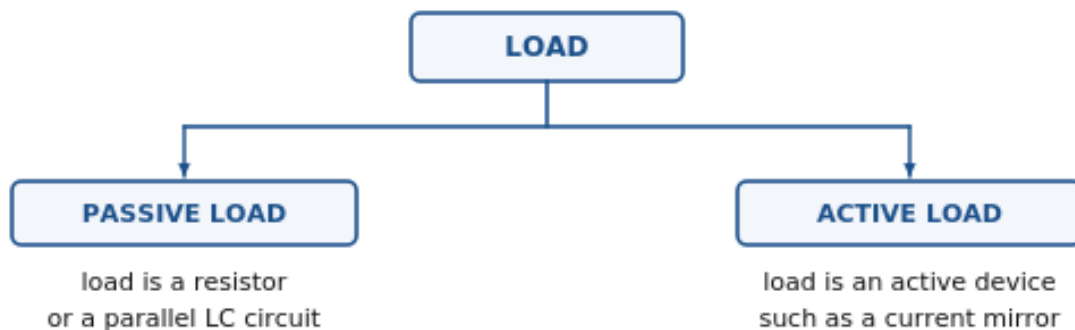
CURRENT MIRROR BIASING CIRCUIT

NOTE

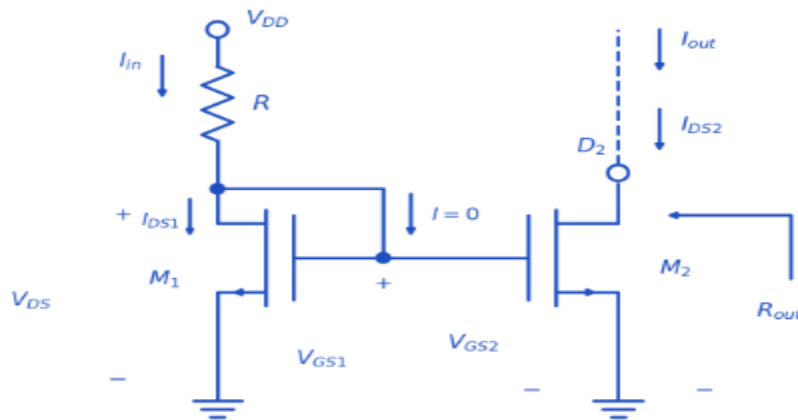
The current mirror is represented by a constant current source symbol.

★ (ii) The current mirror is used as an **ACTIVE LOAD** in IC amplifiers to achieve a high voltage gain. The output resistance of the current mirror acts as the load.

★ The voltage gain of an amplifier is proportional to the load resistance. Loads in amplifiers are of two types:



MOSFET CURRENT MIRROR



MOSFET current mirror

NOTE

M_1 and M_2 are identical MOSFETs.

★ For M_1 , the drain is shorted to the gate, i.e. $V_{DS} = V_{GS}$. Therefore, M_1 will always be in the **saturation region**, since the condition for a MOSFET to be in saturation is:

$$V_{DS} > V_{GS} - V_T$$

NOTE

- Saturation region (MOSFET) $\equiv$ active region (BJT).
- A MOSFET in which the drain is shorted to the gate will always be in the saturation region.

★ As $V_{GS2} = V_{GS1} \Rightarrow I_{DS2} = I_{DS1}$.

★ For a MOSFET the gate current $I_G = 0$, because the insulator SiO_2 is present between the gate and the channel.

$$I_{in} = I_{DS1} ;$$

$$I_{out} = I_{DS2}$$

$$I_{out} = I_{in}$$

NOTE

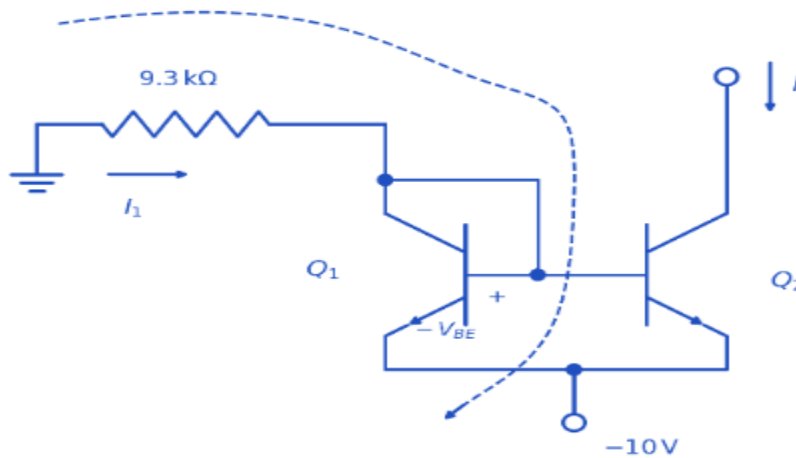
The output resistance of the MOSFET current mirror is the resistance seen between the drain of M_2 and ground. It is the same as the dynamic drain resistance of M_2 .

$$R_{\text{out}} = r_{\text{ds}} \text{ of } M_2 = \frac{1}{\lambda I_{\text{out}}}$$

λ = channel length modulation parameter

PROBLEM 4

In the circuit shown, the transistors have a large β and $V_{\text{BE}} = 0.7 \text{ V}$. Calculate the current I : (a) if Q_1 and Q_2 are identical; (b) if the emitter junction area of Q_1 is half that of Q_2 .



SOLUTION

By KVL:

$$\begin{aligned} 0 &= 9.3I_1 + V_{\text{BE}} - 10 \\ \Rightarrow I_1 &= \frac{10 - V_{\text{BE}}}{9.3} = \frac{10 - 0.7}{9.3} \\ \Rightarrow I_1 &= 1\text{mA} \end{aligned}$$

(a) If Q_1 and Q_2 are identical, the circuit becomes a current mirror. So:

$$I = I_1 = 1\text{mA}$$

(b) For a BJT, the currents are proportional to the area of the emitter junction. Since the emitter junction area of Q_2 is two times the area of Q_1 , the current in Q_2 will be two times the current in Q_1 :

$$I = 2I_1 = 2\text{mA}$$

NOTE

For a BJT, the emitter current can be calculated using the diode equation.

$$I_E = I_{ES}(e^{V_{BE}/V_T} - 1) \approx I_{ES} e^{V_{BE}/V_T}$$

I_{ES} : reverse saturation current of the emitter junction J_E

$$I_{ES} = \frac{A_e D_B}{W_B} \times \frac{n_i^2}{N_B}$$

- D_B : diffusion constant of the minority carrier in the base
- W_B : base width
- N_B : base doping
- A : area of the emitter junction J_E

Hence $I_{ES} \propto A$ and $I_E \propto I_{ES}$, so:

$$I_E \propto A$$

NOTE

All the other currents I_C and I_B are also proportional to the area, since in a BJT the current depends on I_E .

$$V_{BE} = V_T \ln \left(\frac{I_E}{I_{ES}} \right)$$

Also,

$$I_C = \alpha I_E + I_{CO} \approx \alpha I_E$$

$$I_C \approx \alpha I_{ES} e^{V_{BE}/V_T}$$

$$I_C \approx I_S e^{V_{BE}/V_T} ;$$

I_S : saturation current of the BJT (10^{-16} A to 10^{-13} A)

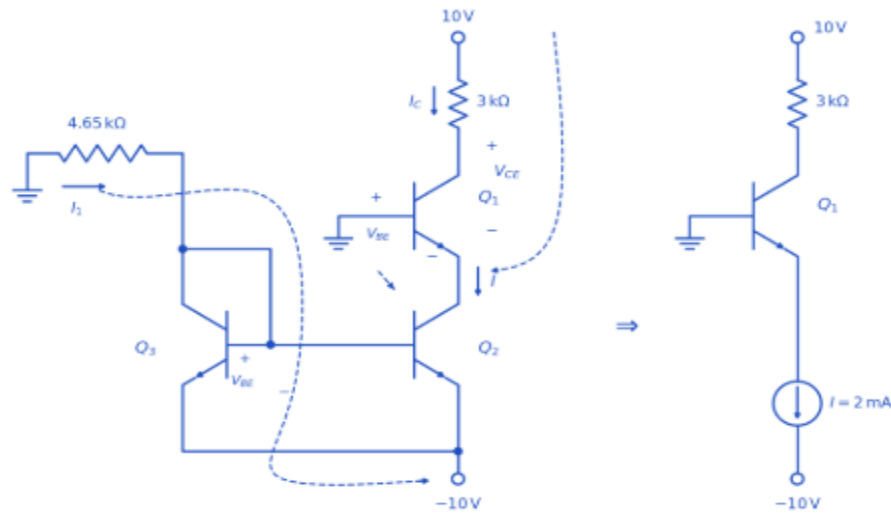
NOTE

If the BJT has the Early effect (base width modulation), then:

$$I_C = [I_S e^{V_{BE}/V_T}] \times \left(1 + \frac{V_{CE}}{V_A} \right)$$

PROBLEM 5

In the circuit shown, the transistors have a large β and $V_{BE} = 0.7$ V. Calculate the operating point of transistor Q_1 .



Q_2 and Q_3 form a current mirror which biases Q_1

SOLUTION

KVL in loop 1:

$$4.65I_1 + V_{BE} - 10 = 0$$

$$I_1 = \frac{10 - V_{BE}}{4.65} = \frac{10 - 0.7}{4.65} = 2\text{mA}$$

★ As a current mirror is formed, $I = I_1 = 2\text{ mA}$, so $I_C = I = 2\text{ mA}$.

KVL in loop 2:

$$10 = 3I_C + V_{CE} - V_{BE}$$

$$V_{CE} = 10 + V_{BE} - 3I_C$$

$$= 10 + 0.7 - 6 = 4.7\text{V}$$

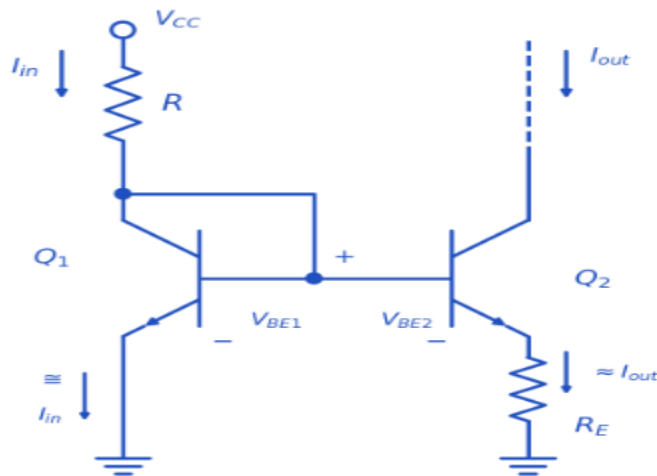
$$Q_1(4.7\text{V}, 2\text{mA})$$

WIDLAR CURRENT SOURCE

NOTE

- If an output current in mA is needed, then a basic current mirror can be used.
- If a smaller output current, i.e. in μA , is needed, then the basic current mirror cannot be used: to achieve a current in μA a large resistor ($\text{M}\Omega$) would be needed, which again requires a greater chip area. Hence this becomes the limitation of the basic current mirror.

★ If a smaller current, i.e. in μA , is needed, then a **Widlar current source** can be used.



WIDLAR CURRENT SOURCE

NOTE

Q_1 and Q_2 are transistors having a large β , hence the base currents can be ignored.

$$V_{BE1} = V_{BE2} + I_{out} \cdot R_E \dots (i)$$

As $V_{BE1} > V_{BE2}$:

$$I_{out} \ll I_{in}$$

NOTE

For a BJT, I_C and I_E depend exponentially on V_{BE} . Hence a small change in V_{BE} can create a large change in current. This property of the BJT is used in the Widlar current source to achieve a smaller output current.

As $V_{BE} = V_T \ln\left(\frac{I_E}{I_{ES}}\right)$:

For Q_1 : $V_{BE1} = V_T \ln\left(\frac{I_{in}}{I_{ES}}\right) \dots (2)$

For Q_2 : $V_{BE2} = V_T \ln\left(\frac{I_{out}}{I_{ES}}\right) \dots (3)$

From equation (i):

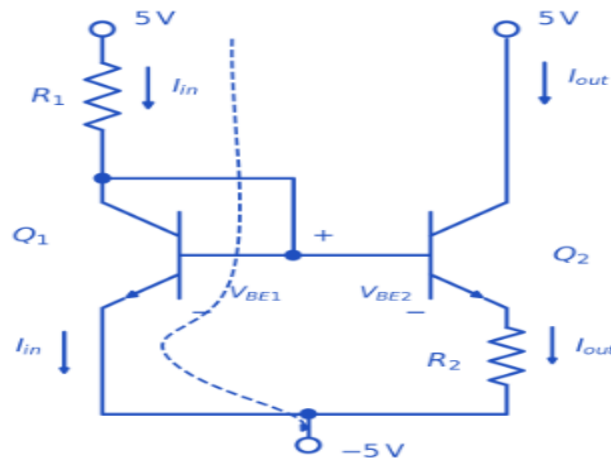
$$I_{out} \cdot R_E = V_{BE1} - V_{BE2} = V_T \ln\left(\frac{I_{in}}{I_{ES}}\right) - V_T \ln\left(\frac{I_{out}}{I_{ES}}\right)$$

$$I_{out} \cdot R_E = V_T \ln \left(\frac{I_{in}}{I_{out}} \right)$$

expression used to find R_E in a Widlar current source

PROBLEM 6

In the circuit shown, the transistors have a large β and Q_1 has $V_{BE} = 0.7$ V. If $I_{in} = 1$ mA and $I_{out} = 12$ μ A, calculate the values of R_1 and R_2 .



SOLUTION

By KVL:

$$5 = I_{in} R_1 + V_{BE1} - 5$$

$$R_1 = \frac{10 - V_{BE}}{I_{in}} = \frac{10 - 0.7}{1\text{mA}}$$

$$R_1 = 9.3k\Omega$$

For the Widlar current source:

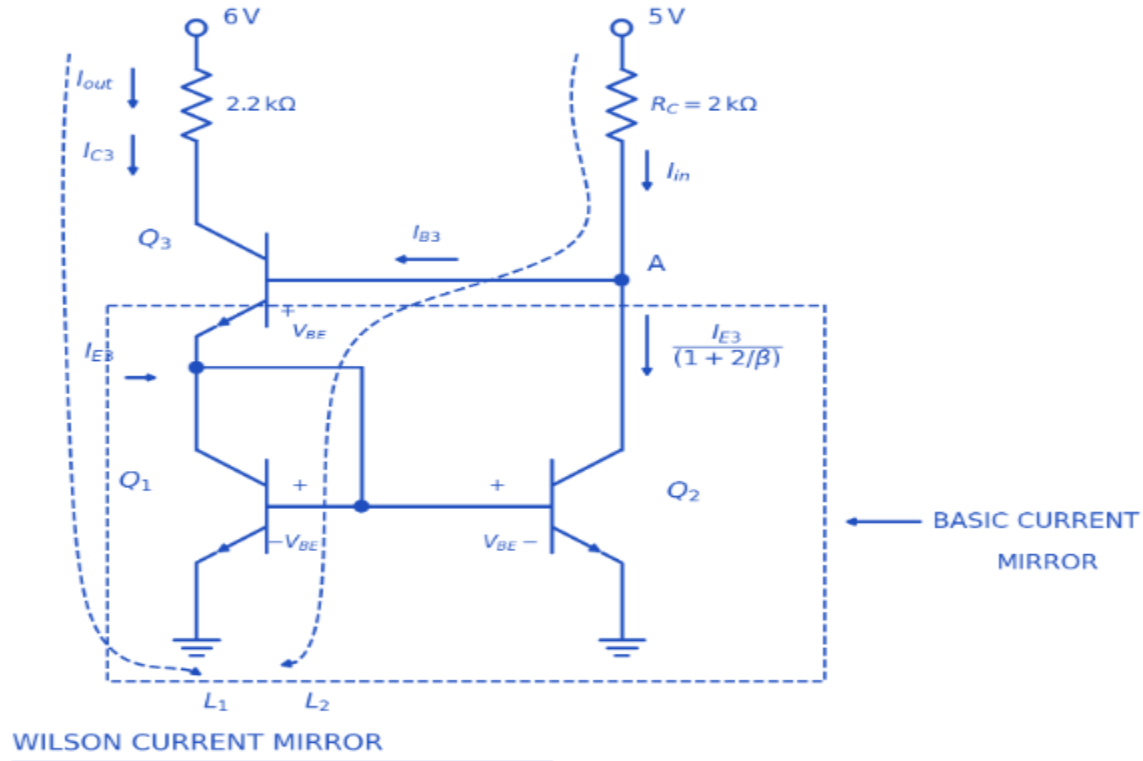
$$I_{out} R_2 = V_T \ln \left(\frac{I_{in}}{I_{out}} \right)$$

$$R_2 = \frac{V_T}{I_{out}} \ln \left(\frac{I_{in}}{I_{out}} \right) = \frac{0.026}{12 \times 10^{-6}} \ln \left(\frac{1 \times 10^{-3}}{12 \times 10^{-6}} \right)$$

$$R_2 = 9.58k\Omega$$

PROBLEM 7

In the circuit shown the transistors are identical. (a) Find an expression for I_{out} in terms of I_{in} . (b) If $\beta = 100$ and $V_{BE} = 0.7$ V, calculate the output current. (c) If the resistance R_C is varied from 1 k Ω to 4 k Ω , calculate the range of the output current.



WILSON CURRENT MIRROR

SOLUTION

(a) Q_1 and Q_2 form a basic current mirror. Applying KCL at node A:

$$I_{in} = I_{B3} + \frac{I_{E3}}{1 + \frac{2}{\beta}} \quad \text{As } \left\{ I_E = I_B + I_C = I_C \left(1 + \frac{1}{\beta} \right) \right\}$$

$$I_{in} = \frac{I_{C3}}{\beta} + \frac{I_{C3} \left(1 + \frac{1}{\beta} \right)}{\left(1 + \frac{2}{\beta} \right)}$$

$$I_{in} = I_{C3} \left[\frac{1}{\beta} + \frac{1 + \beta}{2 + \beta} \right]$$

$$= I_{out} \left[\frac{2 + \beta + \beta(1 + \beta)}{\beta(2 + \beta)} \right]$$

$$= I_{\text{out}} \left[\frac{2 + \beta(2 + \beta)}{\beta(2 + \beta)} \right]$$

$$I_{\text{in}} = I_{\text{out}} \left[\frac{2}{\beta(2 + \beta)} + 1 \right]$$

Hence:

$$I_{\text{out}} = \frac{I_{\text{in}}}{1 + \frac{2}{\beta(2 + \beta)}}$$

NOTE

In a Wilson current mirror $I_{\text{out}} \approx I_{\text{in}}$ even if the transistors have a smaller value of β .

The Wilson mirror circuit has a greater output resistance than the basic mirror circuit: $R_{\text{out}} = \frac{\beta}{2} r_o$.

(b) Given $\beta = 100$ and $V_{\text{BE}} = 0.7 \text{ V}$. By the Wilson current mirror, $I_{\text{out}} \approx I_{\text{in}}$. KVL in the input loop:

$$5 = I_{\text{in}} R_C + V_{\text{BE}} + V_{\text{BE}} \Rightarrow I_{\text{in}} = \frac{5 - 2V_{\text{BE}}}{R_C}$$

$$I_{\text{in}} = 1.8 \text{ mA} \Rightarrow I_{\text{out}} = 1.8 \text{ mA}$$

(c) When $1 \text{ k}\Omega < R_C < 4 \text{ k}\Omega$:

$$\text{When } R_C = 1 \text{ k}\Omega: I_{\text{in}} = \frac{5 - 2V_{\text{BE}}}{R_C} = \frac{5 - 1.4}{1 \text{ k}\Omega} = 3.6 \text{ mA} \approx I_{\text{out}}$$

When $R_C = 4 \text{ k}\Omega$:

$$I_{\text{in}} = \frac{5 - 1.4}{4 \text{ k}\Omega} = 0.9 \text{ mA} \approx I_{\text{out}}$$

NOTE

If $I_{\text{out}} = 3.6 \text{ mA}$, then the voltage drop across the $2.2 \text{ k}\Omega$ would exceed the 6 V supply. As a result, the V_{CE} of Q_3 becomes negative, which indicates that Q_3 is in the saturation region. (For an NPN transistor V_{CE} cannot be negative.)

Hence I_{out} should be calculated by considering Q_3 in the saturation region, i.e. by taking $V_{\text{CE3}} = V_{\text{CE(sat)}} = 0.2 \text{ V}$.

KVL in loop 2:

$$6 = 2.2 I_{\text{out}} + V_{\text{CE3}} + V_{\text{BE1}}$$

$$I_{out} = \frac{6 - V_{CE3} - V_{BE1}}{2.2} = \frac{6 - 0.2 - 0.7}{2.2}$$

$$I_{out} = 2.318\text{mA}$$

When R_C is varied between $1\text{ k}\Omega$ and $4\text{ k}\Omega$, I_{out} will lie in the range:

$$0.9\text{mA} \leq I_{out} \leq 2.318\text{mA}$$

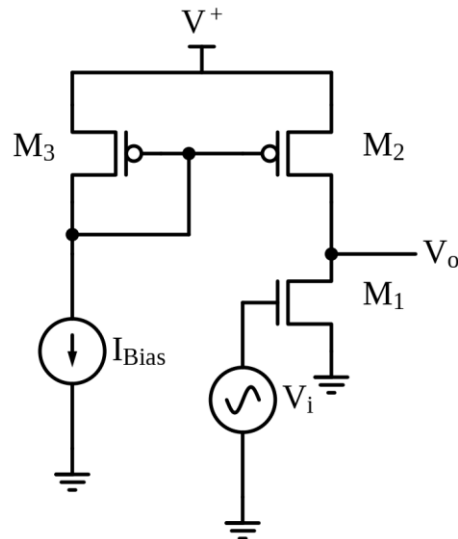
NOTE

In the above circuit, I_{out} and I_{in} become unequal when Q_3 is in the saturation region. Hence the circuit will not act as a current mirror.

For proper operation of a current mirror, all transistors must remain in the **ACTIVE REGION** (in a MOSFET mirror, all MOSFETs should be in the **SATURATION REGION**).

Q. Calculate the small-signal voltage gain of the amplifier shown in Figure 1, if the MOSFETs have:

- $\mu_n C_{ox} = 80\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox} = 40\text{ }\mu\text{A/V}^2$
- $(W/L)_N = 15$, $(W/L)_P = 30$
- $\lambda_n = 0.01\text{ V}^{-1}$, $\lambda_p = 0.02\text{ V}^{-1}$
- $I_{Bias} = 0.2\text{ mA}$



CMOS common-source amplifier (M_1) with PMOS current-mirror load (M_2, M_3)

Solution:

- (i) N-channel MOSFET M_1 is a **common-source amplifier**.
- (ii) P-channel MOSFETs M_2 and M_3 form a **current mirror**, which is used as the **active load** for M_1 .

(iii) The output resistance of the current mirror ($R_{out} = r_{ds2}$) acts as the load of M_1 .

Redrawing the circuit for small-signal analysis:

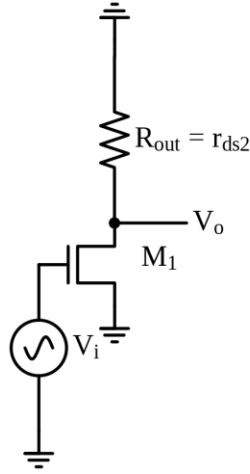


Figure 2: Small-signal equivalent – current mirror replaced by its output resistance r_{ds2}

$$A_v = V_o / V_i = -g_{m1} R_L''$$

$$R_L'' = R_{out} \parallel r_{ds1} = r_{ds2} \parallel r_{ds1}$$

$$A_v = -g_{m1} (r_{ds2} \parallel r_{ds1})$$

Calculation:

Since M_2 and M_3 have the same (W/L), the mirror copies the bias current exactly:

$$I_{DSN} = I_{SDP} = I_{Bias} = 0.2 \text{ mA}$$

Transconductance of M_1 :

$$\begin{aligned} g_{m1} &= \sqrt{2k_n I_{DS}} = \sqrt{2 \times \mu_n C_{ox} (W/L) \times I_{DS}} \\ &= \sqrt{2 \times 80 \times 10^{-6} \times 15 \times 0.2 \times 10^{-3}} \end{aligned}$$

$$g_{m1} = 0.693 \text{ mA/V}$$

Output resistance of M_1 (NMOS):

$$r_{ds1} = 1 / (\lambda_n I_{DSN}) = 1 / (0.01 \times 0.2 \times 10^{-3})$$

$$r_{ds1} = 500 \text{ k}\Omega$$

Output resistance of M_2 (PMOS):

$$r_{ds2} = 1 / (\lambda_p I_{SDP}) = 1 / (0.02 \times 0.2 \times 10^{-3})$$

$$r_{ds2} = 250 \text{ k}\Omega$$

Voltage gain:

$$r_{ds2} \parallel r_{ds1} = (250 \times 500) / (250 + 500) = 166.67 \text{ k}\Omega$$

$$A_v = -0.693 \times 10^{-3} \times 166.67 \times 10^3$$

$$A_v \approx -115.5$$

The negative sign shows a 180° phase shift between input and output.

Note:

- When a current mirror is used as an **active load**, a high voltage gain can be achieved, because the output resistance of the current mirror is high.
- The above circuit is called a **CMOS common-source amplifier**, in which NMOS is used as the amplifier and the PMOS current mirror is used as the load.
- NMOS is preferred as the amplifier over PMOS because it has greater transconductance (g_m), since electron mobility is higher than hole mobility ($\mu_n > \mu_p$).